Electrostatic is a combination of 2 words electro and statics elctro means electric charge and static means at rest. electrostatics, the study of electromagnetic phenomena that occur when there are no moving charges—i.e., after a static equilibrium has been established. Charges reach their equilibrium positions rapidly, because the electric force is extremely strong.

Summary of Lecture 22 – ELECTROSTATICS I

According to Coulomb's law, the force of attraction or repulsion between two charged bodies is directly proportional to the product of their charges and inversely proportional to the square of the distance between them.

1. Like charges repel, unlike charges attract. But by how much? Coulomb's Law says that this depends both upon the strength of the two charges and the distance between them. In mathematical terms, $F \propto \frac{q_1 q_2}{r^2}$ which can be converted into an equality, $F = k\frac{q_1 q_2}{r^2}$. The constant of proportionality will take different values depending upon the units we choose. In the MKS system, charge is measured in Coulombs (C) and $k = \frac{1}{4\pi\varepsilon_0}$ with $\varepsilon_0 = 8.85\times10^{-12}\,C^2/Nm^2$ and hence $k = 8.99\times10^9\,Nm^2/C^2$.

Yeh situation gravity jaisi hi hai, lekin yahan force mass ki wajah se nahi, electric charges ki wajah se hoti hai.

2. The situation is quite similar to that of gravity, except that electric charges and not masses are the source of force. In vector form, $\vec{F}_{12} = \frac{1}{4\pi\varepsilon_0}\frac{q_1 q_2}{r_{12}^2}\hat{r}_{12}$ is the force exerted by 2 on 1, where the unit vector is $\hat{r}_{12} = \frac{\vec{r}_{12}}{r_{12}}$. On the other hand, $\vec{F}_{21} = \frac{1}{4\pi\varepsilon_0}\frac{q_1 q_2}{r_{12}^2}\hat{r}_2$ is the force exerted by 1 on 2. By Newton's Third Law, $\vec{F}_{12} = -\vec{F}_{21}$. For many charges, the force on charge 1 is given by, $\vec{F}_1 = \vec{F}_{12} + \vec{F}_{13} + \vec{F}_{14} + \cdots$

Yani jitni force 2 ne 1 par lagayi, utni hi 1 ne 2 par lagayi, lekin ulte direction mein.

Charge har kisam ke amount mein nahi hota, ye sirf choti choti fixed units mein hota hai jaise e, 2e, 3e, etc.

3. Charge is quantized. This means that charge comes in certain units only. So the size of a charge can only be 0, $\pm e, \pm 2e, \pm 3e, \cdots$ where $e = 1.602\times10^{-19}\,C$ is the value of the charge present on a proton. By definition we call the charge on a proton positive. This makes the charge on an electron negative.

Charge na kabhi banaya ja sakta hai aur na khatam. Kisi bhi reaction ya event ke pehle aur baad total charge hamesha same rehta hai.

4. Charge is conserved. This means that charge is never created or destroyed. Equivalently, in any possible situation, the total charge at an earlier time is equal to the charge at a later time. For example, in any of the reactions below the initial charge = final charge:

$e^- + e^+ \to \gamma + \gamma$ (electron and positron annihilate into neutral photons)

$\pi^0 \to \gamma + \gamma$ (neutral pion annihilates into neutral photons)

$H^2 + H^2 \to H^3 + p$ (two deuterons turn into tritium and proton)

Field wo cheez hai jo space mein kisi bhi point par value batati hai.

5. **Field** : this a quantity that has a definite value at any point in space and at any time. The simplest example is that of a scalar field, which is a *single number* for any value of x, y, z, t. Examples: temperature inside a room $T(x, y, z, t)$, density in a blowing wind $\rho(x, y, z, t)$, $\cdots$ There are also *vector fields,* which comprise of three numbers at each value of x, y, z, t. Examples: the velocity of wind, the pressure inside a fluid, or even a sugarcane field. In

Scalar field: sirf aik value hoti hai (jaise temperature)
Vector field: direction aur value dono hoti hain (jaise hawa ka rukh ya electric field)

every case, there are 3 numbers: $\vec{V}(x,y,z,t)=\{V_1(x,y,z,t),V_2(x,y,z,t),V_3(x,y,z,t)\}$.

Electric field woh force hoti hai jo kisi choti test charge q_0 par lagti hai.

6. The electric field is also an example of a vector field, and will be the most important for our purpose. It is defined as the force on a unit charge. Or, since we don't want the charge to disturb the field it is placed in, we should properly define it as the force on a "test" charge q_0, $\boxed{E\equiv\frac{F}{q_0}}$. Here q_0 is very very small. The electric field due to a point charge can be calculated by considering two charges. The force between them is $\boxed{F=\frac{1}{4\pi\varepsilon_0}\frac{qq_0}{r^2}}$ and so $\boxed{E=\frac{F}{q_0}=\frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}}$. A way to visualize E fields is to think of lines starting on positive charges and ending on negative charges. The number of lines leaving/entering gives the amount of charge.

Yani agar kisi charge se doosri jagah par kisi test charge par force lagti hai, to woh electric field kehlata hai.

Lines positive charge sy nikilti hain or Negative charge par khatam hoti hain

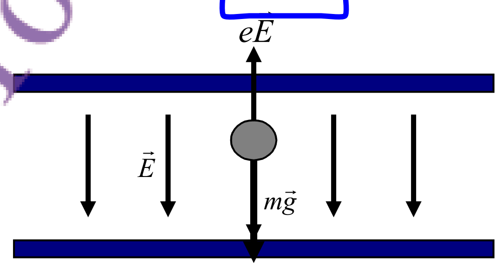
7. Typical values for the maginitude of the electric field E :

McQs

Inside an atom-	10^{11}	N/C
Inside TV tube-	10^{5}	N/C
In atmosphere-	10^{2}	N/C
Inside a wire-	10^{-2}	N/C

Agar ek charged particle par gravity ka force aur electric field ka force barabar ho jaaye, to hum charge *q* nikal sakte ha

8. Measuring charge. One way to do this is to balance the gravitational force pulling a charged particle with mass m with the force exerted on it by a known electric field (see below). For equilibrium, the two forces must be equal and so $\boxed{mg=qE}$. The unknown charge q can then be found from $\boxed{q=\frac{mg}{E}}$.



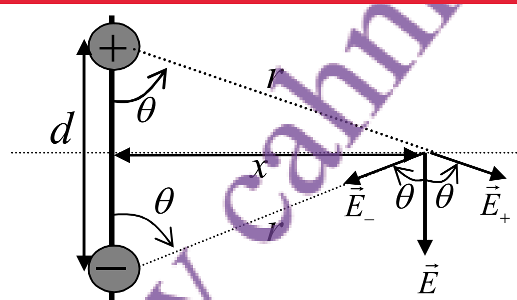
Agar bohot se charges hon, to har charge ki electric field alag alag calculate kar ke sab ko add kr do

9. Given several charges, one can find the total electric field at any point as the sum of the fields produced by the charges at that point individually, $\vec{E}=\vec{E}_1+\vec{E}_2+\vec{E}_3+\cdots$ or $\boxed{\vec{E}=\sum_i\vec{E}_i=k\sum_i\frac{q_i}{r_i^2}\hat{r}_i}$ $(i=1,2,3,\cdots)$. Here $\hat{r}_i$ is the unit vector pointing from the charge to the point of observation.

Agar do barabar magnitude wale charges (ek positive aur ek negative) d faaslay par hon, to unki electric fields opposite directions mein hoti hain. Horizontal components add hote hain Vertical components cancel ho jaate hain Final electric field downward hoti hai.

10. Let us apply the principle we have just learned to a system of two charges $+q$ and $-q$ which are separated by a distance d (see diagram). Then, $\vec{E} = \vec{E}_+ + \vec{E}_-$. Just to make things easier (not necessary; one can do it for any point) I have taken a point that lies on the x-axis. The magnitudes of the electric field due to the two charges are equal; $E_+ = E_- = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2} = \frac{1}{4\pi\varepsilon_0}\frac{q}{x^2+(d/2)^2}$. The vertical components cancel out, and the net electric field is directed downwards with magnitude $\boxed{E = E_+\cos\theta + E_-\cos\theta = 2E_+\cos\theta}$.
From the diagram below you can see that $\cos\theta = \frac{d/2}{\sqrt{x^2+(d/2)^2}}$. Substituting this, we find:

$$\boxed{E = 2\frac{1}{4\pi\varepsilon_0}\frac{q}{x^2+(d/2)^2}\frac{d/2}{\sqrt{x^2+(d/2)^2}} = \frac{1}{4\pi\varepsilon_0}\frac{qd}{\left[x^2+(d/2)^2\right]^{3/2}}}.$$



Dipole moment = charge × distance between charges

11. The result above is so important that we need to discuss it further. In particular, what happens if we are very far away from the dipole, meaning $x >> d$? Let us first define the *dipole moment* as the product of the charge × the separation between them $p = qd$.

Then, $$\boxed{E = \frac{1}{4\pi\varepsilon_0}\frac{p}{x^3}\frac{1}{\left[1+(d/2x)^2\right]^{3/2}} = \frac{1}{4\pi\varepsilon_0}\frac{p}{x^3}\left[1+\left(\frac{d}{2x}\right)^2\right]^{-3/2} = \frac{1}{4\pi\varepsilon_0}\frac{p}{x^3}}.$$ In the above, $d/2x$ has been neglected in comparison to 1. So finally, we have found that for $x >> d$,

$$\boxed{E = \frac{1}{4\pi\varepsilon_0}\frac{p}{x^3}}.$$

Agar point bohot door ho (x >> d), to field simplify ho jaati hai

Jab ek dipole ko uniform electric field mein rakha jaye, to us par torque (τ) lagta hai.

12. It is easy to find the torque experienced by an electric dipole that is placed in a uniform electric field: The magnitude is $\boxed{\tau = F\frac{d}{2}\sin\theta + F\frac{d}{2}\sin\theta = Fd\sin\theta}$ and the direction is perpendicular and into the plane. Here θ is the angle between the dipole and the electric field. So $\boxed{\tau = (qE)d\sin\theta = pE\sin\theta}$.

F=qE
P=qd

electrostatic is defined as force per unit charge
E = F / q

Agar charges kisi area mein continuously phailay hoon (na ke sirf point charges), to humein us area ko choti choti pieces mein divide karna padta hai. Hr choti piece ka apna electric field hoga, aur total electric field un sab ka sum hoga: E = ∑ ΔE
Ya agar infinite small pieces hon to: E= ∫dE
Electric field vector hota hai aur use x, y, z directions mein tod ke likh sakte hain.

Summary of Lecture 23 – ELECTROSTATICS II

1. In the last lecture we learned how to calculate the electric field if there are any number of point charges. But how to calculate this when charges are continuously distributed over some region of space? For this, we need to break up the region into little pieces so that each piece is small enough to be like a point charge. So, $\vec{E} = \Delta\vec{E}_1 + \Delta\vec{E}_2 + \Delta\vec{E}_3 + \cdots$, or $\vec{E} = \sum \Delta\vec{E}_i$ is the total electric field. Remember that $\vec{E}$ is a vector that can be resolved into components, $\vec{E} = E_x\hat{i} + E_y\hat{j} + E_z\hat{k}$. In the limit where the pieces are small enough, we can write it as an integral, $\vec{E} = \int d\vec{E}$ (or $E_x = \int dE_x,\ E_y = \int dE_y,\ E_z = \int dE_z$)

charge density is the amount of electric charge per unit length, surface area, or volume.

2. Charge Density: when the charges are continuously distributed over a region - a line, the surface of a material, or inside a sphere - we must specify the *charge density*. Depending upon how many dimensions the region has, we define:

   jab charge kisi line par ho (a) For linear charge distribution: represented by $dq = \lambda ds$ — λ --- dq/ds

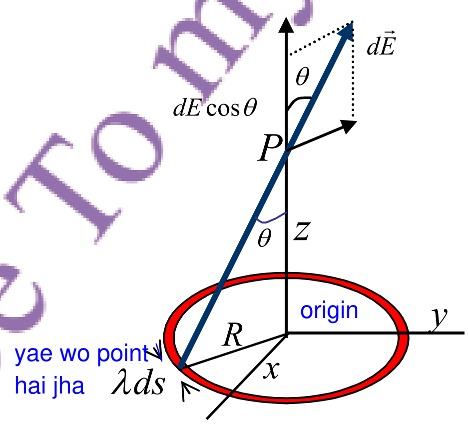
   jab charge kisi surface par ho (b) For surface charge distribution: $dq = \sigma dA$ — σ ---- dq/da

   jab charge kisi volume par ho (c) For volume charge distribution: $dq = \rho dV$ — ρ --- dq/dv

   Linear Charge Density (λ):
   The amount of charge per unit length. It's measured in Coulombs per meter (C/m).
   The amount of charge per unit surface area. It's measured in Coulombs per square meter (C/m²).
   Volume Charge Density (ρ):
   The amount of charge per unit volume. It's measured in Coulombs per cubic meter (C/m³).

   The dimensions of λ, σ, ρ are determined from the above definitions.

3. As an example of how we work out the electric field coming from a continuous charge distribution, let us work out the electric field from a uniform ring of charge at the point P.



The small amount of charge λds gives rise to an electric field whose magnitude is

$$dE = \frac{1}{4\pi\varepsilon_0}\frac{\lambda ds}{r^2} = \frac{\lambda ds}{4\pi\varepsilon_0\left(z^2+R^2\right)}$$

The component in the z direction is $dE_z = dE\cos\theta$ with $\cos\theta = \frac{z}{r} = \frac{z}{\left(z^2+R^2\right)^{1/2}}$.

So $dE_z = \dfrac{z\lambda ds}{4\pi\varepsilon_0\left(z^2+R^2\right)^{3/2}}$. Since s, which is the arc length, does not depend upon z or R,

Total electric field in z direction

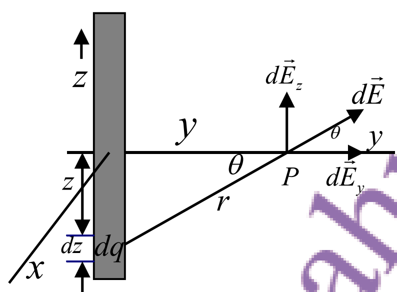
$$E_z = \frac{z\lambda}{4\pi\varepsilon_0\left(z^2+R^2\right)^{3/2}}\int ds = \frac{z\lambda(2\pi R)}{4\pi\varepsilon_0\left(z^2+R^2\right)^{3/2}} = \boxed{\frac{qz}{4\pi\varepsilon_0\left(z^2+R^2\right)^{3/2}}}.$$ Answer!!

Agr point bhot door ho (z >> R), to field simplify ho jati hai Yani ring ek point charge jaisi lgti hai.

Note that if you are very far away, the ring looks like a point: $\boxed{E_z = \dfrac{1}{4\pi\varepsilon_0}\dfrac{q}{z^2},\ (z >> R)}$.

4. As another example, consider a continuous distribution of charges along a wire that lies along the z-axis, as shown below. We want to know the electric field at a distance x from the wire. By symmetry, the only non-cancelling component lies along the y-axis.

Jab wire par charge ho, to us se har point par electric field hoti hai.
Formula:
E = λ / (2πε₀r)
Yeh field wire ke ird gird hoti hai, aur direction radial hoti hai.



Applying Coulomb's law to the small amount of charge λdz along the z axis gives,

$$dE = \frac{1}{4\pi\varepsilon_0}\frac{dq}{r^2} = \frac{1}{4\pi\varepsilon_0}\frac{\lambda dz}{y^2+z^2}$$

the component along the y direction is $dE_y = dE\cos\theta$. Integrating this gives,

$$E_y = \int dE = \int_{z=-\infty}^{z=\infty}\cos\theta dE = 2\int_{z=0}^{z=\infty}\cos\theta dE = \frac{\lambda}{2\pi\varepsilon_0}\int_{z=0}^{z=\infty}\cos\theta\frac{dz}{y^2+z^2}.$$

The rest is just technical: to solve the integral, put $z = y\tan\theta \Rightarrow dz = y\sec^2\theta d\theta$. And so,

$E = \dfrac{\lambda}{2\pi\varepsilon_0 y}\displaystyle\int_{\theta=0}^{\theta=\pi/2}\cos\theta d\theta = \dfrac{\lambda}{2\pi\varepsilon_0 y}$. Now, we could have equally well taken the x axis. The only thing that matters is the distance from the wire, and so the answer is better written as:

$$\boxed{E = \frac{\lambda}{2\pi\varepsilon_0 r}.}$$

flux.......
the rate of flow of electric field through a given surface
the dot product of electric filed and area of surface is called flux

5. The **flux** of any vector field is a particularly important concept. It is the measure of the "flow" or penetration of the field vectors through an imaginary fixed surface. So, if there is a uniform electric field that is normal to a surface of area A, the flux is $\boxed{\Phi = EA.}$ More generally, for any surface, we divide the surface up into little pieces and take the

formula of flux

Flux ka matlab ha electric field ka kisi surface se guzarna component of the electric field normal to each little piece $\boxed{\Phi_E = \sum \vec{E}_i \cdot \Delta\vec{A}_i}$. If the pieces are made small enough, then in this limit, $\boxed{\Phi = \int \vec{E} \cdot d\vec{A}}$.

$A = 4\pi r^2$

Jab charge q sphere ke centre mein ho to har direction mein barabar field hoti hai.
Surface area of sphere = $4\pi r^2$

5. Let us apply the above concept of flux to calculate the flux leaving a sphere which has a charge at its centre. The electric field at any point on the sphere has magnitude equal to $E = \boxed{\frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}}$ and it is directed radially outwards. Let us now divide up the surface of the sphere into small areas. Then $\boxed{\Phi = \sum E\Delta A = E\sum \Delta A = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}\left(4\pi r^2\right)}$. So we end up with the important result that the flux leaving this closed surface is $\boxed{\Phi = \frac{q}{\varepsilon_0}}$.

Definition of Gauss's law:
Gauss's law states that the electric flux Φ across any closed surface is proportional to the net electric charge q enclosed by the surface that is, Φ = q/ε0, where ε0 is the electric permittivity of free space and has a value of 8.854 × 10^−12 square coulombs per newton per square metre

Kisi bhi band (closed) surface se nikalne wala total electric flux us surface ke andar mojood total charge ke barabar hota hai, ε_0 se divide karne ke baad.

6. **Gauss's Law**: the total electric flux leaving a closed surface is equal to the charge enclosed by the surface divided by ε_0. We can express this directly in terms of the mathematics we have learned, $\boxed{\Phi \equiv \int \vec{E} \cdot d\vec{A} = \frac{q_{enclosed}}{\varepsilon_0}}$. Actually, we have already seen why this law is equivalent to Coulomb's Law in point 5 above, but let's see it again. So, applying Gauss's Law to a sphere containing charge, $\varepsilon_0 \int \vec{E} \cdot d\vec{A} = \varepsilon_0 \int E\,dA = q_{enclosed}$. If the surface is a sphere, then E is constant on the surface and $\varepsilon_0 E \int dA = q$ and from this $\varepsilon_0 E\left(4\pi r^2\right) = q \Rightarrow \boxed{E = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}}$. This is Coulomb's Law again, but the power of Gauss's law is that it holds for any shape of the (closed) surface and for any distribution of charge.

Agar charge sirf sphere ke surface par ho, to Sphere ke andar koi charge nahi HOTA or Electric field = 0 Sphere ke bahar total charge q hota hai $E = (1 / 4\pi\varepsilon_0) \times (q / r^2)$ yani jaise poora charge centre mein ho.

7. Let us apply Gauss's Law to a hollow sphere that has charges only on the surface. At any distance r from the centre, Gauss's Law is $\varepsilon_0 E\left(4\pi r^2\right) = q_{enclosed}$. Now, if we are inside the sphere then $\boxed{q_{enclosed} = 0}$ and there is $\boxed{\text{no electric field}}$. But if we are outside, then the total charge is $\boxed{q_{enclosed} = q \text{ and } E = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}}$, which is as if all the charge was concentrated at the centre.

8. Unfortunately, it will not be possible for me to prove Gauss's Law in the short amount of time and space available but the general method can be outlined as follows: take any volume and divide it up into little cubes. Each little cube may contain some small amount of charge. Then show that for each little cube, Gauss's Law follows from Coulomb's Law. Finally, add up the results. For details, consult any good book on electromagnetism.

• Kisi bhi volume ko chhoti chhoti cubes mein divide karo
• Har cube mein thoda sa charge ho sakta hai
• Har cube ke liye Coulomb's Law se show karo ke Gauss's Law apply hota hai
• Sb cubes ka result add kar do

Gravitational potential energy is the energy an object has due to its position above Earth, energy due to its height
Formula=> U = mgh

Electric potential energy is a measure of the potential energy between two charged particles.
.SI unit: joule (J)
Symbol: U_E
Formula: U_E=kq_1 q_2/r

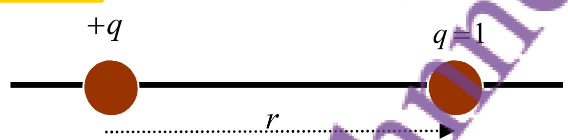
Summary of Lecture 24 – ELECTRIC POTENTIAL ENERGY

Potential energy •••• Jab hum kisi cheez ko upar uthate hain (jaise wazan), to humein gravity ke khilaf kaam karna padta hai. Yehi kaam potential energy ki shakal mein store ho jata hai.

1. You are already familiar with the concept of gravitational potential energy. When you lift a weight, you have to do work against the downwards pull of the Earth. That work is stored as potential energy. Suppose a force $\vec{F}$ acts on something and displaces it by $d\vec{s}$. Then the work done is $\vec{F}\cdot d\vec{s}$. The work done in going from point a to point b (call it W_{ab}) is then got by adding together the little bits of work, $\boxed{W_{ab}=\int_a^b \vec{F}\cdot d\vec{s}}$. The change in potential energy is defined as $\boxed{\Delta U = U_b - U_a = -W_{ab}}$. Remember always that we can only define the potential U at a point if the force is conservative.

Electric Potential Energy:
Agar 2 charges hoon, jaise +q aur +1, aur unke darmiyan distance r ho, to unko paas laane ke liye work karna padta hai, kyunki woh ek dosre ko repel karte hain. Formula: U = (1 / 4πε₀) × (q / r)

2. The electrostatic force is conservative and can be represented by a potential. Let us see how to calculate the potential. So consider two charges separated by a distance as below.



Let us take the point a very far from the fixed charge q, and the unit charge at the point b to be at a distance R from q. Then the work you did in bringing the unit charge from infinity

q ki value put ye constant ko bahir answer

to R is, $W_{ab}=\int_\infty^R(-qE)dr=-\frac{1}{4\pi\varepsilon_0}\int_\infty^R dr\frac{q}{r^2}=-\frac{1}{4\pi\varepsilon_0}q\left(\frac{1}{R}-\frac{1}{\infty}\right)=-\frac{q}{4\pi\varepsilon_0}\frac{1}{R}$. Since the charges repel each other, it is clear that you had to do work in pushing the two charges closer together. So where did the negative sign come from? Answer: the force you exert on the unit charge is directed *towards* the charge q, i.e. is in the negative direction. This is why $\boxed{\vec{F}\cdot d\vec{s}=(-qE)dr. \text{ Now } \Delta U = U(R)-U(\infty)=\frac{q}{4\pi\varepsilon_0}\frac{1}{R}}$. If we take the potential at ∞ to be zero, then the electric potential due to a charge q at the point r is $\boxed{U(r)=\frac{q}{4\pi\varepsilon_0}\frac{1}{r}}$.

Remember that we know how to calculate the force given the potential $\boxed{F=-\frac{dU}{dr}}$. Apply this here and you see that $\boxed{F=\frac{q}{4\pi\varepsilon_0}\frac{1}{r^2}}$, (which also has the correct repulsive sign).

3. From the above, it is quite obvious that the potential energy of two charges q_1,q_2 is, $\boxed{U(r)=\frac{1}{4\pi\varepsilon_0}\frac{q_1q_2}{r}}$. Compare this with the formula for gravitational energy $\boxed{U(r)=-G\frac{m_1m_2}{r}}$. What is the difference? From here, you can see that the gravitational force is always negative (which means attractive), whereas the electrostatic force can be both attractive

•) Agr charges opposite hoon (jaise +q aur -q), to energy negative hogi iska matlab attractive force hai.
•) Agr charges same hoon (jaise +q aur +q), to energy positive matlab repulsive force hai.

Gravitational energy hmesha negative hoti hai (hamesha attractive).
Electric energy positive bhi ho sakti hai aur negative bhi (repel ya attract dono ho sakta hai).

or repulsive because we have both + and - charges in nature.

Electric potential ka matlab hota hai 1 unit charge ko electric field mein le jaane ki energy.

3. The electric potential (or simply potential) is the energy of a unit charge in an electric field.

So, in our MKS units, the unit of potential is $\boxed{1\ \dfrac{\text{Joule}}{\text{Coulomb}} = 1\ \text{Volt}}$. Another useful unit is "electron volt" or eV. The definition is:

One electron - volt = energy gained by moving one electron charge through one Volt

$$= \left(1.6\times10^{-19}\,C\right)\times 1V = 1.6\times10^{-19}\,J$$

It is useful to note that

$$\boxed{\begin{array}{ll} 1\ \text{Kev} = 10^{3}eV & \text{(kilo-electron-volt)} \\ 1\ \text{Mev} = 10^{6}eV & \text{(million-electron-volt)} \\ 1\ \text{Gev} = 10^{9}eV & \text{(giga-electron-volt)} \\ 1\ \text{Tev} = 10^{12}eV & \text{(tera-electron-volt)} \end{array}}$$

Positive charges hamesha low potential ki taraf move karte hain.
Negative charges high potential ki taraf jate hain

4. Every system seeks to minimize its potential energy (that is why a stone falls down!). So, positive charges accelerate toward regions of lower potential, but negative charges accelerate toward regions of higher potential. Note that only the potential difference matters - even if a charge is placed in a region where there is a high potential, it will not want to move unless there is some other place where the potential is higher/lower.

Note karo ke sirf potential difference matter karta hai chahe charge kisi aisi jagah ho jahan potential zyada ho, wo tab tak move nahi karega jab tak koi doosri jagah ho jahan potential zyada ya kam ho.

5. Given a system of charges, we can always compute the force - and hence the potential - that arises from them. Here are some important general statements:

a)Potentials are more positive in regions which have more positive charge. Jahan zyada positive charge hota hai, wahan potential zyada hota hai

b)The electric potential is a scalar quantity (a scalar field, actually). Scalar quantity because direction nahi hoti

c)The electric potential determines the force through $F = -\dfrac{dU}{dr}$, and hence the electric field because $F = qE$. Force aur potential ka relation

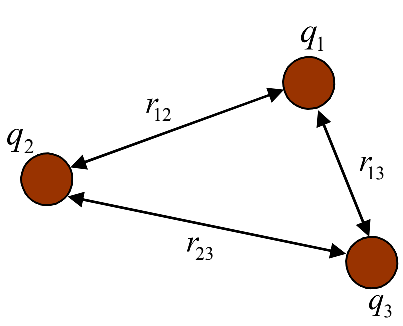
d)The electric potential exists only because the electrostatic force is conservative. Electric potential tabhi hota hai jab electrostatic force conservative ho

Agar kai charges hain, to total potential kaise nikalein?

6. To compute the potential at a point, the potentials arising from charges 1, 2, $\cdots$ N must be added up:

Sab charges ka potential add karna hota hai

$$\boxed{V = V_1 + V_2 + \cdots V_N = \sum_{i=1}^{N} V_i = \frac{1}{4\pi\varepsilon_0}\sum_{i=1}^{N}\frac{q_i}{r_i}}.$$ Here r_i is the distance of the i'th charge from the point where the potential is being calculated or measured. As an example, the potential from the three charges is:

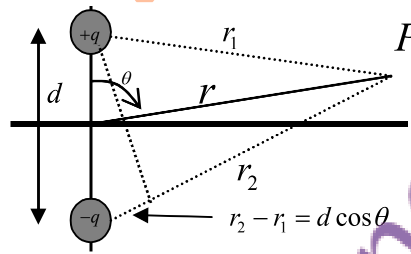


$$\boxed{V(r) = \frac{1}{4\pi\varepsilon_0}\frac{q_1 q_2}{r_{12}} + \frac{1}{4\pi\varepsilon_0}\frac{q_1 q_3}{r_{13}} + \frac{1}{4\pi\varepsilon_0}\frac{q_2 q_3}{r_{23}}}.$$

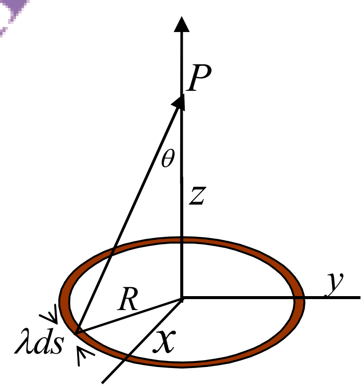
Dipole ka potential:
Agar hum 2 charges ka system lein (+q aur -q) jo ek choti distance d se alag hain, to unka potential P point par hota ha

7. Let us apply these concepts to the dipole system considered earlier. With two charges, $\boxed{V_P = V_1 + V_2 = \frac{1}{4\pi\varepsilon_0}\left(\frac{q}{r_1} + \frac{-q}{r_2}\right) = \frac{q}{4\pi\varepsilon_0}\frac{r_2 - r_1}{r_1 r_2}}$. We are particularly interested in the situation where $r >> d$. from the diagram you can see that $r_2 - r_1 \approx d\cos\theta$ and that $r_1 r_2 \approx r^2$. Hence, $\boxed{V \approx \frac{q}{4\pi\varepsilon_0}\frac{d\cos\theta}{r^2} = \frac{1}{4\pi\varepsilon_0}\frac{p\cos\theta}{r^2}}$. So we have calculated the potential at any θ with such little difficulty. Note that $V = 0$ at $\theta = \frac{\pi}{2}$.

jab θ = 90° hota hai, to cos(θ) = 0, is liye potential V = 0 ho jata hai.



8. Now let us calculate the potential which comes from charges that are uniformly spread over a ring. This is the same problem as in the previous lecture, but simpler. Give the small amount of potential coming from the small amount of charge $dq = \lambda ds$ some name, $dV = \frac{1}{4\pi\varepsilon_0}\frac{dq}{r^2}$. Then obviously



Total potential

$$\boxed{V = \int dV = \frac{1}{4\pi\varepsilon_0}\int\frac{dq}{r^2} = \frac{1}{4\pi\varepsilon_0}\frac{q}{\sqrt{R^2 + z^2}}.}$$

Ring ke upar uniformly spread charge ka potential:
Agar charge ek ring par uniformly spread ho, to har choti choti length ds par ek choti si charge dq hoti hai.

q = total charge
R = ring ka radius
z = height of point P above the ring

imp: capacitor srlease 360 joule/ sec

A capacitor is an electrical component that stores electrical energy. It consists of two conductive plates separated by a dielectric material, which acts as an insulator. Capacitors are used in various applications, including tuning radios, smoothing voltage, and storing energy in circuits.

Jab 2 conductors (plates) ek doosre se alag rakhein jayein aur unke darmiyan battery lagayi jaye, to unke beech ek potential difference create hota hai. Electric field unhi charges ki wajah se banti hai jo battery ne plates par bheje hote hain.

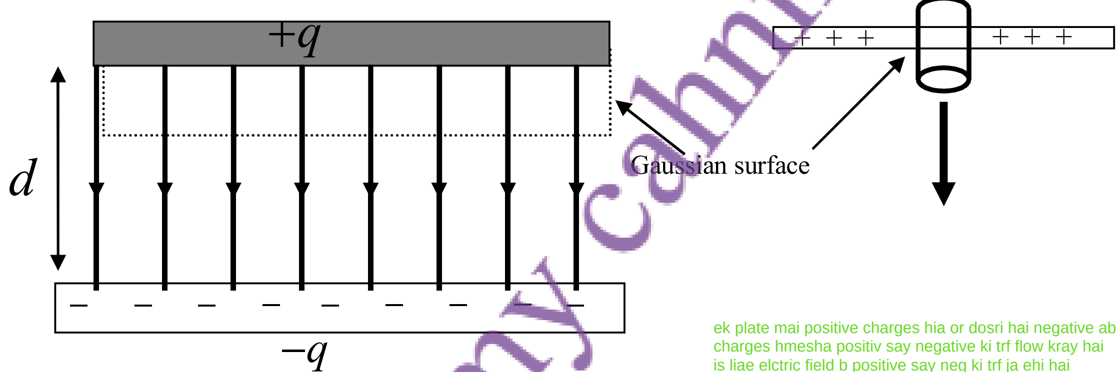
Summary of Lecture 25 – CAPACITORS AND CURRENTS

Current ek flow hota hai electric charge ka kisi conductor (jaise wire) ke through.

1. Two conductors isolated from one another and from their surroundings, form a *capacitor*. These conductors may be of any shape and size, and at any distance from each other. If a potential difference is created between the conductors (say, by connecting the terminals of a battery to them), then there is an electric field in the space between them. The electric field comes from the charges that have been pushed to the plates by the battery. The amount of charge pushed on to the conductors is proportional to the potential difference between the battery terminals (which is the same as between the capacitor plates). Hence, $Q \propto V$. To convert this into an equality, we write $\boxed{Q = CV}$. This provides the definition of capacitance, $\boxed{C = \frac{Q}{V}}$.

how to calculate capacitor

2. Using the above definition, let us calculate the capacitance of two parallel plates separated by a distance d as in the figure below.



ek plate mai positive charges hia or dosri hai negative ab charges hmesha positiv say negative ki trf flow kray hai is liae elctric field b positive say neg ki trf ja ehi hai

Recall Gauss's Law: $\Phi \equiv \int \vec{E} \cdot d\vec{A} = \frac{q_{enclosed}}{\varepsilon_0}$. Draw any Gaussian surface. Since the electric field is zero above the top plate, the flux through the area A of the plate is $\Phi = EA = \frac{Q}{\varepsilon_0}$, where Q is the total charge on the plate. Thus, $\boxed{E = \frac{Q}{\varepsilon_0 A}}$ is the electric field in the gap between the plates. The potential difference is $\boxed{V = \frac{E}{d}}$, and so $\boxed{C = \frac{Q}{V} = \frac{\varepsilon_0 A}{d}}$. You can see that the capacitance will be large if the plates are close to each other, and if the plates have a large area. We have simplified the calculation here by assuming that the electric field is strictly directed downwards. This is only true if the plates are infinitely long. But we can usually neglect the side effects. Note that any arrangement with two plates forms a capacitor: plane, cylindrical, spherical, etc. The capacitance depends upon the geometry, the size of plates and the gap between them.

Important points: ✨
⭐ Capacitance ka size plates ke area aur unke darmiyan faaslay (distance) par depend karta hai.
⭐ Charges hamesha positive se negative ki taraf flow karte hain, is liye electric field bhi usi direction mein hoti hai.

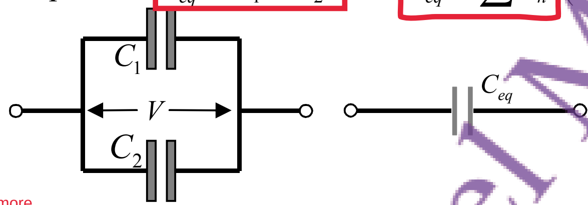
Voltage: All capacitors have the same voltage
Charge: Total charge is the sum of individual charges
Capacitance: Total capacitance increases with adding more
Effective Effect: Increases effective plate area (higher capacitance)

Capacitors in Parallel:

3. One can take two (or more) capacitors in various ways and thus change the amount of charge they can contain. Consider first two capacitors connected in parallel with each other. The same voltage exists across both. For each capacitor, $q_1 = C_1V$, $q_2 = C_2V$ where V is the potential between terminals a and b. The total charge is:

$$\boxed{Q = q_1 + q_2 = C_1V + C_2V = (C_1 + C_2)V}$$

Now, let us define an "effective" or "equivalent" capacitance as $C_{eq} = \frac{Q}{V}$. Then we can immediately see that for 2 capacitors $\boxed{C_{eq} = C_1 + C_2,}$ and $\boxed{C_{eq} = \sum C_n}$ (for n capacitors).



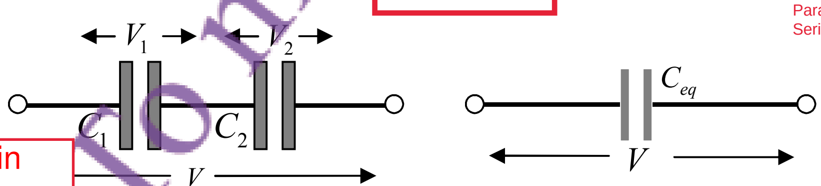
Voltage: Voltage drops add up
Charge: All capacitors have the same charge
Capacitance: Total capacitance decreases with adding more
Effective Effect: Decreases effective plate area (lower capacitance)

4. We can repeat the analysis above when the capacitors are put in series. Here the difference is that now we must start with $V = V_1 + V_2$, where V_1 and V_2 are the voltages across the two. Clearly the same charge had to cross both the capacitors. Hence,

$$\boxed{V = V_1 + V_2 = \frac{Q}{C_1} + \frac{Q}{C_2} = Q\left(\frac{1}{C_1} + \frac{1}{C_2}\right).}$$

From our definition, $C_{eq} = \frac{Q}{V}$, it follows that $\boxed{\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2}}$. The total capacitance is now less than if they were in parallel. In general, $\boxed{\frac{1}{C_{eq}} = \sum \frac{1}{C_n}}$ (for n capacitors).

Parallel m capacitance bas jor dete hain.
Series m reciprocals ka sum lete hain



energy stored in electric fields

5. When a battery is connected to a capacitor, positive and negative charges appear on the opposite plates. Some energy has been transferred from the battery to the capacitor, and now been stored in it. When the capacitor is discharged, the energy is recovered. Now let us calculate the energy required to charge a capacitor from zero to V volts.

Begin: the amount of energy required to transfer a small charge dq to the plates is $\boxed{dU = \mathrm{v}dq,}$ where v is the voltage at a time when the charge is $q = C\mathrm{v}$. As time goes on, the total charge increases until it reaches the final charge Q (at which point the voltage becomes V). So,

Battery capacitor ko charge karti hai, aur energy store hoti hai electric field m.
capacitor mein energy store hoti hai jo baad mein discharge hone par wapas mil sakti hai.

$$dU = vdq = \frac{q}{C}dq \quad \Rightarrow \quad U = \int dU = \int_0^Q \frac{q}{C}dq = \frac{Q^2}{2C} = \frac{1}{2}CV^2.$$

Total stored energy: U = ½ CV²

But where in the capacitor is the energy stored? Answer, it is present in the electric field in the volume between the two plates. We can calculate the energy density:

$$u = \frac{\text{energy stored in capacitor}}{\text{volume of capacitor}} = \frac{U}{Ad} = \frac{\frac{1}{2}CV^2}{Ad} = \frac{\varepsilon_0}{2}\left(\frac{V}{d}\right)^2 = \frac{1}{2}\varepsilon_0 E^2.$$

In the above we have used $C = \frac{\varepsilon_0 A}{d}$, derived earlier. The important result here is that $u \propto E^2$. Turning it around, wherever there is an electric field, there is energy available.

Important point:
Jahan electric field hota hai,
wahan energy bhi hoti ha.

A dielectric material is a poor conductor of electricity but an efficient supporter of electrostatic fields. It can store electrical charges.

6. **Dielectrics.** Consider a free charge $+Q$. Around it is an electric field, $E = \frac{1}{4\pi\varepsilon_0\varepsilon_r}\frac{Q}{r^2}$.

Now suppose this charge is placed among water molecules. These molecules will polarise, i.e. the centre of positive charge and centre of negative charge will be slightly displaced. The negative part of the water molecule will be attracted toward the positive charge $+Q$. So, in effect, the electric field is weakened by $\frac{1}{\varepsilon_r}$ and becomes, $\frac{1}{4\pi\varepsilon_0\varepsilon_r}\frac{Q}{r^2}$. Here I have introduced a new quantity ε_r called "dielectric constant". This is a number that is usually bigger than one and measures the strength of the polarization induced in the material. For air, $\varepsilon_r = 1.0003$ while $\varepsilon_r \approx 80$ for pure water. The effect of a dielectric is to increase the capacitance of a capacitor: if the air between the plates of a capacitor is replaced by a dielectric, $C = \frac{\varepsilon_0 A}{d} \rightarrow \varepsilon_r \frac{\varepsilon_0 A}{d}$.

dielectric lagane se capacitor zyada energy store kar sakta hai.

Agar ek positive charge +Q ke aas paas water molecules rakh do, to unka center thora shift ho jata hai.
Water ka negative part Q ki taraf attract hota hai => polarization.
Is wajah se overall electric field kam ho jati hai.

Summary of Lecture 26 – ELECTRIC POTENTIAL ENERGY

Electric current ka matlab hai charge ka flow. Agar thoda sa charge dq kisi waqt dt mein flow kare, to current ka formula dq/dt HOTA h or Agar current constant ho to q = I × t

1. Electric current is the flow of electrical charge. If a small amount of charge dq flows in time dt, then the current is $\boxed{i = \frac{dq}{dt}}$. If the current is constant in time, then in time t, the current that flows is $q = i \times t$. The unit of charge is ampere, which is define as:

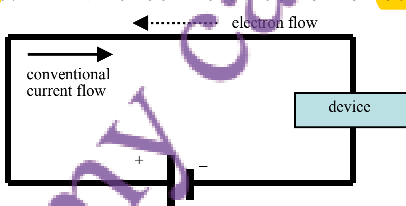
$$\boxed{1 \text{ ampere} = \frac{1 \text{ coulomb}}{\text{second}}}$$

A car's battery supplies upto 50 amperes when starting the car, but often we need to deal with smaller values:

$$\boxed{\begin{aligned} &1 \text{ milliampere} &&= 1 \text{ ma} &&= 10^{-3} A \\ &1 \text{ microampere} &&= 1\ \mu A &&= 10^{-6} A \\ &1 \text{ nanoampere} &&= 1 \text{ n}A &&= 10^{-9} A \\ &1 \text{ picoampere} &&= 1 \text{ p}A &&= 10^{-12} A \end{aligned}}$$

Current ka direction wo hota hai jahan positive charges move karte hain. Lekin asal mein electrons (negative charge) move karte hain, is liye unka flow opposite hota hai.

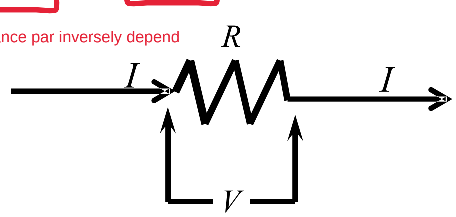
2. The direction of current flow is the direction in which positive charges move. However, in a typical wire, the positive charges are fixed to the atoms and it is really the negative charges (electrons) that move. In that case the direction of current flow is reversed.



Current flow krta h Jab koi cheez (jaise EMF) usay push karti hai circuit mein. EMF ka matlab hota hai potential difference (voltage), na ke koi actual force.

3. Current flows because something forces it around a circuit. That "something" is EMF, electromotive force. But remember that we are using bad terminology and that EMF is not a force - it is actually the difference in electric potentials between two parts of a circuit. So, in the figure below, $V = V_a - V_b$ is the EMF which causes current to flow in the resistor. How much current? Generally, the larger V is , the more current will flow and we expect $\boxed{I \propto V}$. In general this relation will not be completely accurate but when it holds, we say Ohm's Law applies: $\boxed{I = \frac{V}{R}}$. Here $\boxed{R = \frac{V}{I}}$ is called the resistance.

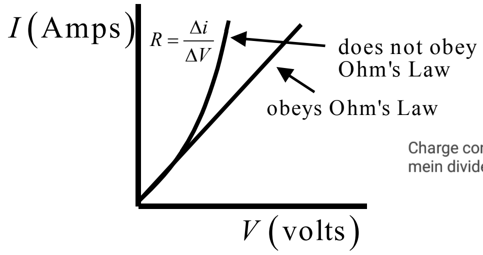
Current, voltage par directly aur resistance par inversely depend karta hai



Ohm's law states that the voltage or potential difference between two points is directly proportional to the current or electricity passing through the resistance, and directly proportional to the resistance of the circuit.

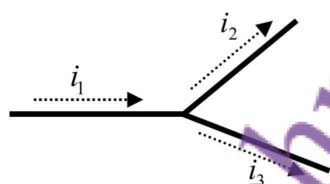
Jab current resistor se guzarta hai to wo garam ho jata hai aur resistance barh jata hai.
Sirf us surat mein Ohm's Law apply hota hai jab current vs voltage ka graph straight line ho.

4. Be careful in understanding Ohm's Law. In general the current may depend upon the applied voltage in a complicated way. Another way of saying this is that the resistance may depend upon the current. Example: when current passes through a resistor, it gets hot and its resistance increases. Only when the graph of current versus voltage is a straight line does Ohm's Law hold. Else, we can only define the "incremental resistance".



Charge conserve hota hai, is liye current bhi hota hai. Jab current 2 branches mein divide hota hai to $i = i_1 + i_2$

5. Charge is always conserved, and therefore current is conserved as well. This means that when a current splits into two currents the sum remains constant, $i_1 = i_2 + i_3$.



Series Resistors:
Current: The same current flows through all resistors in a series circuit.
Voltage: The total voltage across the series is the sum of the individual voltage drops across each resistor.
Total or equivalent Resistance: R_total or R_eq= R1 + R2 + ... + Rn.

6. When resistors are put in series with each other, the same current flows through both. So, $V_1 = iR_1$ and $V_2 = iR_2$. The total potential drop across the pair is $V = V_1 + V_2 = i(R_1 + R_2)$.
$\Rightarrow \boxed{R_{eq} = R_1 + R_2}$ So resistors in series add up.

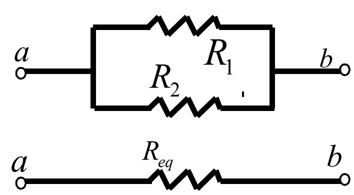
same as lec 25

Parallel Resistors:
Voltage: The same voltage is applied across all resistors in a parallel circuit.
Current: The total current is the sum of the individual currents flowing through each resistor.
Total or equivalent Resistance: 1/R_Total or 1/R_eq = 1/R1 + 1/R2 + ... + 1/Rn. The reciprocal of the total resistance is the sum of the reciprocals of the individual resistances.

7. Resistors can also be put in parallel. This means that the same voltage V is across both. So the currents are $i_1 = \frac{V}{R_1}$, $i_2 = \frac{V}{R_2}$.

Since $i = i_1 + i_2$ it follows that $i = \frac{V}{R_{eq}} = \frac{V}{R_1} + \frac{V}{R_2}$

$$\Rightarrow \boxed{\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} \quad \text{or} \quad R_{eq} = \frac{R_1 R_2}{R_1 + R_2}.}$$



This makes sense: with two possible paths the current will find less resistance than if only one was present.

Jab charge dq ko potential difference V se move karte hain to work dW done HOTA h

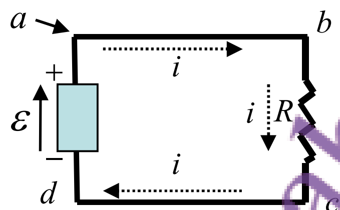
8. When current flows in a circuit work is done. Suppose a small amount of charge dq is moved through a potential difference V. Then the work done is $\boxed{dW = Vdq = V\,idt}$. Hence

$V i dt = i^2 R dt$ (because $i = \frac{V}{R}$). The rate of doing work, i.e. power, is $P = \frac{dW}{dt} = i^2 R$.

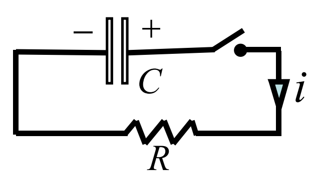
This is an important formula. It can also be written as $P = \frac{V^2}{R}$, or as $P = iV$. The unit of power is: $1 \text{ volt-ampere} = 1 \frac{\text{joule}}{\text{coulomb}} \cdot \frac{\text{coulomb}}{\text{second}} = 1 \frac{\text{joule}}{\text{second}} = 1 \text{ watt}.$

Kirchhoff's second law, also known as Kirchhoff's voltage law (KVL) states that the sum of all voltages around a closed loop in any circuit must be equal to zero.

9. Kirchoff's Law: The sum of the potential differences encountered in moving around a closed circuit is zero. This law is easy to prove: since the electric field is conservative, therefore no work is done in taking a charge all around a circuit and putting it back where it was. However, it is very useful in solving problems. As a trivial example, consider the circuit below. The statement that, starting from any point a we get back to the same potential after going around is: $V_a - iR + \mathcal{E} = V_a$. This says $-iR + \mathcal{E} = 0$.



10. We can apply Kirchoff's Law to a circuit that consists of a resistor and capacitor in order to see how current flows through it. Since $q = VC$, we can see that $\frac{q}{C} + iR = 0$. Now differentiate with respect to time to find $\frac{1}{C}\frac{dq}{dt} + \frac{di}{dt}R = 0$, or $\frac{di}{dt} = -\frac{1}{RC}i$. This equation has solution: $i = i_0 e^{-\frac{t}{RC}}$. The product RC is called the time constant τ, and it gives the time by which the current has fallen to $1/e \approx 1/2.7$ of the initial value.



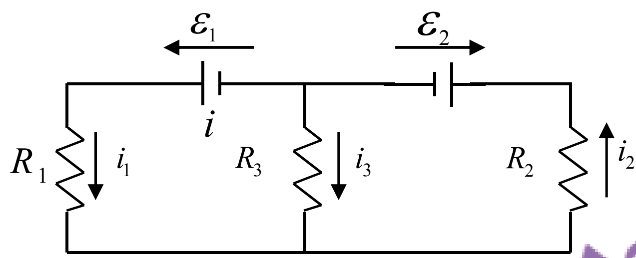
A reminder about the exponential function, $e^x \equiv 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \cdots$

From this, $\frac{d}{dx} e^x = 1 + \frac{2x}{2!} + \frac{3x^2}{3!} + \frac{4x^3}{4!} + \cdots = e^x$ Similarly, $\frac{d}{dx} e^{-x} = -e^{-x}$

11. Circuits often have two or more loops. To find the voltages and currents in such situations, it is best to apply Kirchoff's Law. In the figure below, you see that there are 3 loops and you can see that:

Teen loops wale circuit mein Loop equations likh ke currents i_1, i_2, i_3 calculate kiye jaate hain.

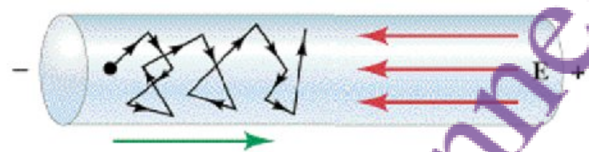
$$i_1 + i_3 = i_2$$
$$\mathcal{E}_1 - i_1R_1 + i_3R_3 = 0$$
$$-i_3R_3 - i_2R_2 + \mathcal{E}_2 = 0$$



Make sure that you understand each of these, and then check that the solution is:

$$\boxed{i_1 = \frac{\mathcal{E}_1(R_2+R_3)-\mathcal{E}_2R_3}{R_1R_2+R_2R_3+R_1R_3} \qquad i_2 = \frac{\mathcal{E}_1R_3-\mathcal{E}_2(R_1+R_3)}{R_1R_2+R_2R_3+R_1R_3} \qquad i_3 = \frac{-\mathcal{E}_1R_2-\mathcal{E}_2R_1}{R_1R_2+R_2R_3+R_1R_3}}$$

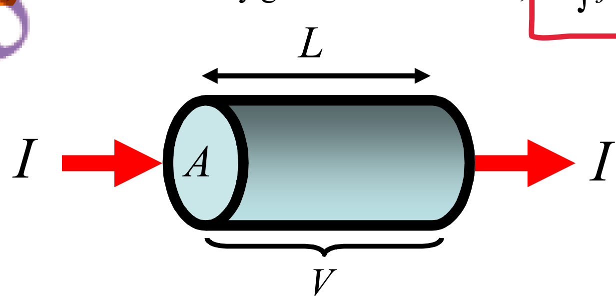
12. A charge inside a wire moves under the influence of the applied electric field and suffers many collisions that cause it to move on a highly irregular, jagged path as shown below.



Jab ek charge wire ke andar hota hai, to woh electric field ke influence mein move karta hai. Lekin uska raasta irregular hota hai kyunke wo bahut collisions karta hai. Phir bhi average mein charge drift velocity se move karta hai right side ki taraf.

Nevertheless, it moves on the average to the right at the "drift velocity" (or speed).

13. Consider a wire through which charge is flowing. Suppose that the number of charges per unit volume is n. If we multiply n by the crossectional area of the wire A and the length L, then the charge in this section of the wire is $\boxed{q=(nAL)e}$. If the drift velocity of the charges is v_d, then the time taken for the charge to move through the wire is $\boxed{t=\frac{L}{v_d}}$. hence the current is $\boxed{i=\frac{q}{t}=\frac{nALe}{L/v_d}=nAev_d}$. From this we can calculate the drift velocity of the charges in terms of the measured current, $\boxed{v_d=\frac{i}{nAe}}$. The current density, which is the current per unit crossectional area is defined as $\boxed{j=\frac{i}{A}=nev_d}$ If j varies inside a volume, then we can easily generalize and write, $\boxed{i=\int \vec{j}\cdot d\vec{A}}$.



Jab charge wire mein flow karta hai, to electrons ki motion random hoti hai lekin overall woh electric field ki taraf drift karte hain — isy drift velocity (v_d) khty h.

Drift velocity is the average velocity with which electrons 'drift' in the presence of an electric field

v_d = drift velocity
I = current flow
n = free electron density
e = charge of an electron
A = cross sectional area

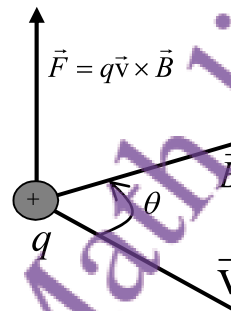
importna thing to remember
jb b hm dot product ko khtam krtya hi hmary pas cos thitha aata hai or
jb b hm cross product ko khtam krtay hai hmaray pas sin thitha aata hai

magnetic field is defined as the region around a magnet or moving electric charge where a magnetic force can be detected.
Magnetic field is represented by B

Summary of Lecture 27 – THE MAGNETIC FIELD

• Jitna charge zyada hoga aur jitni tezi se move karega, utni zyada magnetic force lagegi.
• Force ka rukh hamesha motion aur magnetic field dono ke perpendicular hota hai.
•Jab velocity aur magnetic field parallel hoti hain (θ = 0), to force zero hoti hai. Maximum tab hoti hai jab dono perpendicular

1. The magnetic field exerts a force upon any charge that moves in the field. The greater the size of the charge, and the faster it moves, the larger the force. The direction of the force is perpendicular to both the direction of motion and the magnetic field. If θ is the angle between $\vec{v}$ and $\vec{B}$, then $\boxed{F = qvB\sin\theta}$ is the magnitude of the force. This vanishes when $\vec{v}$ and $\vec{B}$ are parallel $(\theta = 0)$, and is maximum when they are perpendicular.



ab charge ka alag field hai magnet ka alag fell hai ab agr magnetic field mai charge ko rkh dia jaye to jitna bra chrge ho ga utni hi zada force lgay gi

2. The unit of magnetic field that is used most commonly is the *tesla*. A charge of one coulomb moving at 1 metre per second perpendicularly to a field of one tesla experiences a force of 1 newton. Equivalently,

wo force jo chagr ko move krvaty hoye lg rhi hai

Tesla magnetic field ka unit hai, aur is ka matlab hai ek coulomb charge pe 1 newton force lagay jab wo 1 m/s se move kare.

$$\boxed{1\text{ tesla} = 1\frac{\text{newton}}{\text{coulomb}\cdot\text{meter/second}} = 1\frac{\text{newton}}{\text{ampere}\cdot\text{meter}} = 10^4\text{ gauss (CGS unit)}}$$

In order to have an appreciation for how much a tesla is, here are some typical values of the magnetic field in these units:

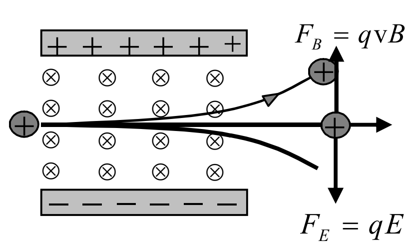
Earth's surface	10^{-4} T
Bar magnet	10^{-2} T
Powerful electromagnet	1 T
Superconducting magnet	5 T

Jab electric aur magnetic field dono hoon, to total force dono ka vector sum hota hai. Isay Lorentz Force kehte hain.

3. When both magnetic and electric fields are present at a point, the total force acting upon a charge is the vector sum of the electric and magnetic forces, $\boxed{\vec{F} = q\vec{E} + q\vec{v} \times \vec{B}}$. This is known as the *Lorentz Force*. Note that the electric force and magnetic force are very different. The electric force is non-zero even if the charge is stationary, and it is in the same direction as $\vec{E}$.

4. The Lorentz Force can be used to select charged particles of whichever velocity we want. In the diagram below, particles enter from the left with velocity v. They experience a force due to the perpendicular magnetic field, as well as force downwards because of an electric field. Only particles with speed $v = E/B$ are undeflected and keep going straight.

Sirf wo charged particles jinki speed E/B ke barabar ho, seedha move karte hain bina deflection ke.



$qE = qvB \Rightarrow v = \dfrac{E}{B}$ velocity selector !!

Magnetic field bohat strong ho sakta hai (itna ke hathi uthaye), lekin yeh kisi particle ki energy ko kam ya zyada nahi kar sakta. Kyun? Kyunki magnetic force hamesha velocity ke perpendicular hoti hai — aur jab force aur motion perpendicular hon, to koi kaam (work) nahi hota.

5. A magnetic field can be strong enough to lift an elephant, but it can never increase or decrease the energy of a particle. Proof: suppose the magnetic force $\vec{F}$ moves a particle through a displacement $d\vec{r}$. Then the small amount of work done is,

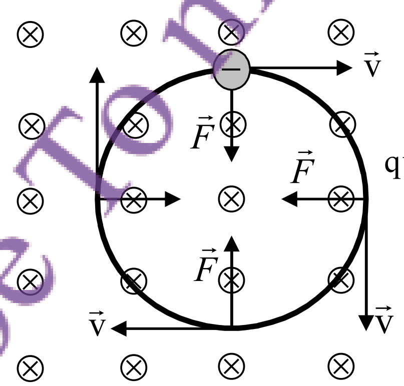
$$dW = \vec{F}\cdot d\vec{r} = q(\vec{\mathrm{v}}\times\vec{\mathrm{B}})\cdot d\vec{r}$$
$$= q(\vec{\mathrm{v}}\times\vec{\mathrm{B}})\cdot\frac{d\vec{r}}{dt}dt$$
$$= q(\vec{\mathrm{v}}\times\vec{\mathrm{B}})\cdot\vec{\mathrm{v}}\,dt = 0.$$

means perpendicular

Basically the force and direction of force are orthogonal, and hence there can be no work done on the particle or an increase in its energy.

Magnetic field charged particle ko ghoomne par majboor karta hai. Speed change nahi hoti, sirf direction change hoti hai.

6. A magnetic field bends a charged particle into a circular orbit because the particle feels a force that is directed perpendicular to the magnetic field. As we saw above, the particle cannot change its speed, but it certainly does change direction! So it keeps bending and bending until it makes a full circle. The radius of orbit can be easily calculated: the magnetic and centrifugal forces must balance each other for equilibrium. So, $\boxed{\mathrm{q}\mathrm{v}B = m\frac{\mathrm{v}^2}{r}}$ and we find that $\boxed{r = \frac{m\mathrm{v}}{qB}}$. A strong B forces the particle into a tighter orbit, as you can see. We can also calculate the angular frequency, $\boxed{\omega = \frac{\mathrm{v}}{r} = \frac{qB}{m}}$. This shows that a strong B makes the particle go around many times in unit time. There are a very large number of applications of these facts.



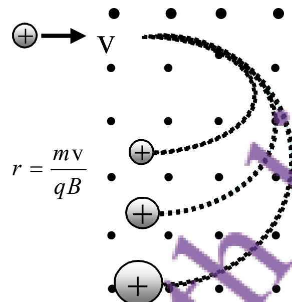
$$\mathrm{q}\mathrm{v}B = m\frac{\mathrm{v}^2}{r} \Rightarrow r = \frac{m\mathrm{v}}{qB}.$$

Magnetic field Earth ko solar wind (dangerous charged particles) se bachata hai. Yeh particles magnetic field ke through deflect ho jaate hain aur Van Allen belt mein trap ho jaate hain is se zameen safe rehti hai.

7. The fact that a magnetic field bends charged particles is responsible for shielding the earth from harmful effects of the "solar wind". A large number of charged particles are released from the sun and reach the earth. These can destroy life. Fortunately the earth's magnetic field deflects these particles, which are then trapped in the "Van Allen" belt around the earth.

Jab ions magnetic field se guzarte hain to unka rasta bend hota hai. Jo ions zyada heavy hote hain wo kam bend hote hain, aur jo halkay hote hain wo zyada bend hote hain. Yehi principle mass spectrometer mein use hota hai.

8. The mass spectrometer is an extremely important equipment that works on the above principle. Ions are made from atoms by stripping away one electron. Then they pass through a velocity selector so that they all have the same speed. In a beam of many different ions, the heavier ones bend less, and lighter ones more, when they are passed through a B field.



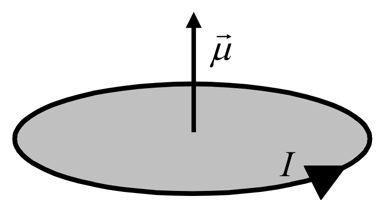
Wire mein current ka matlab hai charges ka flow. Jab wire magnetic field mein hota hai, to current par bhi magnetic force lagta hai.

9. A wire carries current, and current is flowing charges. Since each charge experiences a force when placed in a magnetic field, you might expect the same for the current. Indeed, that is exactly the case, and we can easily calculate the force on a wire from the force on individual charges. Suppose N is the total number of charges and they are moving at the average (or drift) velocity $\vec{v}_d$. Then the total force is $\boxed{\vec{F} = Ne\vec{v}_d \times \vec{B}.}$ Now suppose that the wire has length L, crossectional area A, and it has n charges per unit volume. Then clearly $\boxed{N = nAL,}$ and so $\boxed{\vec{F} = nALe\vec{v}_d \times \vec{B}.}$ Remember that the current is the charge that flows through the wire per unit time, and so $\boxed{nAe\vec{v}_d = \vec{I}}$ We get the important result that the force per unit length on the wire is $\boxed{\vec{F} = \vec{I} \times \vec{B}.}$

Definition of Magnetic moment
A magnetic moment, also known as a magnetic dipole moment, is a measure of the strength and orientation of a magnet or object that produces a magnetic field. It's a vector quantity, meaning it has both magnitude and direction.

Jab current kisi loop (gol shape) mein circulate karta hai, to wo magnetic field produce karta hai.

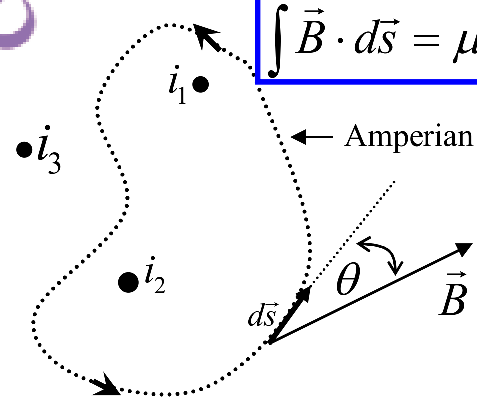
10. A current that goes around a loop (any shape) produces a magnetic field. We define the *magnetic moment* as the product of current and area, $\vec{\mu} = IA\hat{z}$. Here A is the area of the loop and I the current flowing around it. The direction is perpendicular to the plane of the loop, as shown.



11. Magnetic fields are produced by currents. Every small bit of current produces a small amount of the B field. Ampere's Law, illustrated below, says that if one goes around a loop (of any shape or size) then the integral of the B field around the loop is equal to the enclosed current. In the loop below $I = I_1 + I_2$. Here I_3 is excluded as it lies outside.

$$\boxed{\int \vec{B} \cdot d\vec{s} = \mu_0 I_{enclosed}}$$

Har choti si current bhi magnetic field banati hai. Ampere's Law ke mutabiq agar hum magnetic field ke ird gird loop banayein to us loop ka B field ka integral enclosed current ke barabar hota hai.



Ampère's Law
"The magnetic field produced by an electric current is directly proportional to the intensity of the electric current and the constant of probability (permeability of free space)."

The line integral of the magnetic field B around a closed loop is equal to μ_0 times the total current I passing through the loop.
$\oint B \cdot dl = \mu_0 I$

Jab ek lambe wire ke bahar radius *r* par loop banate hain aur usme se current *I* guzar raha hota hai, to magnetic field *B* gol gol chakkar lagata hai.

Ampere's Law (outside wire):

12. Let us apply Ampere's Law to a circular loop of radius r outside an infinitely long wire carrying current I through it. The magnetic field goes around in circles, and so $\vec{B}$ and $d\vec{s}$ are both in the same direction. Hence, $\boxed{\int \vec{B} \cdot d\vec{s} = B\int ds = B(2\pi r) = \mu_0 I.}$ We get the important result that $\boxed{B = \frac{\mu_0 I}{2\pi r}}$.

Yani jitna zyada current ya kam radius ho, utna zyada magnetic field.

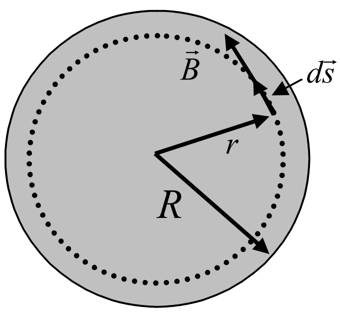
Ampere's Law (inside wire):

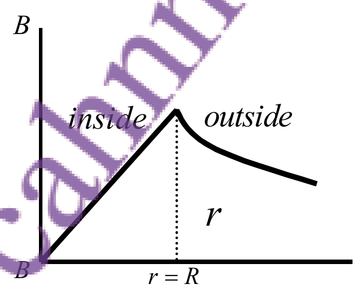
13. Assuming that the current flows uniformly over the crossection, we can use Ampere's Law to calculate the magnetic field at distance r, where r now lies inside the wire.

$$\boxed{B(2\pi r) = \mu_0 I\left(\frac{\pi r^2}{\pi R^2}\right) \Rightarrow B = \frac{\mu_0 I r}{2\pi R^2}}$$

Agar point wire ke andar ho (r < R), aur current uniformly distribute ho, to field ka formula hota hai:
$B = (\mu_0 I r) / (2\pi R^2)$
Yani field linearly badhti hai andar ke liye (r tak proportional), aur wire ke bahar same pehle wale formula se milti hai.

Here is a sketch of the B field inside and outside the wire as a function of distance r.





Graph mein dikhaya gaya ke wire ke andar B linearly barhta hai, aur wire ke bahar inverse relation ke sath girta hai (1/r).

Left Diagram (Wire cross-section):
Yeh ek gol wire dikhayi gayi hai jisme se current guzar raha hai.
r: Current point ka distance wire ke center se hai (andar ka point).
R: Wire ka total radius hai.
B:Magnetic field tangential direction mein hai, yani gol gol.
ds:Ampere's loop ka chhota part (line element) hai jahan field calculate kar rahe hain.

Right Graph (B vs r):
Yeh graph magnetic field B ka distance r ke saath relation dikhata hai.
Inside (r < R): B linearly barhta hai, yani jese jese center se door jaate ho, B barhta hai.
r = R: Yahan B maximum hota hai (boundary of wire).
Outside (r > R): B ka value gradually kam hota hai (1/r ke mutabiq).

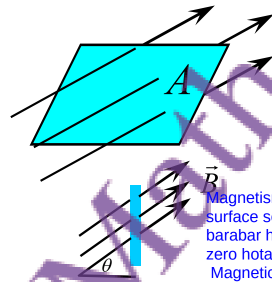
Electromagnetic or magnetic induction is the production of an electromotive force across an electrical conductor in a changing magnetic field mtlb Jab kabhi magnetic flux kisi coil ya conductor ke through change hota hai, to usmein automatically ek EMF (electromotive force) induce hoti hai. Agar circuit closed ho, to current bhi flow karta hai.

Summary of Lecture 28 – ELECTROMAGNETIC INDUCTION

Magnetic flux ka matlab hai magnetic field ka kisi area se guzarne wala hissa.

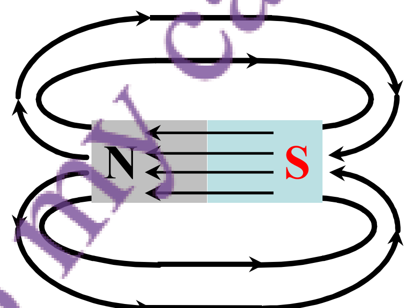
1. Earlier we had defined the flux of any vector field. For a magnetic field, this means that flux of a uniform magnetic field (see figure) is $\boxed{\Phi = B_{\perp} A = BA\cos\theta}$. If the field is not constant over the area then we must add up all the little pieces of flux: $\boxed{\Phi_B = \int \vec{B}\cdot d\vec{A}}$. The dimension of flux is magnetic field × area, and the unit is called weber, where $\boxed{1 \text{ weber} = 1 \text{ tesla}\cdot\text{metre}^2}$.

Agar field uniform na ho, to chhoti chhoti surfaces ka flux add karte hain



Magnetism ka ek basic rule hai ke kisi bhi closed surface se jo magnetic flux nikalta hai, woh usi ke barabar hota hai jo andar aata hai yani net flux zero hota hai.
Magnetic monopole (jaise electric charge) nahi hotay. Magnet ke north aur south poles sirf current ke effects hote hain, koi alag magnetic charge nahi hota.

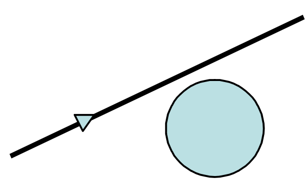
2. A fundamental law of magnetism states that the net flux through a closed surface is always zero, $\Phi_B = \oint \vec{B}\cdot d\vec{A} = 0$. Note that this is very different from what you learned earlier in electrostatics where the flux is essentially the electric charge. There is no such thing as a magnetic charge! What we call the magnetic north (or south) pole of a magnet are actually due to the particular electronic currents, not magnetic charges. In the bar magnet below, no matter which closed surface you draw, the amount of flux leaving the surface is equal to that entering it.



Bar magnet ke field lines close loop mein hain jitna flux bahar jaata hai, utna hi wapas aata hai.

Example : A sphere of radius R is placed near a long, straight wire that carries a steady current I. The magnetic field generated by the current is B. Find the total magnetic flux passing through the sphere.

Ek sphere jo current-carrying wire ke paas rakha hai, usmein total magnetic flux zero hoga, kyunke flux andar aur bahar same hai.



Faraday's law of induction, a fundamental principle of electromagnetism, states that the induced electromotive force (EMF) in a circuit is directly proportional to the rate of change of magnetic flux through that circuit. In simpler terms, a changing magnetic field can induce an EMF, which in turn can drive an electric current.

Answer: zero, of course!

Faraday's Law of Induced EMF kehta hai ke jab kisi coil mein magnetic flux change hota hai to ek EMF induce hoti hai.

3. Faraday's Law for Induced EMF: when the magnetic flux changes in a circuit, an electro-motive force is induced which is proportional to the rate of change of flux. Mathematically, $\boxed{\varepsilon = -\frac{d\Phi_B}{dt}}$ where ε is the induced emf. If the coil consists of N turns, then $\boxed{\varepsilon = -N\frac{d\Phi_B}{dt}}$.

How does the flux through a coil change? Consider a coil and magnet. We can:

a) move the magnet,

b) change the size and shape of the coil by squeezing it,

c) move the coil.

Flux ko badalnay ke tareeqay:
a) Magnet ko move karna
b) Coil ka size ya shape change karna
c) Coil ko move karna

In all cases, the flux through the coil changes and $\frac{d\Phi_B}{dt}$ is non-zero leading to an induced emf.

Jab bhi flux change hota hai to EMF induce hoti hai.

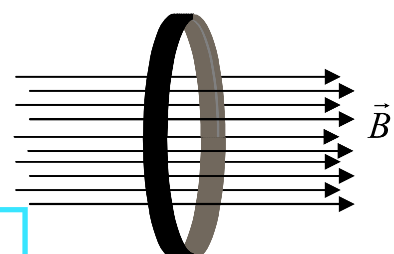
Example: A flexible loop has a radius of 12cm and is in a magnetic field of strength 0.15T. The loop is grasped at points A and B and stretched until it closes. If it takes 0.20s to close the loop, find the magnitude of the average induced emf in it during this time.

Area change hone se flux change hua => EMF induce hui.

Solution: Here the loop area changes, hence the flux. So the induced emf is:

$$\varepsilon = -\frac{d\Phi_B}{dt} \approx -\left[\frac{\text{final flux} - \text{initial flux}}{\text{time taken}}\right] = -\left[\frac{0-\pi(0.12)^2 \times 0.15}{0.2}\right] = 0.034 \text{ Volts.}$$

Example : A wire loop of radius 0.30m lies so that an external magnetic field of +0.30T is perpendicular to the loop. The field changes to -0.20T in 1.5s. Find the magnitude of the average induced EMF in the loop during this time.



Solution: Again, we will find the initial and final fluxes first and then divide by the time taken for the change.

Use $\Phi = BA = B\pi r^2$ to calculate the flux.

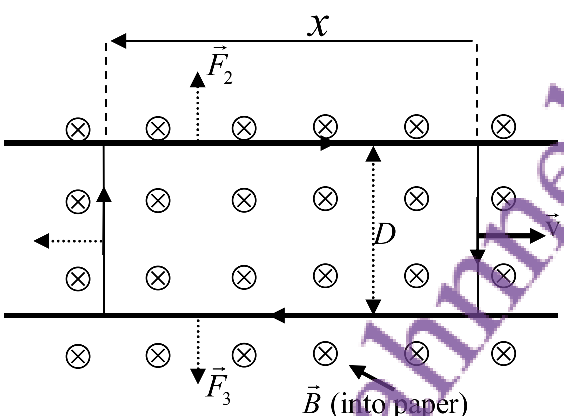
$$\Phi_i = 0.30 \times \pi \times (0.30)^2 = 0.085 \text{ Tm}^2$$

$$\Phi_f = -0.20 \times \pi \times (0.30)^2 = -0.057 \text{ Tm}^2$$

$$\varepsilon = \frac{\Delta\Phi}{\Delta t} = \frac{\Phi_f - \Phi_i}{\Delta t} = \frac{0.085 - 0.057}{1.5} = 0.095 \text{ V}$$

4. Remember that the electromotive force is not really a force but the difference in electric potentials between two points. In going around a circular wire, where the electric field is constant as a function of angle, the emf is $\varepsilon = E(2\pi r)$. More generally, for any size or shape of a closed circuit, $\varepsilon = \int \vec{E}\cdot d\vec{s}$. So Faraday's Law reads: $\int \vec{E}\cdot d\vec{s} = -\frac{d\Phi_B}{dt}$.

Example: A conducting wire rests upon two parallel rails and is pulled towards the right with speed v. A magnetic field B is perpendicular to the plain of the rails as shown below. Find the current that flows in the circuit.



Solution: As the wire is pulled to the right, the area of the circuit increases and so the flux increases. Measure x as above so that $x = vt$, i.e. x keeps increasing as we pull. The flux at any value of x is $\Phi_B = BDx$, and so,

$$\varepsilon = -\frac{d\Phi_B}{dt} = -\frac{d(BDx)}{dt} = -BD\frac{dx}{dt} = BDv.$$

To calculate the current, we simply use Ohm's Law: $I = \frac{\varepsilon}{R} = \frac{BDv}{R}$. Here R is the resistance of the circuit.

Example: Find the power dissipated in the above circuit using I^2R, and then by directly calculating the work you do by pulling the wire.

Solution: Clearly $I^2R = \frac{B^2D^2v^2}{R}$ is the power dissipated, as per usual formula. Now let us calculate the force acting upon the piece of wire (of length D) that you are pulling. From the formula for the force on a wire, $\vec{F} = I\,\vec{L}\times\vec{B}$, the magnitude is $F = IBD = \frac{B^2D^2v}{R}$. So, the power is $P = Fv = \frac{B^2D^2v^2}{R}$. This is exactly the value calculated above!

Lenz's Law: The induced current always flows in a direction that opposes the change in magnetic flux that caused it.

5. **Lenz's Law :** The direction of any magnetic induction effect is such as to oppose the cause of the effect. Imagine a coil wound with wire of finite resistance. If the magnetic field decreases, the induced EMF is positive. This produces a positive current. The magnetic field produced by the current opposes the decrease in flux. Of course, because of finite resistance in loop, the induced current cannot completely oppose the change in flux.

Lenz's Law: Jab bhi magnetic flux mein tabdili hoti hai, to induced current us tabdili ke khilaaf direction mein flow karta hai.

Example: Agar magnetic field kam ho rahi ho, to induced current aisi direction mein flow karega jo is kam hone wali field ko oppose kare.

Direct current

DC is an electrical current that flows continuously in one direction. Examples: DC is produced by batteries, solar panels, and some types of power supplies. Use in electronics: DC is often used in electronic devices like smartphones, laptops

Alternating current

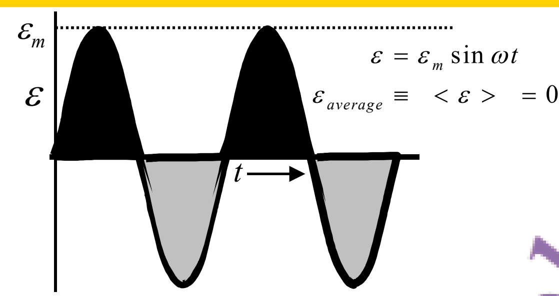
AC is electrical current that periodically changes direction, usually back and forth in a sinusoidal pattern. Examples: The electricity that powers homes and businesses is usually AC. Use in appliances: AC is used in various appliances like refrigerators, washing machines, and fans.

AC wo current hota hai jo wire ke through ek direction mein flow karne ke baad opposite direction mein flow karta hai bar bar reverse hota hai. Most common form sinusoidal hoti hai, jisme voltage ya current ka graph sine wave jaisa hota hai.

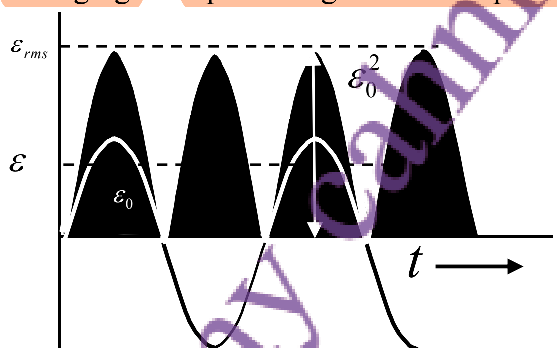
Summary of Lecture 29 – ALTERNATING CURRENT

1. Alternating current (AC) is current that flows first in one direction along a wire, and then in the reverse direction. The most common AC is sinuisoidal in which the current (and voltage) follow a sine function, as in the graph below. The average value is zero because the current flows for the same time in one direction as in the other.

Average value of AC = 0, kyunke equal time dono directions mein flow karta hai.



However, the square of any AC wave is always positive. Thus its average is not zero. As we shall see, upon averaging the square we get half the square of the peak value.



The calculation follows: if there is a sine wave of amplitude (height) equal to one and frequency ω, $(\omega = 2\pi / T$, T=time period), then the squared amplitude is $\sin^2 \omega t$ and its average is:

$$\boxed{< \sin^2 \omega t > \equiv \frac{1}{T}\int_0^T dt \sin^2 \omega t = \frac{1}{T}\int_0^T dt (\frac{1-\cos 2\omega t}{2}) = \frac{1}{T}\int_0^T dt \frac{1}{2} = \frac{1}{2}}$$

Taking the square root gives the root mean square value as $1/\sqrt{2}$ of the maximum value. Of course, it does not matter whether this is of the voltage or current:

$$\boxed{\varepsilon_{rms} = \sqrt{\frac{\varepsilon_m^2}{2}} = \frac{\varepsilon_m}{\sqrt{2}} = 0.707\varepsilon_m, \text{ and } I_{rms} = \sqrt{\frac{I_m^2}{2}} = \frac{I_m}{\sqrt{2}} = 0.707 I_m.}$$

Exactly the same results are obtained for cosine waves. This is what one expects since the difference between sine and cosine is only that one starts earlier than the other.

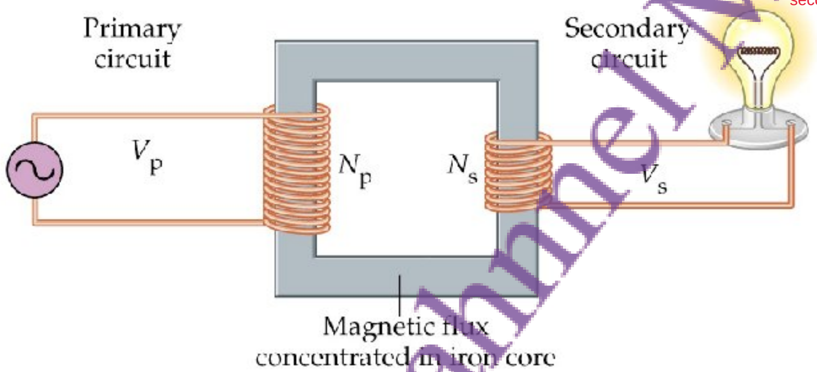
AC generate hoti hai jab ek coil magnetic field mein rotate karti hai.

2. AC is generated by a coil rotating in a magnetic field. We know from Faraday's Law, $\boxed{\varepsilon = -\frac{d\Phi_B}{dt}}$, that a changing magnetic flux gives rise to an emf. Imagine a magnetic field and a coil of area A, rotating with frequency ω so that the flux through the coil at any instant of time is $\boxed{\Phi_B = BA\cos\omega t}$. Then the induced emf is $\boxed{\varepsilon = BA\omega\sin\omega t}$.

jab magnetic flux change hota hai, to EMF induce hoti hai. Agar coil area A aur frequency ω se rotate ho rahi ho, to induced EMF hoti hai

3. AC is particular useful because *transformers* make it possible to step up or step down voltages. The basic transformer consists of two coils - the primary and secondary - both wrapped around a core (typically iron) that enhances the magnetic field.

Transformers AC ka voltage step up ya step down karne ke liye use hotay hain. Do coils hoti hain primary aur secondary iron core ke around.



Suppose that the flux in the core is Φ and that its rate of change is $\frac{\partial\Phi}{\partial t}$. Call the number of turns in the primary and secondary N_p and N_s respectively. Then, from Faraday's law, the primary emf is $\boxed{\varepsilon_p = -N_p\frac{\partial\Phi}{\partial t}}$ and the secondary emf is $\boxed{\varepsilon_s = -N_s\frac{\partial\Phi}{\partial t}}$. The ratio is, $\boxed{\frac{\varepsilon_p}{\varepsilon_s} = \frac{N_p}{N_s}}$. So, the secondary emf is $\boxed{\varepsilon_s = \frac{N_s}{N_p}\varepsilon_p}$. If $\frac{N_s}{N_p}$ is less than one, then it is called a step-down transformer because the secondary voltage is less than the input voltage. Else, it is a step-up transformer. Both types are used.

Ns < Np ho to step-down transformer, warna step-up

Lossless transformer m input power = output power

4. If this is a lossless transformer (and good transformers are 99% lossless), then the input power must equal the output power, $\boxed{\varepsilon_p I_p = \varepsilon_s I_s}$. From above, this shows that the ratio of currents is $\boxed{I_s = \frac{N_p}{N_s}I_p}$.

Magnetic flux through loop proportional hota hai current k is proportionality constant ko inductance L kehte hain

5. Whatever the shape or size of a current carrying loop, the magnetic flux that passes through it is proportional to the current, $\boxed{\Phi \propto I}$. The inductance L (called the self-inductance if

there is only one coil) is the constant of proportionality in the relation $\Phi = LI$. The unit of inductance is called Henry, 1 Henry= 1 Tesla metre^2 / Ampere. Note that inductance, like capacitance, is purely geometrical and depends only upon the shape and sizes of wires. It does not depend on the current.

6. Let us calculate the inductance of a long coil wound with n turns per unit length. As calculated earlier, the B field is $B = \mu_0 nI$, and the flux passing through N turns of the coil of length l and area A is $N\Phi_B = (nl)(BA) = \mu_0 n^2 IAl$. From the definition, we find:

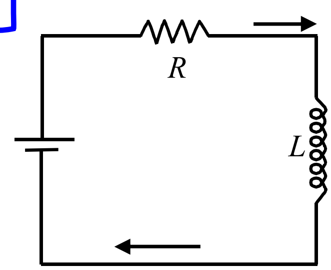
$$L = \frac{N\Phi_B}{I} = \frac{\mu_0 n^2 lIA}{I} = \mu_0 n^2 lA \quad \text{(inductance of long solenoid).}$$

Example: Find the inductance of a coil with 3500 turns, length 10 cm, and radius 5cm.

Solution: $L = 4\pi \cdot 10^{-7}\frac{T\cdot m}{A}\cdot 3500^2 \cdot \frac{\pi(0.05m)^2}{0.10m} = 1.21\frac{T\cdot m^2}{A} = 1.21H$

7. When the current changes through an inductor, by Faraday's Law it induces an emf equal to $-\frac{d\Phi_B}{dt} = -\frac{d(LI)}{dt} = -L\frac{dI}{dt}$.

Jab current change hota hai through an inductor, to EMF induce hoti hai



Let us use this to calculate the current through the circuit shown here, where a resistor is present. Then, by using Kirchoff's Law, $\varepsilon = IR + L\frac{dI}{dt}$ or $\varepsilon I = I^2R + LI\frac{dI}{dt}$. In words: the power expended by the battery equals energy dissipated in the resistor + work done on inductor.

Battery ka power = resistor + inductor mein energy

8. Before we go on to the AC case, suppose that we suddenly connect a battery to an inductor so that the voltage suddenly increases from zero to ε across the circuit. Then, $L\frac{dI}{dt} + IR = \varepsilon$ has solution: $I(t) = \frac{\varepsilon}{R}\left(1 - e^{-t/\tau}\right)$ where $\tau = \frac{L}{R}$. You can easily see that this true using $\frac{dI}{dt} = \frac{\varepsilon}{R}\frac{1}{\tau}e^{-t/\tau}$. The solution shows that the current increases from zero to a maximum of $\frac{\varepsilon}{R}$, i.e. that given by Ohm's Law. We need to pay a little attention to the "time constant" τ, which is the time after which the constant approaches 63% of its final value. Units: $[\tau] = \frac{[L]}{[R]} = \frac{\text{henry}}{\text{ohm}} = \frac{\text{volt.second/ampere}}{\text{ohm}} = \left(\frac{\text{volt}}{\text{ampere.ohm}}\right)\text{second} = \text{second}.$

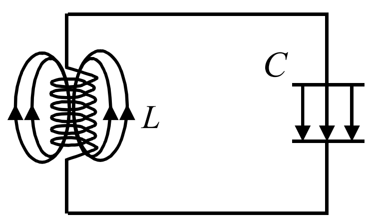
Note that when $t=\tau$, then $I = \frac{\varepsilon}{R}\left(1-e^{-1}\right) = \frac{\varepsilon}{R}\left(1-0.37\right) = \frac{\varepsilon}{R}0.63$

9. When we pass current through an inductor a changing magnetic field is produced. This, by Faraday's Law, induces an emf across the coil. So work has to be done to force the current through. How much work? The power, or rate of doing work, is emf $\times$ current. Let U_B be the work done in passing current I. Then, $\frac{dU_B}{dt} = \left(L\frac{dI}{dt}\right)I = LI\frac{dI}{dt}$. Let us integrate $dU_B = LI\,dI$. Then, $\int_0^{U_B} dU_B = \int_0^I LIdI \quad \Rightarrow \quad U_B = \frac{1}{2}LI^2$. This is an important result. It tells us that an inductor L carrying current I requires work $\frac{1}{2}LI^2$. By conservation of energy, this is also the energy stored in the inductor. Compare this result with the result $U_E = \frac{1}{2}CV^2$ for a capacitor. Notice that $C \leftrightarrow L$ and $V \leftrightarrow I$.

10. Let us use the result derived earlier for the inductance of a solenoid and the magnetic field in it, $L = \mu_0 n^2 lA$ and $B = \mu_0 nI$. Putting this into $U_B = \frac{1}{2}LI^2$ gives $U_B = \frac{1}{2}(\mu_0 n^2 lA)(B/\mu_0 n)^2$. Divide the energy by the volume of the solenoid, $\frac{U_B}{\text{volume}} = \frac{B^2}{2\mu_0}$. This directly gives the energy density (energy per unit volume) contained in a magnetic field.

11. **Electromagnetic oscillations in an LC circuit.**

We know that energy can be stored in a capacitor as well as in an inductor. What happens when we connect them up together and put some charge on the capacitor? As it discharges, it creates a current that transfers energy to the inductor. The total energy remains constant, of course.



Capacitor aur inductor ko jod kar energy oscillate karti hai.

This means that the sum $U = U_B + U_E = \frac{1}{2}LI^2 + \frac{1}{2}\frac{q^2}{C}$ is constant. Differentiate this:

$$0 = \frac{dU}{dt} = \frac{d}{dt}\left(\frac{1}{2}LI^2 + \frac{1}{2}\frac{q^2}{C}\right). \text{ Since } I = \frac{dq}{dt}, \text{ and } \frac{dI}{dt} = \frac{d^2q}{dt^2}, \quad \therefore \quad \frac{d^2q}{dt^2} + \frac{1}{LC}q = 0$$

To make this look nicer, put $\omega^2 \equiv \frac{1}{LC} \Rightarrow \frac{d^2q}{dt^2} + \omega^2 q = 0$. We have seen this equation many times earlier. The solution is: $q = q_m \cos\omega t$. The important result here is that the charge, current, and voltage will oscillate with frequency $\boxed{\omega = \sqrt{\frac{1}{LC}}}$. This oscillation will go on for forever if there is no resistance in the circuit.

Agar resistance na ho to oscillation hamesha chalta rahega.

Gauss's Law (Electric Field): Electric flux closed surface pe total charge se related hota hai.
Gauss's Law (Magnetic Field): Magnetic flux ka total closed surface pe zero hota hai (koi magnetic monopole nahi hota).
Faraday's Law: Changing magnetic field se electric field (emf) induce hoti hai.
Ampere's Law: Current se magnetic field banti hai.

Summary of Lecture 30 – ELECTROMAGNETIC WAVES

1. Before the investigations of James Clerk Maxwell around 1865, the known laws of electromagnetism were:

    a) Gauss' law of electricity: $\int \vec{E} \cdot d\vec{A} = \frac{q}{\varepsilon_0}$ (integral is over any closed surface)

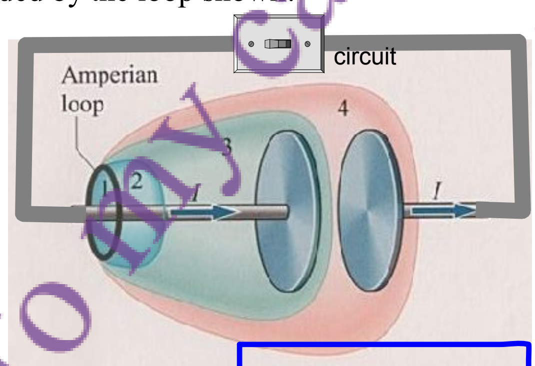
    b) Gauss' law of magnetism: $\int \vec{B} \cdot d\vec{A} = 0$ (integral is over any closed surface)

    c) Faraday's law of induction: $\int \vec{E} \cdot d\vec{s} = -\frac{d\Phi_B}{dt}$ (integral is over any closed loop)

    d) Ampere's law: $\int \vec{B} \cdot d\vec{s} = \mu_0 I$ (integral is over any closed loop)

2. But Maxwell realized that the above 4 laws were not consistent with the conservation of charge, which is a fundamental principle. He argued that if you take the space between two capacitors (see below) and take different surfaces 1,2,3,4 then applying Ampere's Law gives an inconsistency: $\left[\int \vec{B} \cdot d\vec{s}\right]_{(1,2,4)} \neq \left[\int \vec{B} \cdot d\vec{s}\right]_3$ because obviously charge cannot flow in the gap between plates. So Ampere's Law gives different results depending upon which surface is bounded by the loop shown!

Maxwell ne dekha ke agar capacitor ke plates ke darmiyan space consider karein to Ampere's Law inconsistent lagta hai, kyunki current plates ke darmiyan nahi flow karta phir bhi magnetic field exist karti hai.



Maxwell modified Ampere's law as follows: $\int \vec{B} \cdot d\vec{s} = \mu_0 \left(I + I_d\right)$ where the "displacement current" is $I_d = \varepsilon_0 \frac{d\Phi_E}{dt}$. Let's look at the reasoning that led to Maxwell's discovery of the displacement current. The current that flows in the circuit is $I = \frac{dQ}{dt}$ But the charge on the capacitor plate is $Q = \varepsilon_0 EA$. Hence, $I = \frac{d}{dt}\left(\varepsilon_0 EA\right) = \varepsilon_0 \frac{d(EA)}{dt} = \varepsilon_0 \frac{d\Phi_E}{dt} \equiv I_D$ In words, the changing electric field in the gap acts as source of the magnetic field in just the same way as the current in the outside wires. This is really the most important point - a magnetic field may have two separate reasons for existence - flowing charges or changing electric fields.

3. The famous Maxwell's equations are as follows:

$$\boxed{\begin{aligned}
&a)\ \int \vec{E}\cdot d\vec{S} = \frac{Q}{\varepsilon_0}\\
&b)\ \int \vec{E}\cdot d\vec{\ell} = -\frac{d\Phi_B}{dt}\\
&c)\ \int \vec{B}\cdot d\vec{S} = 0\\
&d)\ \int \vec{B}\cdot d\vec{\ell} = \mu_0\left(I + \varepsilon_0\frac{d\Phi_E}{dt}\right)
\end{aligned}}$$

Magnetic field sirf real current se nahi, balkay changing electric field (displacement current) se bhi generate hoti hai. Ye Maxwell ka major contribution tha jo electromagnetic waves ko explain karta hai

Together with the Lorentz Force $\boxed{\vec{F} = q\left(\vec{E} + \vec{v}\times\vec{B}\right)}$ they provide a complete description of all electromagnetic phenomena, including waves.

4. Electromagnetic waves were predicted by Maxwell and experimentally discovered many years later by Hertz. Note that for these waves:

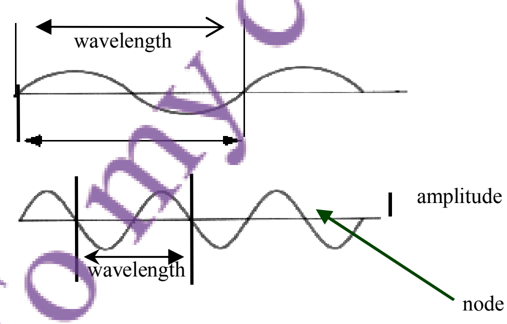
a) Absolutely no medium is required - they travel through vacuum.

b) The speed of propagation is c for all waves in the vacuum.

c) There is no limit to the amplitude or frequency.

In waves ko travel karne ke liye kisi medium ki zarurat nahi, matlab vacuum mein bhi travel karti hain.
Inki propagation speed vacuum mein c hoti hai (light ki speed).
Amplitude aur frequency ki koi limit nahi hoti.

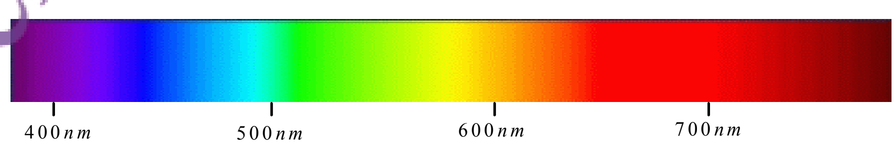
A wave is characterized by the amplitude and frequency, as illustrated below.



Example: Red light has $\lambda = 700$ nm. The frequency ν is calculated as follows:

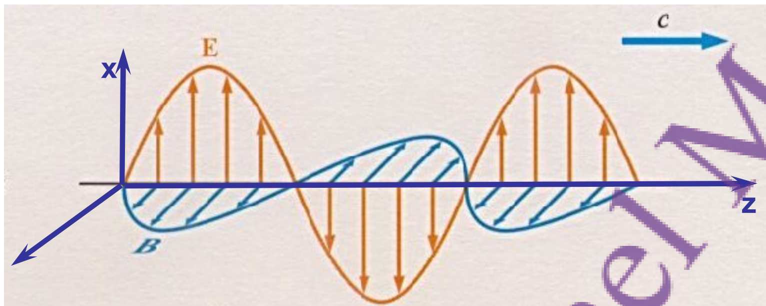
$$\boxed{\nu = \frac{3.0\times10^{8}\,m/\sec}{7\times10^{-7}\,m} = 4.29\times10^{14}\,Hertz}$$

By comparison, the electromagnetic waves inside a microwave oven have wavelength of 6 cm, radio waves are a few metres long. For visible light, see below. On the other hand, X-rays and gamma-rays have wavelengths of the size of atoms and even much smaller.



Wave Types and Wavelengths:
Microwave waves ki wavelength cm scale ki hoti hai.
Radio waves meters mein hoti hain.
X-rays aur gamma-rays atoms ke size ke barabar ya usse chhoti wavelength rakhte hain.

5. We now consider how electromagnetic waves can travel through empty space. Suppose an electric field (due to distant charges in an antenna) has been created. If this changes then this creates a changing electric flux $\frac{d\Phi_E}{dt}$ which, through $\int \vec{B}\cdot d\vec{s} = \mu_0\varepsilon_0 \frac{d\Phi_E}{dt}$, creates a changing magnetic flux $\frac{d\Phi_B}{dt}$. This, through $\boxed{\int \vec{E}\cdot d\vec{s} = -\frac{d\Phi_B}{dt}}$, creates a changing electric field. This chain of events in free space then allows a wave to propagate.

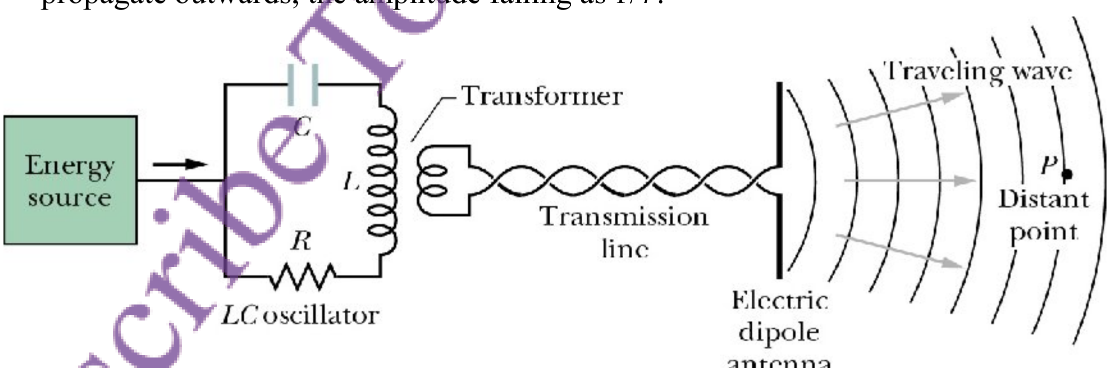


Electromagnetic Waves ka Space mein Travel Karna:
Jab antenna mein electric field banti hai (jo ke antenna ke charges ki wajah se hoti hai), agar yeh field time ke sath change hoti hai to yeh ek changing electric flux banati hai.
Faraday aur Maxwell ke laws ke mutabiq, yeh changing electric flux changing magnetic flux bhi create karta hai, aur yeh changing magnetic flux wapas electric field create karta hai.
Is tarah electric aur magnetic fields ek doosre ko generate karte rehte hain, jisse electromagnetic wave propagate karti hai.
Diagram mein wave z-direction mein travel kar rahi hai, electric field x-direction mein aur magnetic field uske perpendicular y-direction mein hai.
Dono fields phase mein hain aur amplitude Maxwell ke equations ke mutabiq relate hai: Ex = E0 sin(kz - ωt) aur By = B0 sin(kz - ωt).

In the diagram above, an electromagnetic wave is moving in the z direction. The electric field is in the x direction, $\boxed{E_x = E_0\sin(kz-\omega t)}$, and the magnetic field is perpendicular to it, $\boxed{B_y = B_0\sin(kz-\omega t)}$. Here $\omega = kc$. From Maxwell's equations the amplitudes of the two fields are related by $E_0 = cB_0$. Note that the two fields are in phase with each other.

6. The production of electromagnetic waves is done by forcing current to vary rapidly in a small piece of wire. Consider your mobile phone, for example. Using the power from the battery, the circuits inside produce a high frequency current that goes into a "dipole antenna" made of two small pieces of conductor. The electric field between the two oppositely charged pieces is rapidly changing and so creates a magnetic field. Both fields propagate outwards, the amplitude falling as $1/r$.

Electromagnetic Waves ka Production:
EM waves tab banti hain jab current wire mein rapidly change hota hai, jaise mobile phone ki antenna mein.
High frequency current wire ke do chhote conductors ke darmiyan electric field banata hai, jo rapidly change hota hai.
Yeh changing electric field magnetic field bhi banata hai, aur dono fields outward propagate karti hain.
Power jo wave mein carry hoti hai amplitude ka square hoti hai aur distance ke sath 1/r ke proportional kam hoti hai.
Power direction ke hisaab se sin²θ/r² ke proportional hoti hai, jisme θ angle hai, jiska matlab hai power unequal distribution hoti hai direction mein, max power θ=π/2 pe hoti hai aur min power θ=0 pe.



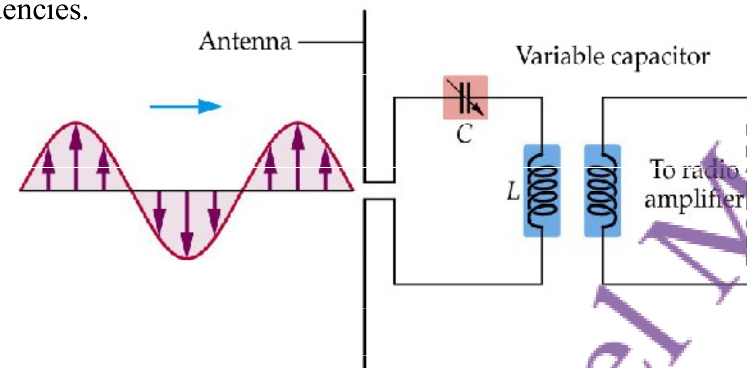
The power, which is the square of the amplitude, falls off as $I(\theta) \propto \frac{\sin^2\theta}{r^2}$. The $\sin^2\theta$ dependence shows that the power is radiated unequally as a function of direction. The maximum power is at $\theta = \pi/2$ and the least at $\theta = 0$.

EM waves electric aur magnetic fields ke mutual generation se space mein propagate karti hain aur yeh waves rapidly changing current ke zariye produce hoti hain.

Electromagnetic wave reception (Antenna ka role):

Jab EM wave antenna se takrati hai, uska *electric field electrons ko hilata hai jisse current generate hota hai. Ye current amplify hota hai (signal barhta hai), aur variable capacitor system ko different frequencies pe tune karta hai.

7. The reception of electromagnetic waves requires an antenna. The incoming wave has an electric field that forces the electrons to run up and down the antenna wire, i.e. it produces a tiny electric current. This current is then amplified (increased in amplitude) electronically. This is schematically indicated below. Here the variable capacitor is used to tune to different frequencies.



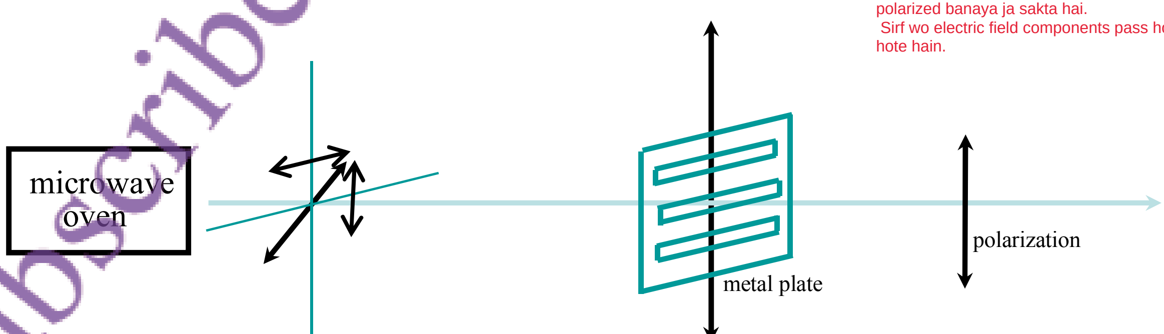
Polarization of EM waves:
EM wave ka electric field hamesha wave ke motion ke perpendicular hota hai.
Agar field ek fixed direction mein ho (e.g., $\hat{x}$), to wave linearly polarized kehlati hai.
Most natural sources (sun, candle) unpolarized light dete hain (random orientation).
Polarized wave ka electric field sirf ek direction mein oscillate karta hai.

8. As we have seen, the electric field of a wave is perpendicular to the direction of its motion. If this is a fixed direction (say, $\hat{x}$), then we say that wave is polarized in the x direction. Most sources - a candle, the sun, any light bulb - produce light that is unpolarized. In this case, there is no definite direction of the electric field, no definite phase between the orthogonal components, and the atomic or molecular dipoles that emit the light are randomly oriented in the source. But for a typical linearly polarized plane electromagnetic wave polarized along $\hat{x}$, $E_x = E_0 \sin(kz - \omega t)$, $B_y = \frac{E_0}{c}\sin(kz - \omega t)$ with all other components zero. Of course, it may be that the wave is polarized at an angle θ relative to $\hat{x}$, in which case

$$\boxed{E_x = E_0 \cos\theta \cdot \sin(kz - \omega t), \quad E_y = E_0 \sin\theta \cdot \sin(k z - \omega t), \quad E_z = 0.}$$

9. Electromagnetic waves from an unpolarized source (e.g. a burning candle or microwave oven) can be polarized by passing them through a simple polarizer of the kind below. A metal plate with slits cut into will allow only the electric field component perpendicular to the slits. Thus, it will produce linearly polarized waves from unpolarized ones.

Polarization from unpolarized source:
Unpolarized waves ko metal plate with slits ke through pass karke linearly polarized banaya ja sakta hai.
Sirf wo electric field components pass hote hain jo slits ke perpendicular hote hain.

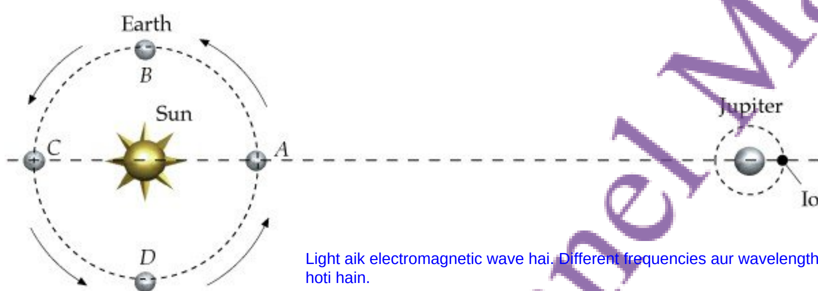


Important points••• EM waves antenna se detect hoti hain, polarization electric field ki direction se define hoti hai, aur unpolarized waves ko polarizer se pass karke polarized banaya ja sakta hai.

Summary of Lecture 31 – LIGHT

Light ki speed bohat zyada hoti hai, lekin infinite nahi hoti. 1675 mein astronomer Roemer ne Jupiter ke moon Io ke eclipse ka time observe karke light ki speed estimate ki takriban 3 × 10⁸ m/s. Ye value modern measurement ke kafi qareeb hai.

1. Light travels very fast but its speed is not infinite. Early attempts to measure the speed using earth based experiments failed. Then in 1675 the astronomer Roemer studied timing of the eclipse of one of Jupiter's moon called Io. In the diagram below Io is observed with Earth at A and then at C. The eclipse is 16.6 minutes late, which is the time taken for light to travel AC. Roemer estimated that $\boxed{c = 3\times10^8 \text{ metres/sec}}$ a value that is remarkably close to the best modern measurement, $\boxed{c = 299792458.6 \text{ metres/sec}}$.



Light aik electromagnetic wave hai. Different frequencies aur wavelengths different colors ke mutabiq hoti hain.

2. Light is electromagnetic waves. Different frequencies ν correspond to different colours. Equivalently, different wavelength λ correspond to different colours. Recall that the product $\boxed{\nu\times\lambda = c}$. In the diagram below you see that visible light is only one small part of the total electromagnetic spectrum. Here nm means nanometres or 10^{-9} metres.

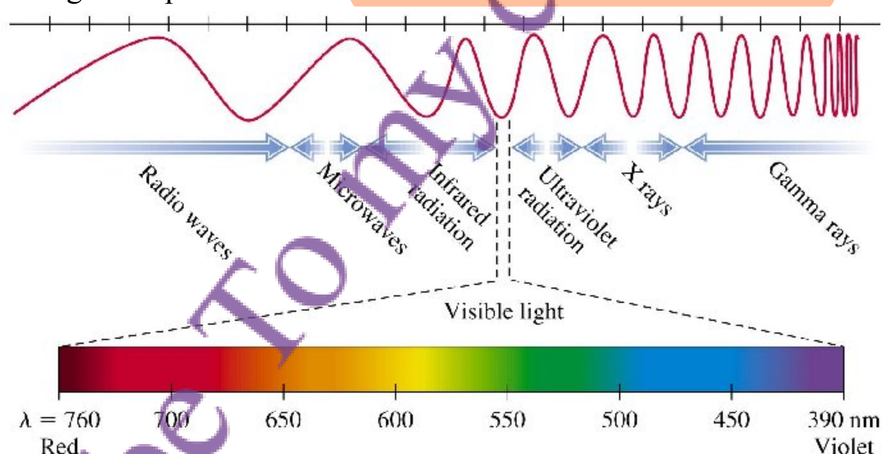
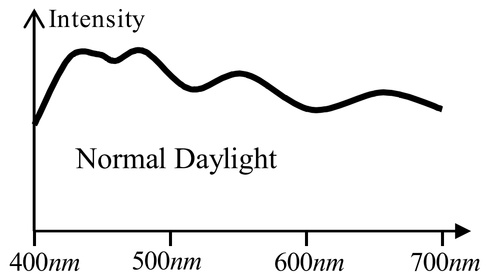


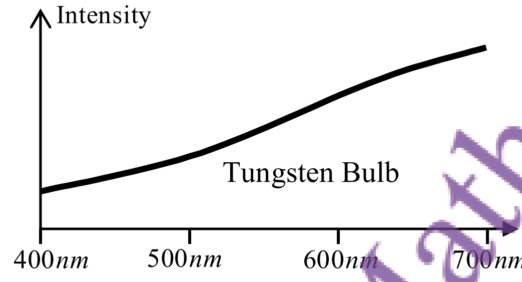
Diagram mein dikhaya gaya hai ke visible light electromagnetic spectrum ka sirf chhota sa hissa hai.

3. If light contained all frequencies with equal strength, it would appear as white to us. course, most things around us appear coloured. That is because they radiate more strongly in one range of frequency than in others. If there is more intensity in the yellow range than the green range, we will see mostly yellow. The sky appears blue to us on a clear day because tiny dust particles high above in the atmosphere reflect a lot of the blue light coming from the sun. In the figure below you can see the hump at smaller wavelengths.

Agar light mein sab frequencies barabar strength ke sath hotin, to wo humein white nazar aati. Lekin cheezein colored is liye lagti hain kyunke wo kuch specific frequency mein zyada radiate karti hain.Agar yellow range mein zyada intensity ho, to humein mostly yellow nazar aata hai. Sky neela lagta hai kyunke upper atmosphere mein dust particles blue light ko zyada scatter karte hain.

Compare this with the spectrum of light emitted from a tungsten bulb. You can see that this is smoother and yellow dominates.





4. What path does light travel upon? If there is no obstruction, it obviously likes to travel on straight line which is the shortest path between any two points, say A and B. Fermat's Principle states that in all situations, light will always take that path for which it takes the least time. As an example, let us apply Fermat's Principle to the case of light reflected from a mirror, as below.

Light jab kisi obstruction ke bagair move karti hai to seedha rasta choose karti hai jo ke do points ke darmiyan shortest path hota hai.
Fermat's Principle kehta hai ke light hamesha aisa rasta choose karti hai jahan se minimum time mein pohonch sake.

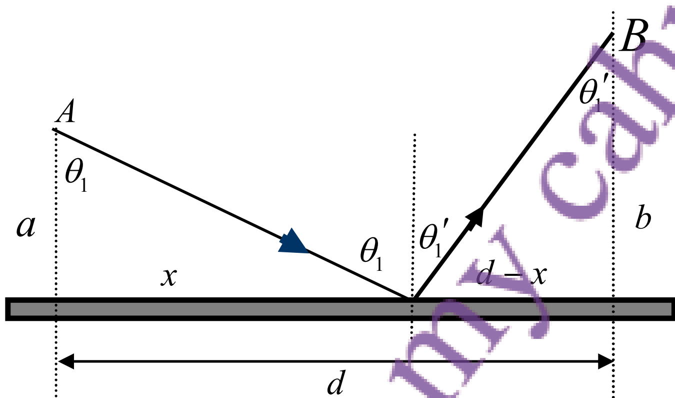


Diagram mein dekha gaya ke Light point A se mirror tak ja rahi hai (angle θ_1), phir wahan se reflect ho ke point B tak jaati hai (angle θ_1').Total distance calculate karke time function banaya gaya. Phir minimum time ke liye is function ka derivative zero set kiya gaya.

The total distance travelled by the ray is $L=\sqrt{a^2+x^2}+\sqrt{b^2+(d-x)^2}$. The time taken is $t=L/c$. To find the smallest time, we must differentiate and then set the derivative to zero,

$$0=\frac{dt}{dx}=\frac{1}{c}\frac{dL}{dx}$$

$$=\frac{1}{2c}\left(a^2+x^2\right)^{-1/2}(2x)+\frac{1}{2c}\left[b^2+(d-x)^2\right]^{1/2}(2)(d-x)(-1)$$

From here we immediately see that $\boxed{\frac{x}{\sqrt{a^2+x^2}}=\frac{d-x}{\sqrt{b^2+(d-x)^2}}}$ From the above diagram, $\sin\theta_1=\sin\theta_1'$. Of course, it is no surprise that the angle of reflection equals the angle of incidence. You knew this from before, and this seems like a very complicated derivation of a simple fact. But it is still nice to see that there is a deeper principle behind it.

Law of Reflection (angle of incidence = angle of reflection)

light reflection mein hamesha fastest path follow karti hai jis m usy minimum time required ho

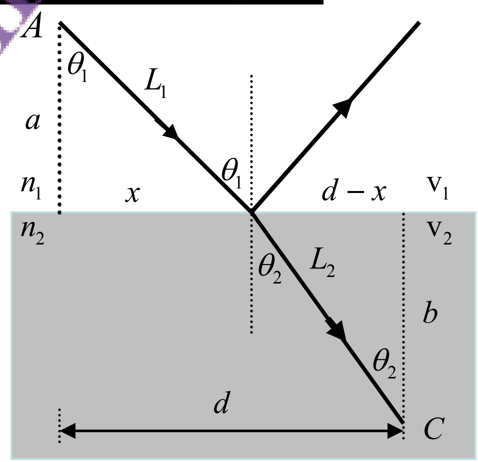
refractive index of an optical medium is the ratio of the apparent speed of light in the air or vacuum to the speed in the medium.
The refractive index determines how much the path of light is bent, or refracted, when entering a material.

5. The speed of light in vacuum is a fixed constant of nature which we usually call c, but in a medium light can travel slower or faster than c. We define the "refractive index" of that medium as: $\text{Refractive index} = \dfrac{\text{Speed of light in vacuum}}{\text{Speed of light in material}}$ (or $n = \dfrac{c}{\mathrm{v}}$). Usually the values of n are bigger than one (e.g. for glass it is around 1.5) but in some special media, its value can be less than one. The value of n also depends on the wavelength (or frequency) of light. This is called dispersion, and it means that different colours travel at different speeds inside a medium. This is why, as in the diagram below, white light gets separated into different colours. The fact that in a glass prism blue light travels faster than red light is responsible for the many colours we see here.

Light ke different colors (frequencies) ki speed alag hoti hai, isliye prism mein rainbow banta hai (blue light slow aur zyada bend hoti hai).

Light jab ek medium se doosre mein jaati hai (jaise air to water), to kuch part reflect hota hai aur kuch part refract hota hai.
Fermat ka principle kehta hai light wo rasta choose karti hai jahan time minimum lage.

6. We can apply Fermat's Principle to find the path followed by a ray of light when it goes from one medium to another. Part of the light is reflected, and part is "refracted", i.e. it bends away or towards the normal. The total time is,



$$t = \frac{L_1}{\mathrm{v}_1} + \frac{L_2}{\mathrm{v}_2},\quad n = \frac{c}{\mathrm{v}} \Rightarrow t = \frac{n_1 L_1 + n_2 L_2}{c} = \frac{L}{c}$$

$$L = n_1 L_1 + n_2 L_2 \quad = n_1\sqrt{a^2 + x^2} + n_2\sqrt{b^2 + (d-x)^2}$$

Fermat's Principle says that the time t must be minimized: $0 = \dfrac{dt}{dx} = \dfrac{1}{c}\dfrac{dL}{dx} = \dfrac{n_1}{2c}\left(a^2 + x^2\right)^{-1/2}(2x) + \dfrac{n_2}{2c}\left[b^2 + (d-x)^2\right]^{1/2}(2)(d-x)(-1)$

And so we get "Snell's Law" $\boxed{n_1 \dfrac{x}{\sqrt{a^2 + x^2}} = n_2 \dfrac{d-x}{\sqrt{b^2 + (d-x)^2}} \text{ or } n_1 \sin\theta_1 = n_2 \sin\theta_2}$. This required a little bit of mathematics, but you can see how powerful Fermat's Principle is!

Snell's Law => $n_1 \sin\theta_1 = n_2 \sin\theta_2$

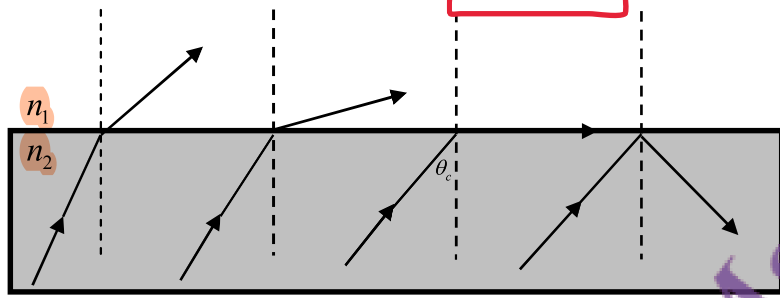
7. Light coming from air into water bends toward the normal. Conversely, light from a source in the water will bend away from the normal. What if you keep increasing the angle with respect to the normal so that the light bends and begins to just follow the surface? This phenomenon is called total internal reflection and θ_c is called the critical

Total Internal Reflection:
Light jab water se air mein jaati hai, aur agar angle critical value se zyada ho jaye, to light surface se bahar nahi nikalti balkay andar hi reflect hoti hai. Isay total internal reflection kehte hain, aur us angle ko critical angle (θc) kehte hain.

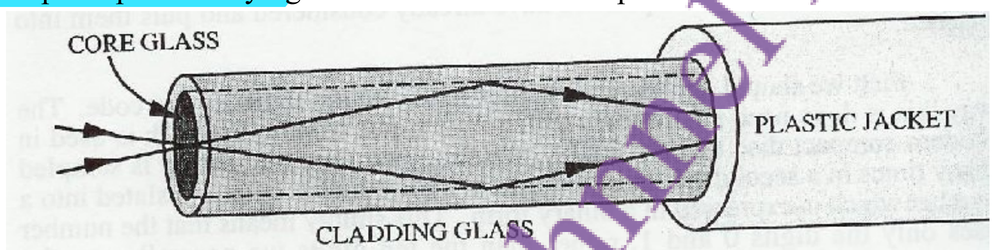
Total internal reflection: It is the phenomenon in which waves arriving at the interface from one medium to another are not refracted into the second medium, but completely reflected back into the first medium.

angle. It obeys: $\boxed{n_1 \sin\theta_c = n_2 \sin 90^0}$ from which $\boxed{\theta_c = \sin^{-1}\frac{n_2}{n_1}}$.

The angle of incidence when the angle of refraction is 90°, is called the critical angle. Total internal reflection only occurs when: light travels from a more dense medium to a less dense medium e.g. from glass to air. the angle of incidence in the more dense medium is greater than the critical angle



8. Fibre optic cables, which are now common everywhere, make use of the total internal reflection principle to carry light. Here is what a fibre optic cable looks like from inside:



Even if the cable is bent, the light will continue to travel along it. The glass inside the cable must have exceedingly good consistency - if it thicker or thinner in any part, the refractive index will become non-uniform and a lot of light will get lost. Optical fibres now carry thousands of telephone calls in a cable whose diameter is only a little bigger than a human hair!

Fibre Optic Cables:
Yeh cables total internal reflection ka use karti hain taake light ko guide karein.
Cable ke andar core glass hota hai jisme light reflect hoti rehti hai due to higher refractive index.
Cladding glass ka refractive index kam hota hai, jo reflection maintain karta hai.
Light bend hone ke bawajood andar travel karti rehti hai agar glass ki quality uniform ho
Agar refractive index uneven ho gaya (glass zyada mota/patla ho gaya), to light leak ho sakti hai
Ye optical fibres bohat thin hote hain (insani baal se bhi patle), aur hazaron telephone calls carry karte hain.

Summary of Lecture 32 – INTERACTION OF LIGHT WITH MATTER

1. In this lecture I shall deal with the 4 basic ways in which light interacts with matter:

a) Emission - matter releases energy as light. Jab matter light energy release karta hai.

b) Absorption - matter takes energy from light. Jab matter light energy ko absorb karta hai.

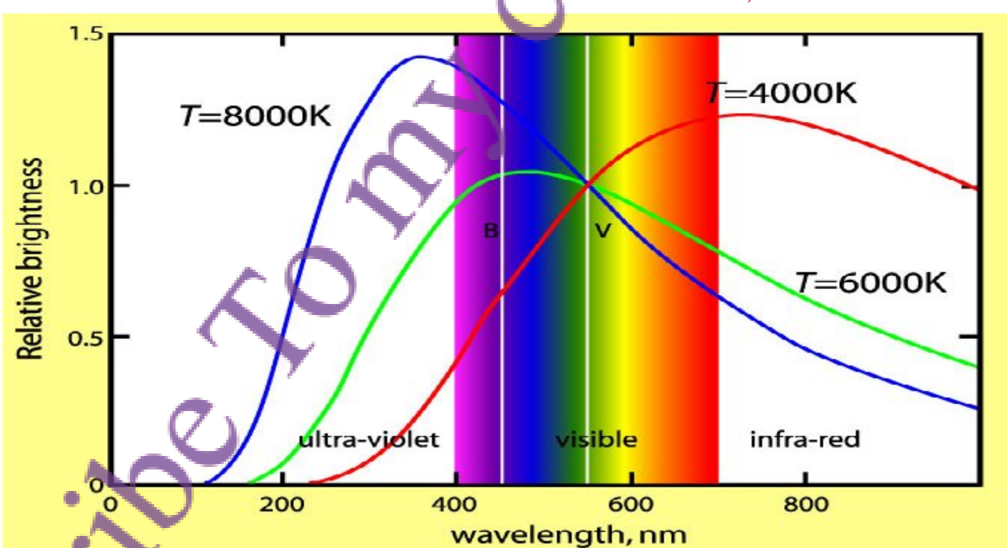
c) Transmission - matter allows light to pass through it. Jab light matter ke through guzarti hai.

d) Reflection - matter repels light in another direction. Jab light matter se takra kar wapas jati hai.

Jab kisi object ka temperature barhta hai (jaise iron rod ya bulb filament), to wo light emit karta hai.

2. When an object (for example, an iron rod or the filament of a tungsten bulb) is heated, it emits light. When the temperature is around 800°C, it is red hot. Around 2500°C it is yellowish-white. At temperatures lower than 800°C, infrared (IR) light is emitted but our eyes cannot see this. This kind of emission is called blackbody radiation. Blackbody radiation is continuous - all wavelengths are emitted. However most of the energy is radiated close to the peak. As you can see in the graph, the position of the peak goes to smaller wavelengths (or higher frequencies) as the object becomes hotter. The scale of temperature is shown in degrees Kelvin (^{o}K). To convert from °C to ^{o}K, simply add 273. We shall have more to say about the Kelvin scale later.

Graph batata hai ke temperature barhne se peak wavelength choti hoti hai (ya frequency barhti hai).



Where exactly does the peak occur? Wien's Law states that $\lambda_{max} T = 2.90 \times 10^{-3}$ m K. We can derive this in an advanced physics course, but for now you must take this as given.

λ_{max} = peak wavelength
T = temperature in Kelvin

Blackbody radiation tab hoti hai jab koi object heat ho kar energy radiate karta hai.

3. In the lecture on electromagnetic waves you had learnt that these waves are emitted when charges accelerate. Blackbody radiation occurs for exactly this reason as well. When a body is heated up, the electrons, atoms, and molecules which it contains undergo violent random motion. Light may emitted by electrons in one atom and absorbed in another. Even an empty box will be filled with blackbody radiation because the sides of the box are made up of material that has charged constituents that radiate energy when they undergo acceleration during their random motion.

Wien's Law un objects ke liye use hoti hai jo light emit karte hain (jaise Sooraj). Ye law batata hai ke object ka temperature aur uski peak wavelength ka aapas mein kya relation hai.

4. When can you use Wien's Law? More generally, when can you expect a body to emit blackbody radiation? Answer: only for objects that emit light, not for those that merely reflect light (e.g. flowers). The Sun and other stars obey Wien's Law since the gases they are composed of emit radiation that is in equilibrium with the other materials. Wien's law allows astronomers to determine the temperature of a star because the wavelength at which a star is brightest is related to its temperature.

Stefan-Boltzmann Law kehta hai ke jitna zyada temperature hoga, utni zyada energy radiate hogi.

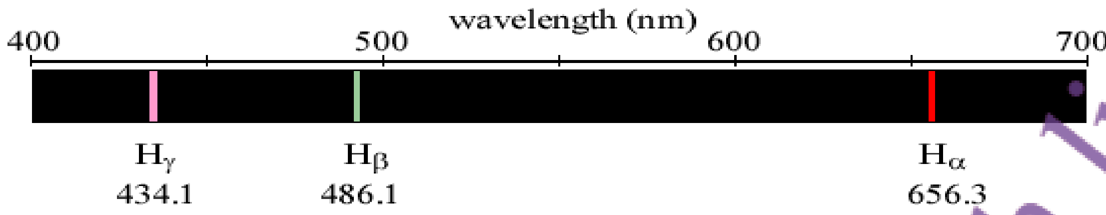
5. All heated matter radiates energy, and hotter objects radiate more energy. The famous Stefan-Boltzman Law, which we unfortunately cannot derive in this introductory course, states that the power radiated per unit area of a hot body is $\boxed{P = \sigma T^4}$, where the Stefan-Boltzman constant is $\boxed{\sigma = 5.67 \times 10^{-8}\ \text{W m}^{-2}\ \text{K}^{-4}}$.

6. Let us apply $P = \sigma T^4$ for finding the temperature of a planet that is at distance R from the sun. The sun has temperature T_{sun} and radius R_{sun}. In equilibrium, the energy received from the sun is exactly equal to the energy radiated by the planet. Now, the total energy radiated by the sun is $\sigma T_0^4 \times 4\pi R_{sun}^2$. But on a unit area of the planet, only $\frac{1}{4\pi R^2}$ of this is received. So the energy received per unit area on the planet is $\sigma T_0^4 \times 4\pi R_{sun}^2 \times \frac{1}{4\pi R^2}$. This must be equal to $\sigma T^4 \Rightarrow \boxed{T = T_0 \sqrt{\frac{R_{sun}}{R}}}$.

Emission Spectrum: Agar gas excite ho jaye, to wo sirf kuch specific rang (wavelengths) mein light emit karti hai.

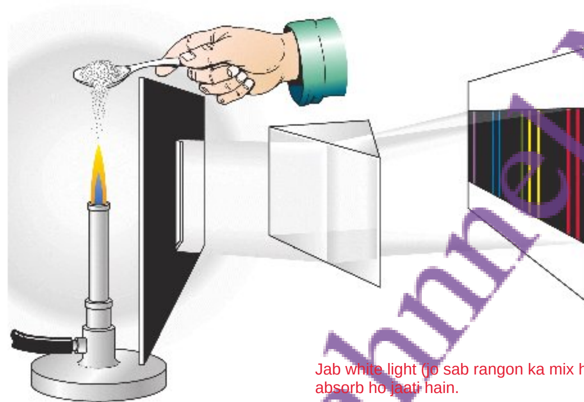
7. The above was for blackbody radiation where the emitted light has a continuous spectrum. But if a gas of identical atoms is excited by some mechanism, then only a few discrete wavelengths are emitted. Each chemical element produces a very distinct pattern of colors called an emission spectrum. So, for example, laboratory hydrogen gas lamps emit 3 lines in the visible region, as you can see below. Whenever we see 3 lines spaced apart in this way, we immediately know that hydrogen gas is present. It is as good as the thumbprint of a man!

Hydrogen gas teen distinct lines emit karti hai visible range mein in lines ko dekh kar hydrogen ki pehchan ho sakti hai.



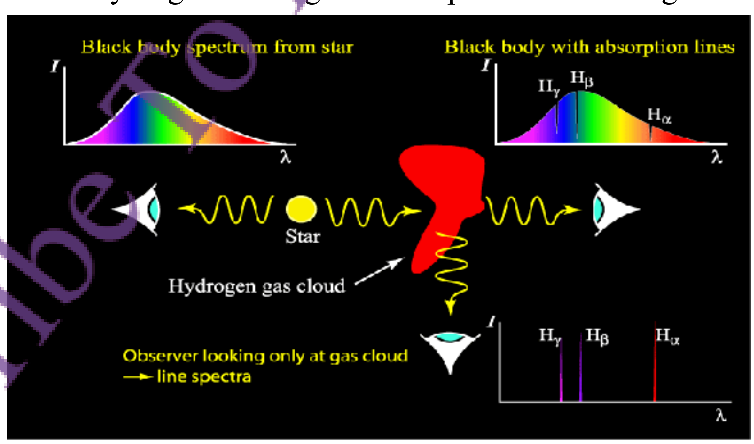
8. But how do we get atoms excited so that they can start revealing their identity? One way is to simply heat material containing those atoms. You saw in the lecture how different colours come from sprinkling different materials on a flame.

Jab kisi atom ko heat dete hain, to wo excite hota hai aur specific rang ka light emit karta hai. Har element ka apna unique color hota hai is tarah wo apni pehchaan reveal karta hai.

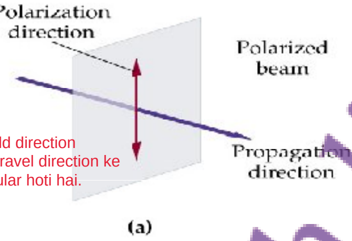


Jab white light (jo sab rangon ka mix hai) kisi gas (jaise hydrogen) se guzarti hai, to kuch specific wavelengths absorb ho jaati hain.

9. Everything that I have said about the emission of light applies exactly to the *absorption* of light as well. So, for example, when white light (which has all different frequencies within it) passes through hydrogen gas, you will see that all wavelengths survive except the three on the previous page. So the absorption spectrum looks exactly the same as the emission spectrum - the same lines are emitted and absorbed. This how we know that there are huge clouds of hydrogen floating in outer space. See the diagram below.

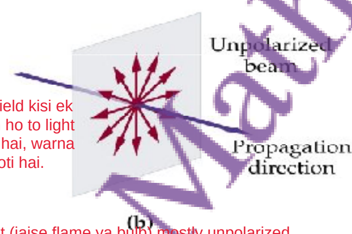


Light ek electromagnetic wave hai jisme electric aur magnetic field hoti hain.



11. As you learned earlier, light is an electromagnetic wave that has an electric field vector. This vector is always perpendicular to the direction of travel, but it can be pointing anywhere in the plane. If it is pointing in a definite direction, we say that the wave is polarized, else it is unpolarized. In general, the light emitted from a source, such as a flame, will be unpolarized and it is equally possible to find any direction of the electric field in the wave. Of course, the magnetic field is perpendicular to both the direction of travel and the electric field.

Electric field direction hamesha travel direction ke perpendicular hoti hai.



Agar electric field kisi ek direction mein ho to light polarized hoti hai, warna unpolarized hoti hai.

Natural light (jaise flame ya bulb) mostly unpolarized hoti hai.

10. The atmosphere contains various gases which absorb light at many different wavelengths. Molecules of oxygen, nitrogen, ozone, and water have their own absorption spectra, just as atoms have their own.

Molecules atmosphere mein alag wavelengths absorb karte hain Jaise oxygen, ozone, aur water ka apna absorption spectrum hota hai.

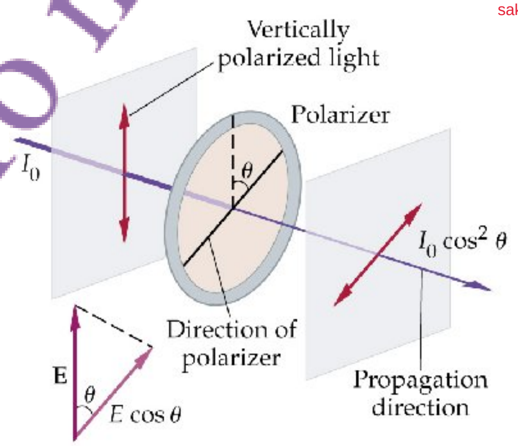
11. All the beauty of colours we see is due to the selective absorption by molecules of certain frequencies. So, for example, *carotene* is a long, complicated molecule that makes carrots orange, tomatoes red, sarson yellow, and which absorbs blue light. Similarly *chlorophyll* makes leaves green and which absorbs red and blue light.

Carotene blue light absorb karta hai (carrots, tomatoes, sarson ka color).

Chlorophyll red aur blue absorb karta hai, is liye leaves green dikhte hain.

12. From unpolarized light we can make polarized light by passing it through a polarizer as shown below.

Unpolarized light ko polarizer se guzarne par polarized banaya ja sakta hai.



Amplitude reduce hoti hai cos(θ) se aur intensity cos²(θ) se.

Each wave is reduced in amplitude by $\cos\theta$, and in intensity by $\cos^2\theta$. The wave that emerges is now polarized in the θ direction.

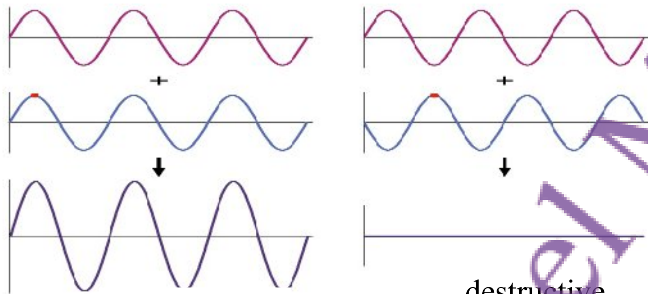
13. We can design materials (crystals or stressed plastics) so that they have different optical properties in the two transverse directions. These are called birefringent materials. They are used to make commonly used liquid crystal displays (LCD) in watches and mobile phones. Birefringence can occur in any material that possesses some asymmetry in its structure where the material is more springy in one direction than another.

Birefringent materials (LCDs mein use hote hain) different directions mein light ko alag treat karte hain.

Interference:
(a) It is the result of inter-action of light coming from two wavefronts different from two coherent sources.
(b) Interference fringes are of the same width
(c) The dark fringes are perfectly dark
(d) All bright bands are of the same intensity.
Diffraction:
a) It is the result of interaction of light coming from different parts of the same wavefront.
(b) Diffraction fringes are not of the same width.
c) The dark fringes are not perfectly dark.
(d) All bright bands are not of the same intensity.

Summary of Lecture 33 – INTERFERENCE AND DIFFRACTION

1. Two waves (of any kind) add up together, with the net result being the simple sum of the two waves. Consider two waves, both of the same frequency, shown below. If they start together (i.e. are *in phase* with each other) then the net amplitude is increased. This is called *constructive interference.* But if they start at different times (i.e. are *out of phase* with each other) then the net amplitude is decreased. This is called *destructive interference.*



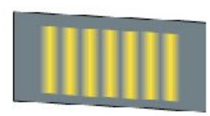
constructive destructive

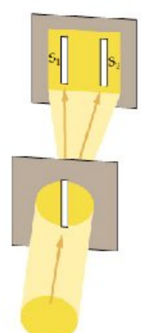
Constructive Interference:
Jab do waves in phase hoti hain (ek hi waqt par start karti hain), to unka amplitude add hota hai amplitude zyada hota hai

Destructive Interference:
Jab do waves out of phase hoti hain (alag waqt par start karti hain), to unka amplitude cancel hota hai net amplitude kam ya zero hoti hai.

In the example above, both waves have the same frequency and amplitude, and so the resulting amplitude is doubled (constructive) or zero (destructive). But interference occurs for any two waves even when their amplitudes and frequencies are different.

2. Although any waves from different sources interfere, if one wants to observe the interference of light then it is necessary to have a *coherent* source of light. Coherent means that both waves should have a fixed phase relative to each other. Even with lasers, it is very difficult to produce coherent light from two separate sources. Observing interference usually requires taking two waves from a single source, with each going along a different path. In the figure, an incoherent light source illuminates the first slit. This creates a uniform and coherent illumination of the second screen. Then waves from the slits S_1 and S_2 meet on the third screen and create a pattern of alternating light and dark fringes.





Coherent Sources:

Interference ke liye waves ka fixed phase relation hona zaroori hai. Isliye hum usually ek hi source se do waves lete hain (e.g., ek slit se light jaake doosri slit tak pohanchti hai).

Fringe Pattern: Jab light S_1 aur S_2 se third screen par milti hai, to phase difference ki wajah se bright aur dark fringes banti hain.

3. Wherever there is a bright fringe, constructive interference has occured, and wherever there is a dark fringe, destructive interference has occured. We shall now calculate where on the third screen the interference is constructive. Take any point on the third screen. Light reaches this point from both S_1 and S_2, but it will take different amounts of time to get there. Hence there will be a phase difference that we can calculate. Look at the diagram below. You can see that light from one of the slits has to travel an extra

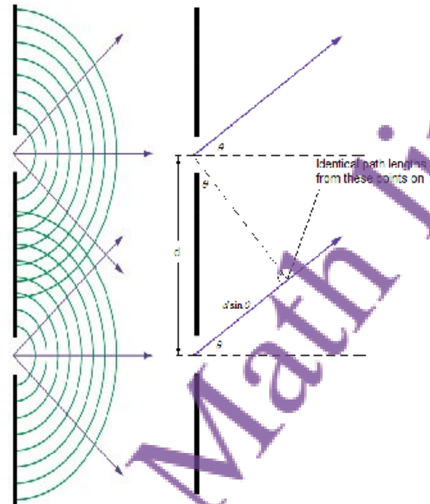
Bright fringe: Constructive interference
Dark fringe: Destructive interference

Constructive Interference Condition:
d·sinθ = nλ (n = 1, 2, 3, ...)
Jab path difference nλ ho to bright fringes banti hain.

distance equal to $d\sin\theta$, and so the extra amount of time it takes is $(d\sin\theta)/c$. There will be constructive interference if this is equal to $T, 2T, 3T, \cdots$ (remember that the time period is inversely related to the frequency, $T = 1/\nu$, and that $c = \lambda\nu$).

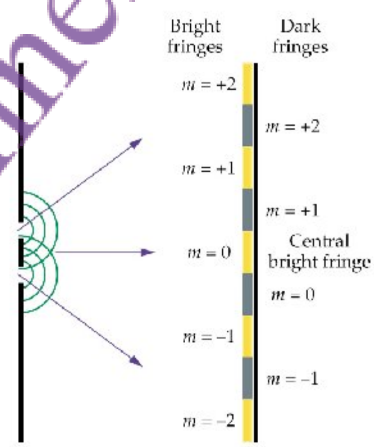
We find that $\dfrac{d\sin\theta}{c} = nT = \dfrac{n}{\nu} \quad \Rightarrow \boxed{d\sin\theta = n\lambda,}$

where $n = 1, 2, 3, \cdots$ What about for destructive interference? Here the waves will cancel each other if the extra amount of time is $\dfrac{T}{2}, \dfrac{3T}{2}, \dfrac{5T}{2}, \cdots$ The condition then becomes $\boxed{d\sin\theta = (n+\frac{1}{2})\lambda.}$



Destructive Interference Condition:
d·sinθ = (n + ½)λ
Jab path difference odd multiple of λ/2 ho to dark fringes banti hain.

4. Example: two slits with a separation of 8.5×10^{-5}m create an interference pattern on a screen 2.3m away. If the $n = 10$ bright fringe above the central is a linear distance of 12 cm from it, what is the wavelength of light used in the experiment?

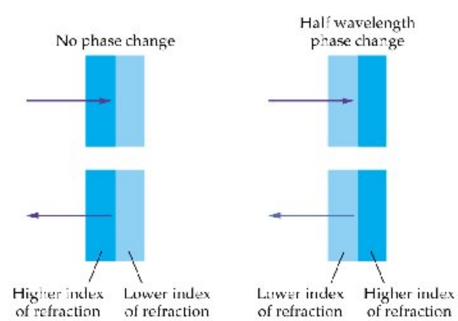


Answer: First calculate the angle to the tenth bright fringe using $y = L\tan\theta$. Solving for θ gives,

$$\boxed{\theta = \tan^{-1}\left(\frac{y}{L}\right) = \tan^{-1}\left(\frac{0.12m}{2.3m}\right) = 3.0^\circ.}$$ From this,

$$\boxed{\lambda = \frac{d}{n}\sin\theta = \left(\frac{8.5\times 10^{-5}}{10}\right)\sin(3.0^\circ) = 4.4\times 10^{-7}m = 440nm \text{ (nanometres).}}$$

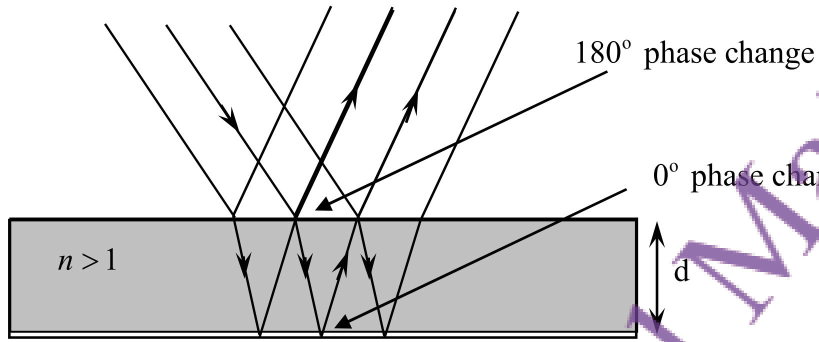
5. When a wave is reflected at the interface of two media, the phase will not change if it goes from larger refractive index to a smaller one. But for smaller to larger, there will be phase change of a half-wavelength. One can show this using Maxwell's equations and applying the boundary conditions, but this will require some more advanced studies. Instead let's just use this fact below.



Jab light lower to higher refractive index se reflect hoti ha half-wavelength phase shift hoti hai. Higher to lower mein no phase shift.

Thin Film Interference: jab light ek thin film (jaise soapy water, oil) par girti hai, to upar wali surface se reflection mein 180° phase change hota hai, aur neeche wali surface se no phase change. Yeh dono reflected rays aapas mein interfere karti hain.

5. When light falls upon a thin film of soapy water, oil, etc. it is reflected from two surfaces. On the top surface, the reflection is with change of phase by π whereas at the lower surface there is no change of phase. This means that when waves from the two surfaces combine at the detector (your eyes), they will interfere.



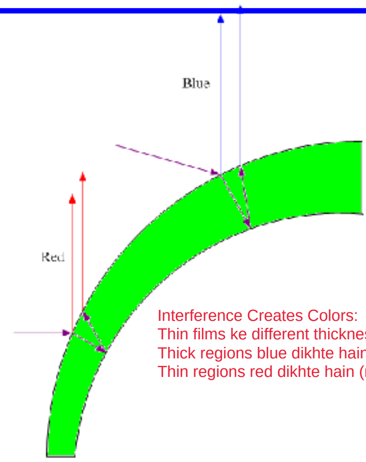
To simplify matters, suppose that you are looking at the thin film almost directly from above. Here n is the index of refraction for the medium. Then,

The condition for destructive interference is: $2nd = m\lambda \quad (m = 0,1,2,...)$

The condition for constructive interference is : $2nd = (m + \frac{1}{2})\lambda \quad (m = 0,1,2,...)$.

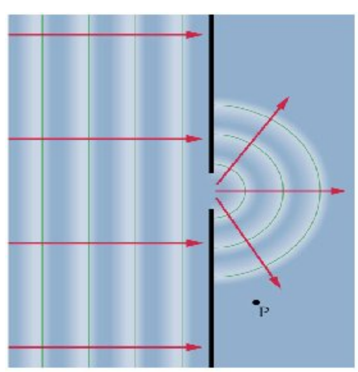
Prove it!

6. Interference is why thin films give rise to colours. A drop of oil floating on water spreads out until it is just a few microns thick. It will have thick and thin portions. Thick portions of any non-uniform thin film appear blue because the long-wavelength red light experiences destructive interference. Thinner portions appear red because the short-wavelength blue light interferes destructively.



Interference Creates Colors:
Thin films ke different thickness regions mein different colors dikhte hain.
Thick regions blue dikhte hain (blue light destructively interfere hoti hai).
Thin regions red dikhte hain (red light zyada interfere nahi karti).

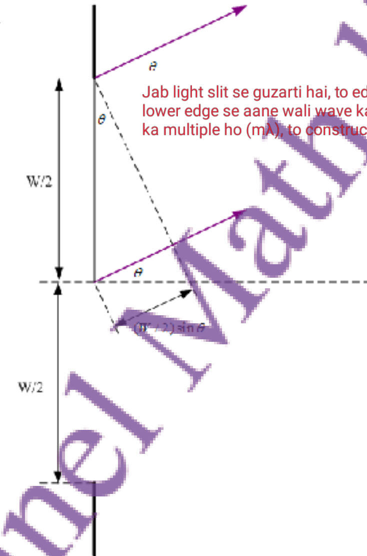
7. **Diffraction :** The bending of light around objects (into what would otherwise be a shadowed region) is known as diffraction. Diffraction occurs when light passes through very small apertures or near sharp edges. Diffraction is actually just interference, with the difference being only in the source of the interfering waves. Interference from a single slit, as in the figure here, is called diffraction.



Light ka bend hona obstacles ke aas-paas is called diffraction. Diffraction hoti hai jab light narrow slit ya sharp edge se guzarti hai. Actually, diffraction bhi interference ka hi ek form hai, jisme wave ek hi source se nikalti hai.

We can get an interference pattern with a single slit provided its size is approximately equal to the wavelength of the light (neither too small nor too large).

8. Let's work out the condition necessary for diffraction of light from a single slit. With reference to the figure, imagine that a wave is incident from the left. It will cause secondary waves to be radiated from the edges of the slit. If one looks at angle θ, the extra distance that the wave emitted from the lower slit must travel is $W\sin\theta$. If this is a multiple of the wavelength λ, then constructive interference will occur. So the condition becomes $\boxed{W\sin\theta = m\lambda \quad \text{with} \quad m = \pm1, \pm2, \pm3}$.. So, even from a single slit one will see a pattern of light and dark fringes when observed from the other side.



Jab light slit se guzarti hai, to edges se secondary waves nikalti hain.Agar lower edge se aane wali wave ka extra path W·sinθ ho aur yeh wavelength λ ka multiple ho (mλ), to constructive interference hoti hai.

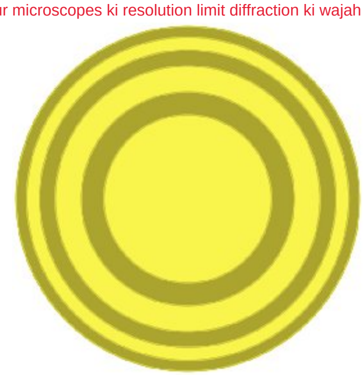
9. Light with wavelength of 511 nm forms a diffraction pattern after passing through a single slit of width 2.2×10^{-6}m. Find the angle associated with (a) the first and (b) the second bright fringe above the central bright fringe.

SOLUTION: For $m = 1$, $\theta = \sin^{-1}\left(\frac{m\lambda}{W}\right) = \sin^{-1}\left(\frac{(1)(511\times10^{-9}m)}{2.20\times10^{-6}m}\right) = 13.4^\circ$

For $m = 2$, $\theta = \sin^{-1}\left(\frac{m\lambda}{W}\right) = \sin^{-1}\left(\frac{(2)(511\times10^{-9}m)}{2.20\times10^{-6}m}\right) = 27.7^\circ$

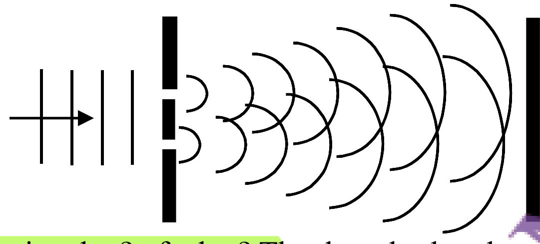
10. Diffraction puts fundamental limits on the capacity of telescopes and microscopes to separate the objects being observed because light from the sides of a circular aperture interferes. One can calculate that the first dark fringe is at $\boxed{\theta_{min} = 1.22\frac{\lambda}{D}}$, where D is diameter of the aperture. Two objects can be barely resolved if the diffraction maximum of one object lies in the diffraction minimum of the second object. Clearly, the larger D is, the smaller the angular diameter separation. We say that larger apertures lead to better resolution.

Telescopes aur microscopes ki resolution limit diffraction ki wajah se hoti hai.

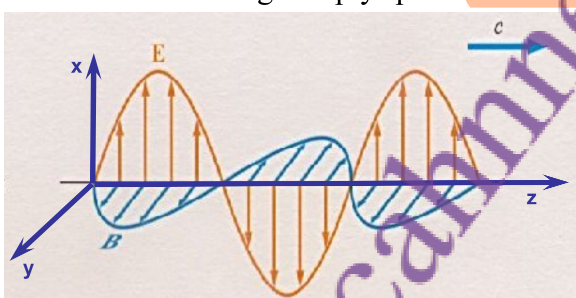


Summary of Lecture 34 – THE PARTICLE NATURE OF LIGHT

1. In the previous lecture I gave you some very strong reasons to believe that light is waves. Else, it is impossible to explain the interference and diffraction phenomena that we see in innumerable situations. Interference from two slits produces the characteristic pattern.



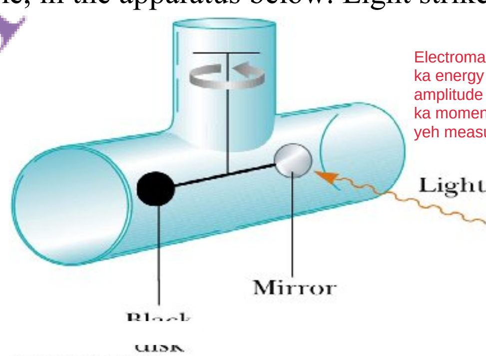
2. Light is waves, but waves in what? of what? The thought that there is some invisible medium (given the name *aether*) turned out to be wrong. Light is actually electric and magnetic waves that can travel through empty space. The electric and magnetic waves



Light waves hai, lekin kis cheez ki waves? Pehle yeh socha jata tha ke koi invisible medium hai (jise aether kaha jata tha), lekin yeh ghalat sabit hua. Light asal mein electric aur magnetic waves ka combination hai jo empty space mein travel karti hain. Electric aur magnetic waves ek dosray ke perpendicular hoti hain aur travel ke direction (yahan z-direction) ke bhi perpendicular hoti hain.

are perpendicular to each other and to the direction of travel (here the z direction).

3. Electromagnetic waves transport linear momentum and energy. If the energy per unit volume in a wave is U then it is carrying momentum p, where $\boxed{p = U/c}$ Waves with large amplitude carry more energy and momentum. For the sun's light on earth the momentum is rather small (although it is very large close to or inside the sun). Nevertheless, it is easily measureable as, for example, in the apparatus below. Light strikes a mirror and rebounds.

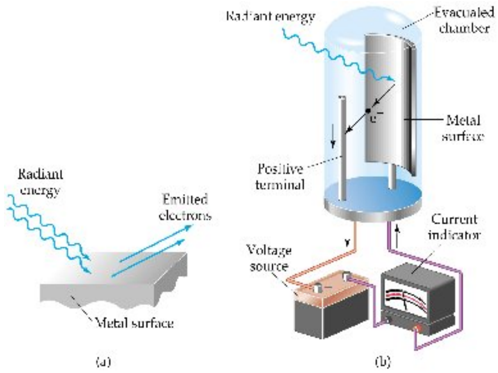


Electromagnetic waves linear momentum aur energy transport karti hain. Agar kisi wave ka energy per unit volume U ho to uska momentum p = U/c hota hai. Jis wave ka amplitude zyada hoga, usme energy aur momentum bhi zyada hogi. Earth par sunlight ka momentum chhota hota hai (lekin sun ke paas yeh bohot zyada hota hai). Phir bhi, yeh measurable hai. Diagram mein light mirror pe strike karti hai aur rebound hoti hai.

Thus the momentum of the light changes and this creates a force that rotates the mirror. The force is quite small - just 5×10^{-6} Newtons per unit area (in metre^2) of the mirror.

Light ki waves hone ka saboot itna strong tha ke jab photoelectric effect observe hua to sabko bohot surprise hua. Neeche diagram mein dikhaya gaya hai ke jab light ek metal surface par girti hai to wo electrons ko nikaal deti hai jo anode ki taraf travel karte hain. Current sirf tab tak flow karta hai jab tak light shine kar rahi ho. Threshold frequency se zyada frequency par, ejected electrons ki tadaad light ki intensity par depend karti hai. Isay photoelectric effect kaha jata hai.

4. So strong was the evidence of light as waves that observation of the photoelectric effect came as a big shock to everybody. In the diagram below, light hits a metal surface and



Photoelectric effect har frequency v par nahi hota; agar v ek certain value se kam ho to effect hota hi nahi. Lekin classical theory ke mutabiq (Maxwell ke electromagnetic wave theory ke mutabiq) electron kisi bhi frequency par eject ho jana chahiye. Agar wave electron ko violently shake kare to wo zarur nikalta.

and knocks out electrons that travel to the anode. A current flows only as long as the light is shining. Above the threshold frequency, the number of electrons ejected depends on the intensity of the light. This was called the photoelectric effect. The following was observed:

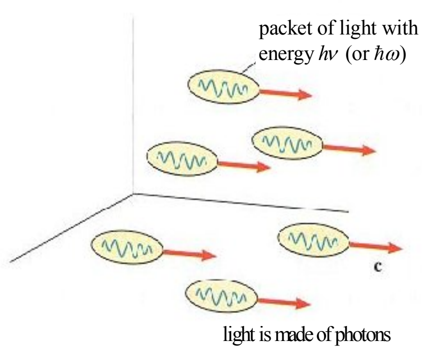
a) The photoelectric effect does not occur for all frequencies ν; it does not occur at all when ν is below a certain value. *But classically (meaning according to the Maxwell nature of light as an electromagnetic wave) electrons should be ejected at any ν. If an electron is shaken violently enough by the wave, it should surely be ejected!*

Dekha gaya ke pehle photoelectrons turant emit hote hain.Lekin classical theory ke mutabiq, pehle electron ko kuch waqt lagna chahiye jab wave surface se takra kar atoms ko ionize kare.

b) It is observed that the first photoelectrons are emitted instantaneously. *But classically the first photoelectrons should be emitted some time after the light first strikes the surface and excites its atoms enough to cause ionization of their electrons.*

Einstein ne kaha ke light continuous wave nahi balkay chhoti packets mein hoti hai, jinhein quanta (plural) ya quantum (singular) kehte hain. Har quantum ki energy hoti hai

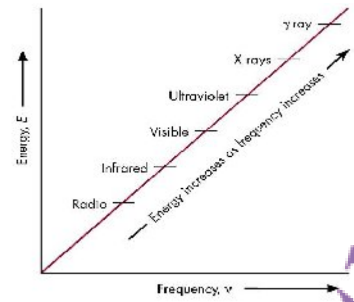
5. Explanation of the photoelectric puzzle came from Einstein, for which he got the Nobel Prize in 1905. Einstein proposed that the light striking the surface was actually made of little packets (called quanta in plural, quantum in singular). Each quantum has an energy energy $\boxed{\varepsilon = h\nu}$ (or $\boxed{\varepsilon = \hbar\omega}$ where $\boxed{\hbar = h/2\pi}$ and $\omega = 2\pi\nu$) where h is the famous Planck's constant, 6.626×10^{-34} Joule-seconds. An electron is kicked out of the metal only when a quantum has energy (and frequency) big enough to do the job. It doesn't matter how many quanta of light - called photons - are fired at the metal. No photoelectrons will be released unless ν is large enough. Furthermore the photoelectrons are released immediately when the photon hits an electron.



Electron sirf tabhi metal se nikalta hai jab quantum ki energy (aur frequency) kaafi hoti hai usay nikalne ke liye. Kitne bhi photons bhejein, agar energy kam hai to photoelectrons release nahi honge. Jab photon electron se takrata hai, photoelectron turant release hota hai.

Microwaves, radio aur TV waves, X-rays, gamma rays waghera yeh sab photons hote hain lekin unki frequencies bohat mukhtalif hoti hain. Kyun ke Planck's constant h bohat chhoti value ka hota hai, is liye har photon ki energy bhi bohat chhoti hoti hai.

6. Microwaves, radio and TV waves, X-rays, γ-rays, etc. are all photons but of very different frequencies. Because Planck's constant h is an extremely small number, the energy of each photon is very small.



7. How many photons do we see? Here is a table that gives us some interesting numbers:

a) Sunny day (outdoors): 10^{15} photons per second enter eye (2 mm pupil).

b) Moonlit night (outdoors): 5×10^{10} photons/sec (6 mm pupil).

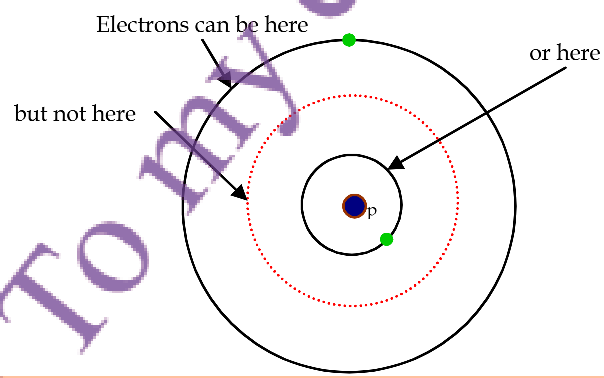
c) Moonless night (clear, starry sky): 10^{8} photons/sec (6 mm pupil).

d) Light from dimmest naked eye star (mag 6.5): 1000 photons/sec entering eye.

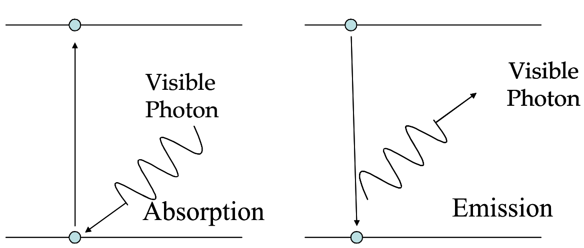
8. Where do photons come from? For this it is necessary to first understand that electrons inside an atom can only be in certain definite energy states. When an electron drops from a state with higher energy to one with lower energy, a photon is released whose energy is exactly equal to the difference of energies. Similarly a photon is absorbed when a photon of just the right energy hits an electron in the lower state and knocks it into a higher state.

Photons kahan se aate hain?

Yeh samajhne ke liye pehle yeh jaana zaroori hai ke electrons sirf kuch specific energy states mein hotay hain. Jab electron high energy state se low energy state mein girta hai, to ek photon release hota hai. Photon ki energy in dono levels ke farq ke barabar hoti hai.
Isi tarah, jab photon kisi electron ko hit karta hai aur usay excite karke upar le jata hai, to photon absorb hota hai.



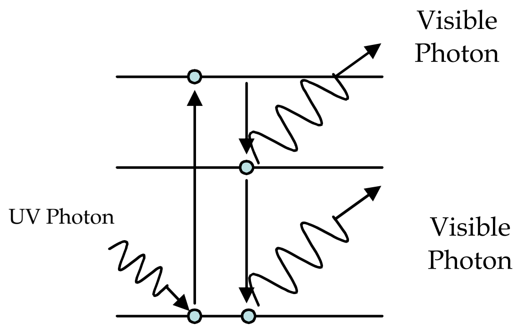
The upper and lower levels can be represented differently with the vertical direction representing energy. The emission and absorption of photons is shown below.

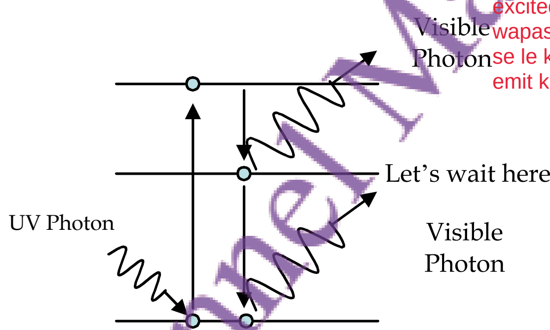


Fluorescence aur phosphorescence do phenomena hain jo kuch materials mein dekhe gaye hain. Jab ye materials kisi particular rang ki light ke source ke samne rakhe jate hain, to wo light emit karte rehte hain, source band hone ke baad bhi. Is wajah se wo andhere mein bhi chamak sakte hain.

9. Fluorescence and phosphorence are two phenomena observed in some materials. When they are exposed to a source of light of a particular colour, they continue to emit light of a different colour even after the source has been turned off. So these materials can be seen to glow even in the dark. In phosphorescence, a high-frequency photon raises an electron to an excited state. The electron jumps to an intermediate states, and then drops after a little while to the ground state. This is fluorescence. Phosphorescence is similar to fluorescence, except that phosphorescence materials continue to give off a secondary glow long (seconds to hours) after the initial illumination that excited the atoms.

Fluorescence: Ek high-frequency photon electron ko excited state mein le jata hai. Phir electron thodi der baad intermediate state se ground state mein wapas aata hai aur light emit karta hai.





Phosphorescence: Ismein bhi electron excited hota hai lekin ground state mein wapas aane mein der lagti hai (seconds se le kar hours tak). Tab tak material light emit karta rehta hai.

10. One of the most important inventions of the 20th century is the laser which is short for:

$$\boxed{\text{LASER} \equiv \textbf{L}\text{ight } \textbf{A}\text{mplification by the } \textbf{S}\text{timulated } \textbf{E}\text{mission of } \textbf{R}\text{adiation}}$$

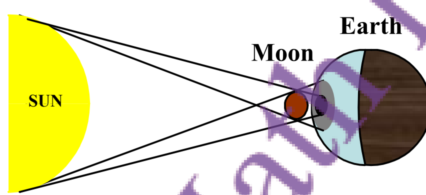
Lasers are important because they emit a very large number of photons all with one single frequency. How is this done? By some means - called optical pumping - atoms are excited to a high energy level. When one atom starts decaying to the lower state, it encourages all the others to decay as well. This is called spontaneous emission of radiation.

Kesy kam krta h ?

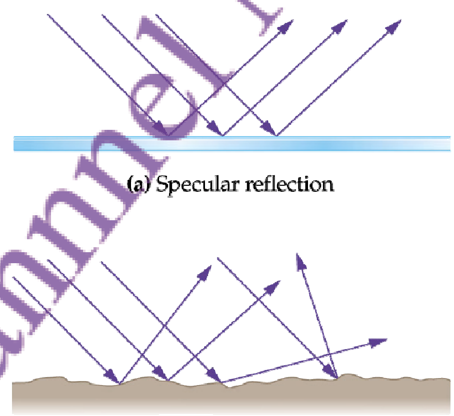
Optical pumping ke zariye atoms ko high energy level tak excite kiya jata hai. Jab ek atom lower level par decay karna start karta hai to baqi atoms bhi encouraged ho kar decay karte hain. Isay kehte hain Spontaneous emission of radiation.

Summary of Lecture 35 – GEOMETRICAL OPTICS

1. In the previous lecture we learned that light is waves, and that waves spread out from every point. But in many circumstances we can ignore the spreading (diffraction and interference), and light can then be assumed to travel along straight lines as rays. This is hown by the existence of sharp shadows, as for case of the eclipse illustrated here.



2. When light falls on a flat surface, the angle of incidence equals the angle of reflection. You can verify this by using a torch and a mirror, or just by sticking pins on a piece of paper in front of a mirror. But what if the surface is not perfectly flat? In that case, as shown in fig. (b), the angle of incidence and reflection are equal at every point, but the normal direction differs from point to point. This is called "diffuse reflection". Polishing a surface reduces the diffusiveness.

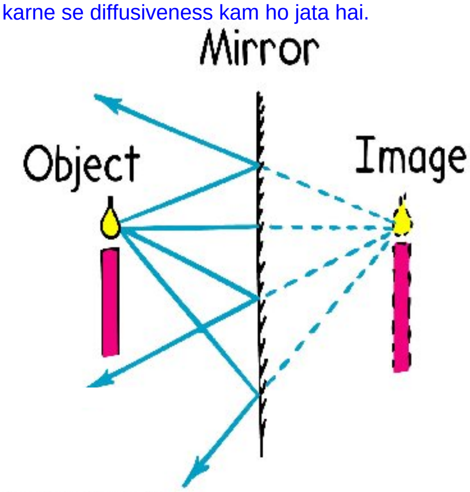


(a) Specular reflection

(b) Diffuse reflection

Jab roshni aik flat surface par girti hai, to angle of incidence barabar hota hai angle of reflection ke. Aap isay torch aur mirror ka use kar ke, ya aik paper par pins laga kar verify kar saktay hain. Agar surface flat na ho, to jaise fig (b) mein dikhaya gaya hai, har point par angle of incidence aur reflection barabar hotay hain, lekin normal direction point to point farq karta hai. Isy diffuse reflection kehtay hain. Surface ko polish karne se diffusiveness kam ho jata hai.

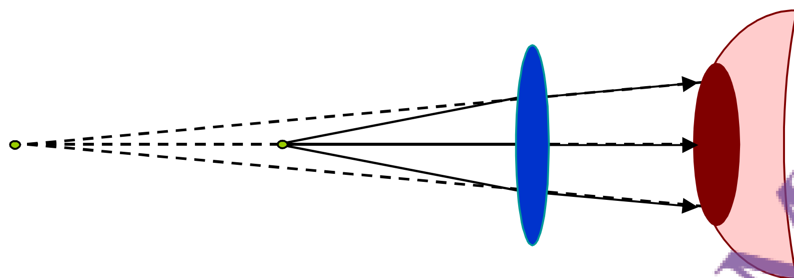
3. If you look at an object in the mirror, you will see its image. It is not the real thing, and that is why it is called a "virtual" image. You can see how the virtual image of a candle is formed in this diagram. At each point on the surface, there is an incident and reflected ray. If we extend each reflected ray backwards, it appears as if they are all coming from the same point. This point is the image of the tip of the flame. If we take other points on the candle, we will get their images in just the same way. This way we will have the image of the whole candle. The candle and its image are at equal distance from the mirror.



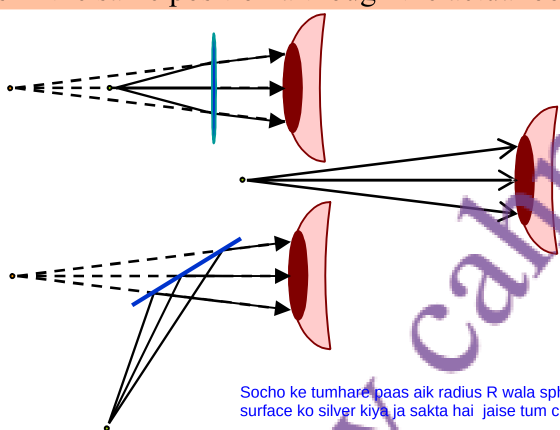
Agar aap mirror mein kisi cheez ko dekhein, to aap uska image dekhte hain. Ye asal cheez nahi hoti, isi liye isay virtual image kehtay hain. Candle ke example mein aap dekh saktay hain ke virtual image kaise banti hai. Har point par incident aur reflected ray hoti hai. Agar reflected rays ko peeche extend kiya jaye, to lagta hai wo aik hi point se aa rahi hain yehi point image ka hota hai. Candle ke har point se reflected rays isi tarah image banati hain. Image aur object mirror se barabar distance par hotay hain.

4. Here is another example of image formation. A source of light is placed in front of a bi-convex lens which bends the light as shown. The eye receives rays of light which seem to originate from a positon that is further away than the actual source.

Yeh image formation ka aik aur example hai. Aik light ka source aik bi-convex lens ke samne rakha gaya hai jo roshni ko bend karta hai jaise ke diagram mein dikhaya gaya hai. Aankh aisi rays receive karti hai jo aise lagta hai jaise wo kisi door ke point se aayi hoon, jabke asal source us se kareeb hota hai.
Ab point ko aur clear banane ke liye, neeche diye gaye tino situations mein virtual image ek hi jagah par hai, halanke asal object teen alag alag jagahon par hai.

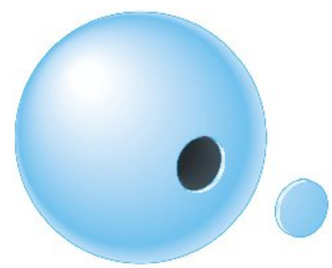


Now just to make the point even more forcefully, in all three situations below, the virtual image is in the same position although the actual object is in 3 different places.

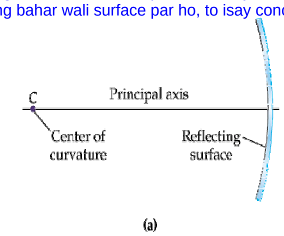


Socho ke tumhare paas aik radius R wala sphere hai, aur tum us ka koi bhi hissa kaat saktay ho. Bahar ya andar wali surface ko silver kiya ja sakta hai jaise tum chaho. Tum is tarah se spherical mirrors bana saktay ho.

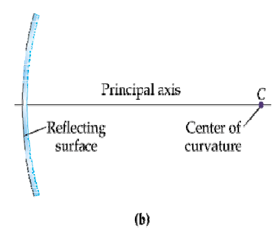
5. Imagine that you have a sphere of radius R and that you can cut out any piece you want. The outside or inside surface can be silvered, as you want. You can make spherical mirrors in this way. These can be of two kinds. In the first case, the silvering can be on the inside surface of the sphere, in which case this is is called a convex spherical mirror. The normal directed from the shiny surface to the centre of the sphere (from which it was cut out from) is called the principal axis, and the radius of curvature is *R*. The other situation is that in which the outside surface is shiny. Again, the principal axis the same, but now the radius of curvature (by definition) is -*R*. What does a negative curvature mean? It means precisely what has been illustrated - a convex surface has a positive and a concave surface has a negative curvature.



Agar silvering andar wali surface par ho, to isay convex spherical mirror kehtay hain.
Agar silvering bahar wali surface par ho, to isay concave spherical mirror kehtay hain.



(a)

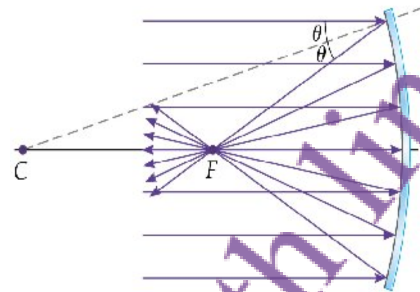


(b)

Concave surface ka radius of curvature -R hota hai. Negative curvature ka matlab hai concave surface, jabke convex surface ka curvature positive hota hai.

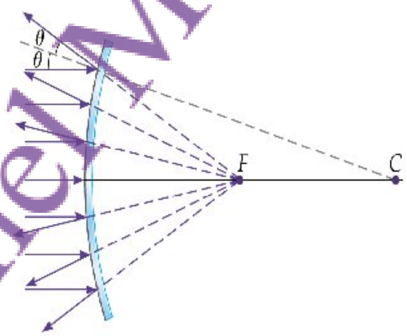
jab parallel rays aik convex mirror se takraati hain, to wo sab aik hi point par converge karti hain, jise focus kehte hain. Figure dekho: har ray ka incident aur reflected angle barabar hota hai, is liye sari rays focus F par ja kar milti hain jo distance R/2 par hota hai. Hum kehte hain ke convex mirror ka focal length f = R/2 hota hai.

6. Now we come to the important point: a beam of parallel rays will reflect off a convex mirror and all get together (or converge) at one single point, called the focus. Now look at the figure. Since the incident and reflected angles are equal for every ray, you can see that the ray will have to pass through the focus at F, at distance $R/2$. We say that the focal length of such a mirror is $f = R/2$.



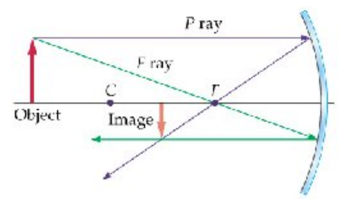
Jab parallel rays left se mirror tak aati hain to wo reflect hone ke baad spread out ho jaati hain (diverge karti hain). Agar hum in rays ko peeche ki taraf trace karein to wo aise lagti hain jaise sab aik hi point se aayi hoon. Is point ko virtual focus kehte hain kyunki asal mein wahan koi light ka source nahi hota. Diagram se dekho ke virtual focus F mirror ke peeche R/2 distance par hai. Is liye concave mirror ka focal length f = –R/2 hota hai.

7. The concave mirror is different. A parallel beam incident from the left is reflected and the rays now spread out (diverge). However, if we were to look at any outgoing ray and extend it backward, all the rays would appear to be coming from one single point. This is the *virtual focus* because there is, in fact, no real source of light there. From the diagram you can see the virtual focus F is at distance $R/2$ *behind* the mirror. We say that the focal length of such a mirror is $f = -R/2$.

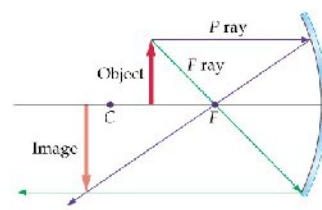


Ab dekhte hain ke concave mirror kaise image banata hai. Aik object ke top se nikli do rays lo unhe P ray aur F ray keh lo aur dekho unka reflection kaise hota hai. Jahan wo rays milti hain, wahi image ka point hota hai.

8. Now let's see how images are formed by concave mirrors. Take two rays that are emitted from the top of an object placed in front of the mirror. Call them the P and F ray, and see what happens to them after reflection. Where the two cross, that is the image point. As you can see the image is inverted and smaller than the actual object if the object is far away (i.e. lies to the left of C) and is larger if close to the mirror. The magnification of size makes this useful as a shaving mirror, among other things.



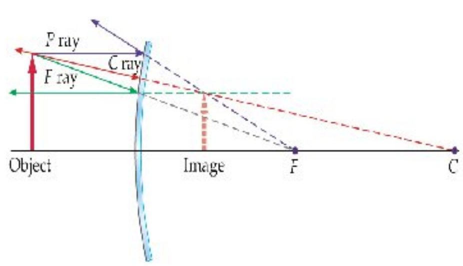
(a)



(b)

Diagram mein dikhaya gaya hai ke image inverted hoti hai aur object se choti hoti hai agar object door ho (C se left mein). Lekin agar object kareeb ho, to image badi hoti hai. Is wajah se yeh mirror shaving mirror ke liye useful hota hai.

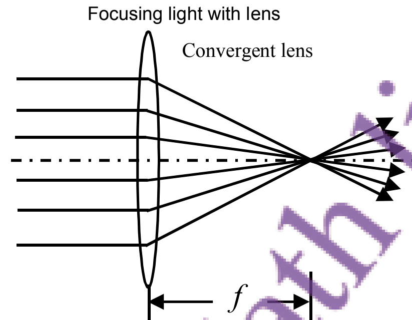
9. Now repeat for a concave mirror. Here the rays will never actually meet so we can have only a virtual object. Take 3 rays - call them P,F, and C - and see how they are reflected from the mirror's surface. Where they meet is the position of the image. The image is always smaller than the actual size of the object. This is obviously useful for driving a car because you can see a wide area.



Yahan rays kabhi bhi waqai mein milti nahi hain, is liye sirf virtual image banti hai. 3 rays lo P, F, aur C aur dekho unka reflection. Jahan ye extend ho kar milti hain, wahi image ka point hai. Image choti hoti hai aur asal object se kam size ki hoti hai. Is tarah ka mirror driving mein useful hota hai jahan wide area dikhana hota hai.

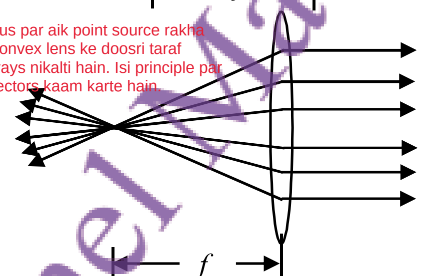
Aik lens sheeshe ka bana hota hai jo ek khas tarah se curved hota hai. Kyun ke sheeshe ka refractive index hawa se zyada hota hai, har ray normal ki taraf bend hoti hai. Yahan aik double convex lens dikhaya gaya hai jo parallel rays ko focus karta hai. Perfect lens mein saari rays aik hi point par milti hain jise focus kehte hain. Focus tak ka faasla focal length kehlata hai.

10. A lens is a piece of glass curved in a definite way. Because the refractive index of glass is bigger than one, every ray bends towards the normal. Here you see a double convex lens that focuses a beam of parallel rays. For a perfect lens, all the rays will converge to one single point that is (again) called the focus. The distance f is called the focal length.



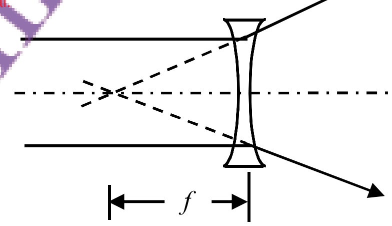
Of course, light can equally travel the other way, so if a point source is placed at the focus of a convex lens then a parallel beam of light will emerge from the other side. This is how some film projectors produce a parallel beam.

Agar focus par aik point source rakha jaye to convex lens ke doosri taraf parallel rays nikalti hain. Isi principle par film projectors kaam karte hain.



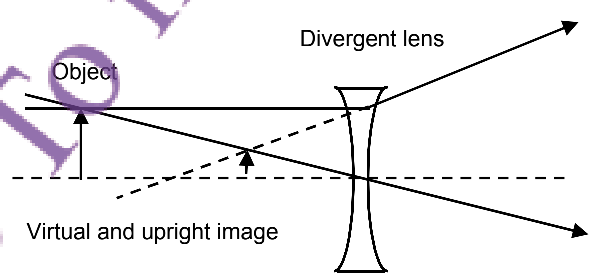
A concave lens does not cause a parallel beam to converge. On the contrary, it makes it diverge, as shown. Note, however, that if the rays are continued backwards, then they appear to come from one single point, which is here the virtual focus. The distance f is the focal length.

Concave lens parallel rays ko converge nahi karta, balkay unhein diverge karta hai. Lekin agar rays ko peeche extend karein to yeh aik virtual point se aati hui lagti hain isay virtual focus kehte hain, aur faasla f hi hota hai.



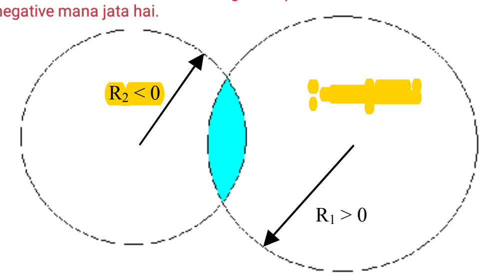
12. Here is how a concave (or divergent) lens forms an image. An observer on the side opposite to the object will see the image upright and smaller in size than the object.

Concave (ya divergent) lens image banata hai. Agar observer object ke opposite side par ho to use image seedhi aur chhoti nazar aati hai.

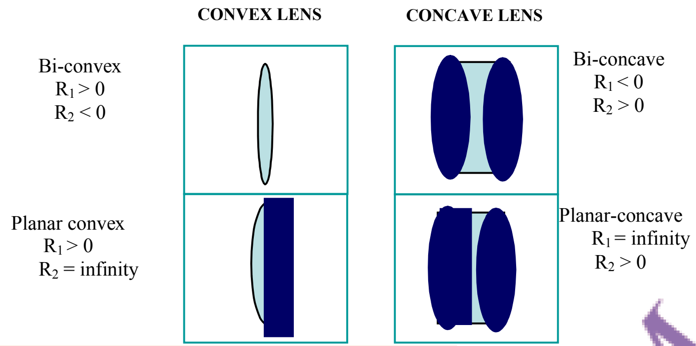


Aik lens ko aise socha ja sakta hai ke do spheres ke tukdon se banaya gaya ho jin ke radii R_1 aur R_2 hain. Dono surfaces roshni ko alag tareeqon se bend karti hain. Convention ke mutabiq, R_2 ko negative mana jata hai.

13. A lens can be imagined as cut out of two spheres of glass as shown, with the spheres having radii R_1 and R_2. Note that they tend to bend light in opposite ways. By convention, R_2 is negative. It is possible to show that the focal length of the lens is $\boxed{\frac{1}{f}=(n-1)\left(\frac{1}{R_1}-\frac{1}{R_2}\right)}$.



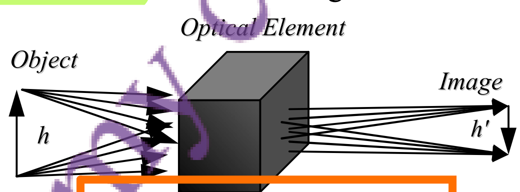
14. The following figure summarizes the shape of some common types of lenses.



Note that a flat surface has infinite radius of curvature. The focal legth of each can be calculated using the previous formula.

15. The "strength" of a lens is measured in *diopters*. If the focal length of a lens is expressed in metres, the diopter of the lens is defined as $D = 1/f$. If the refractive index of the glass in a lens is n, then the diopters due to the first interface D_1 and the second interface D_2 are, $\boxed{D_1 = (n-1)/R_1 \text{ and } D_2 = (1-n)/R_2}$. The total diopter of the lens is $D = D_1 + D_2$.

16. For any optical system - meaning a collection of lenses and mirrors - we can define a magnification factor as a ratio of sizes -- see the diagram below.

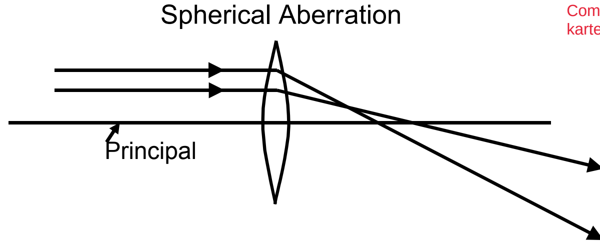


$\boxed{\textit{Lateral Magnification } M = h'/h}$

17. The perfect lens will focus a parallel beam of rays all to exactly the same focus. But no lens is perfect, and every lens suffers from aberration although this can be made quite small by following one lens with another. Below you see an example of "spherical aberration". Rays crossing different parts of the lens do not reach exactly the same focus. This distorts the image. Computers can design lens surfaces to minimize this aberration.

Computers lens design karte waqt aberration minimize karne ki koshish karte hain.

Spherical Aberration



Perfect lens saari rays ko ek hi focus point par le aata hai. Lekin har lens mein kuch aberration hoti hai. Niche diya gaya example spherical aberration ka hai ismein lens ke alag alag parts se guzri hui rays exact same point par focus nahi kartin, jiski wajah se image distort ho jati hai.

Purane zamane mein log samajhtay thay ke heat aik colourless, weightless fluid hai jo kisi cheez ka hissa ban jata hai. Isay phlogiston kehte thay. Lekin yeh ghalat sabit hua, kyunki heat ko create aur destroy kiya ja sakta hai, jabke liquids ko nahi.

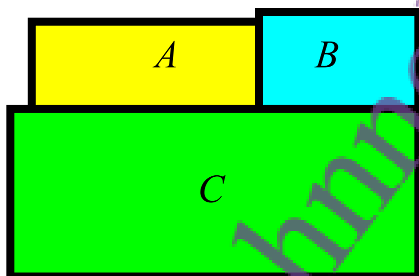
Thermal physics ak branch hai physics ki jo heat, temperature, energy, aur un ke aapas ke taluq ko study karti hai.or Thermal physics batati hai ke heat aur energy systems mein kaise transfer hoti hai aur unka matter par kya asar hota hai.

Summary of Lecture 36 – THERMAL PHYSICS I

1. The ancient view was that heat is a colourless, weightless, fluid which occupies no volume has no smell, etc. This imagined substance - called phlogiston - was supposedly stored in objects and transferred between objects. It took a long time to reject this notion. Why is it wrong? Because (as we will see) heat can be created and destroyed, whereas liquids keep their volume and cannot be created or destroyed.

khayal

2. We are all familiar with an intuitive notion of temperature. We know that hotter things have higher temperature. But let us try to define temperature more rigorously. In the diagram below, the three bodies A,B,C are in contact with each other. After sufficient time passes, one thing will be common to all three - a quantity that we call temperature.



Teen bodies A, B, aur C ko agar ek dosray ke contact mein rakha jaye, to kuch waqt baad sab ki temperature equal ho jati hai. Is property ko temperature kehte hain.

3. Now let us understand heat. Heat is energy, but it is a very special kind of energy: it is that energy which flows from a system at high temprature to a system at low temperature.

Heat energy hai, lekin yeh ek special energy hai jo high temperature se low temperature walay system mein flow karti hai.

Stated in a slightly different way: heat is the flow of internal energy due to a temperature difference. (Note that we do not have to know about atoms, molecules, and the internal composition of a body to be able to define heat - all that will come later).

4. The term "thermal equilibrium" is extremely important to our understanding of heat. If some objects (say, a glass of water placed in the open atmosphere) are put in thermal contact but there is no heat exchange, then we say that the objects are in thermal equilibrium. So, if the glass of water is hot or cold initially, after sufficient time passes it will be in thermal equilibrium and will neither receive nor lose heat to the atmosphere.

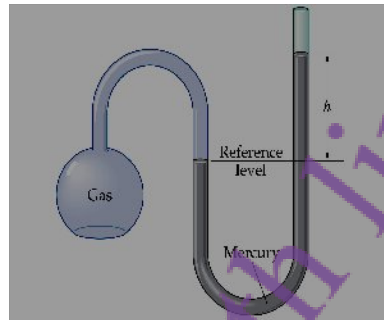
Agar do ya zyada cheezen contact mein hon, aur unke darmiyan heat ka exchange na ho raha ho, to woh thermal equilibrium mein hoti hain. Jaise paani ka glass jo environment ke sath kuch waqt baad balance mein aa jata hai.

Temperature ko thermometric properties se measure karte hain

5. We have an intuitive understanding of temperature, but how do we measure it? Answer: by looking at some physical property that changes when the temperature changes. So, for example, when the temperature rises most things expand, the electrical resistance changes, some things change colour, etc. These are called "thermometric properties".

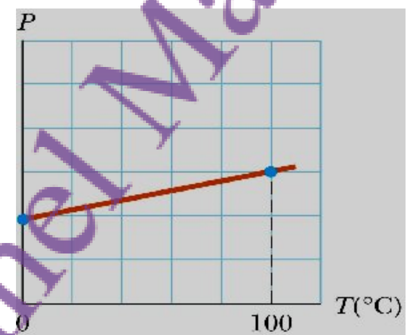
Temperature badhne se cheezein expand karti hain, Electrical resistance change hoti hai, Kuch cheezein colour change karti hain.

6. Let's take a practical example: the constant volume gas thermometer. Here the reference level is kept fixed by raising or lowering the tube on the right side (the tube below is made of rubber or some flexible material). So the gas volume is fixed. The gas pressure is $\rho \times g \times h$, and so the pressure is known from measuring h.



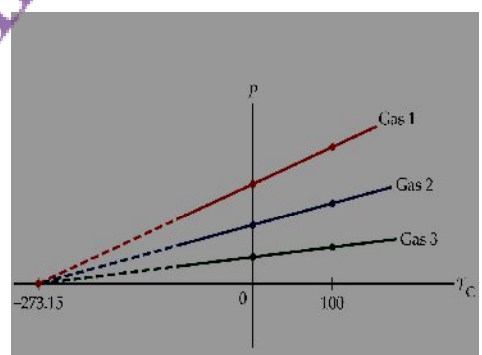
Yahan volume fix hota hai aur pressure measure karke temperature nikalte hain.

To find the temperature of a substance, the gas flask is placed in thermal contact with the substance. When the temperature is high, the pressure is large. From the graph of pressure versus temperature, you can easily read off the temperature. Note that we are using two fixed points, which we call 0^0C and 100^0C. This is called the Centigrade scale, and the two fixed points correspond to the freezing and boiling of water.



Ek limit hoti hai jahan temperature aur zyada kam nahi ho sakta.

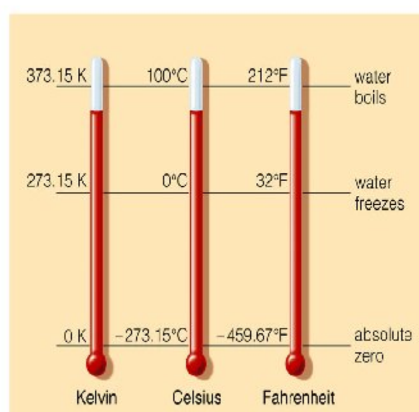
7. There is a temperature below which it is not possible to go. In other words, if you cool and cool there comes a point after which you cannot cool any more! How do we know this? Take different gases and plot how their pressure changes as you cool them down. You can see from the graph that all the lines, when drawn backwards, meet exactly at one point. This point is the absolute zero of temperature and lies at about -273.15^0C. This is also called 0^0K, or zero degrees Kelvin.



Sab gases ki lines backward ja kar isi point par milti hain.

Agar aap 300 saal pehle paida hotay aur temperature ko accurately measure karne ka tareeqa dhoond laitay, to shayad aaj aapke naam ka scale hota!

8. Temperature Scales. If you had been born 300 years ago, and if you had discovered a reliable way to tell different levels of "hotness", then maybe today there would be a temperature scale named after you! The relation between Centigrade, Fahrenheit, and Kelvin scales is illustrated to your right. You can see that absolute zero corresponds to $-460^0 K$, $-273.15^0 C$, and $0^0 K$. The Kelvin scale is the most suited for scientific purposes, and you should be careful to use this in all heat related calculations.



Relation between kelvin, Celsius and Fahrenheit

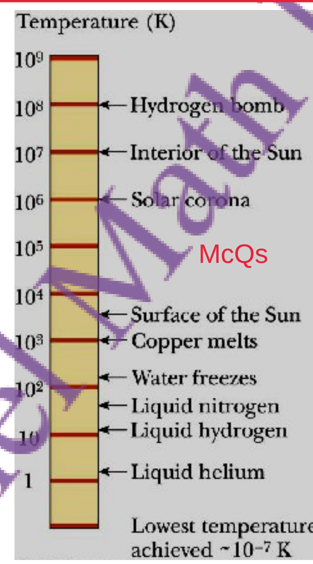
Kelvin scale scientific use ke liye sabse zyada suitable hai.

Conversion between degrees Celsius and degrees Fahrenheit: $T_F = (\frac{9}{5}F^\circ / C^\circ)T_C + 32^\circ F$

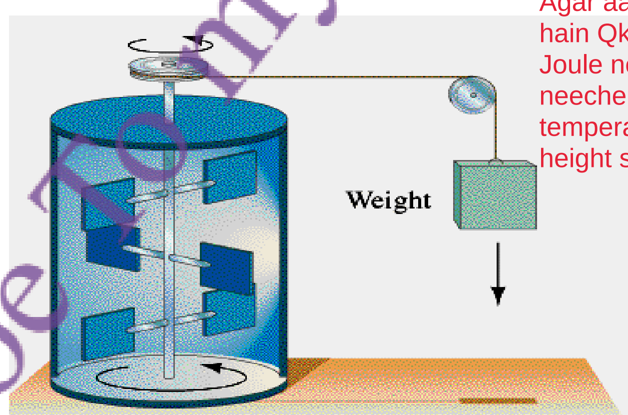
Conversion between degrees Fahrenheit and degrees Celsius: $T_C = (\frac{5}{9}C^\circ / F^\circ)(T_F - 32^\circ F)$

Conversion between Celsius and Kelvin temperatures: $T = T_C + 273.15$

9. How hot is hot, and how cold is cold? Whenever a scientist says something is large or small, it is always relative to something. The hottest thing ever was the universe when it just came into existence (more about this in the last lecture). After this comes the hydrogen bomb (10^8K). The surface of the sun is not so hot, only 5500K. Copper melts around 1000K, water turns to steam at 373K. If you cool further, then all the gases start to solidify. The lowest temperature that has ever been achieved is $10^{-7}K$, which is one tenth of one millionth of one degree! Why not still lower? We shall later why this is not possible.



10. When you rub your hands, they get hot. Mechanical work has been converted into heat. The first person to investigate this scientifically was Joule. In the experiment below, he allowed a weight to drop. This turned a paddle that stirred up the water and caused the temperature to rise. The water got hotter if the weight was released from a greater height.



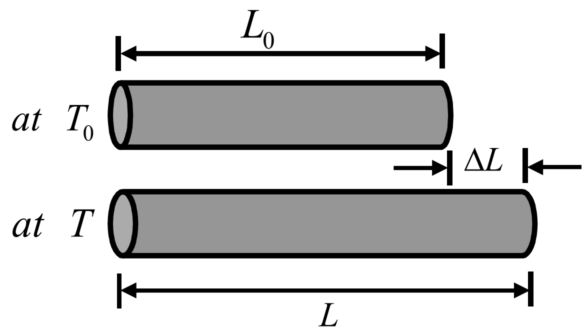
Agar aap haath ragartay ho, woh garam ho jaatay hain Qk mechanical work heat mein badal jaata hai. Joule ne yeh scientifically prove kiya Ek weight ko neeche gira kar paddle ghumaaya, jis sy pani ka temperature zyada ho gya or Agar weight zyada height se gira hota, to pani aur zyada garam hota.

Joule established the units for the mechanical equivalent of heat. The units we use today are: 1 calorie (1 cal) raises the temperature of 1 g of water by 1 °C

1 cal = 4.186 Joule, 1 kilocalorie (1 kcal) = 1000 cal

Remember also that joule is the unit of work: when a force of one newton acts through a distance of one metre, the work done is one joule.

11. Let's now consider one important effect of heat - most things expand when heated. Of course, our world is 3-dimensional but if there is a thin long rod then the most visible effect of heating it is that the rod increases in length. By how much?



Jab kisi rod ko garam kiya jata hai to woh lambi ho jati hai. Uska pehla length L_0 hota hai (at temperature T_0) Garam hone ke baad length L ho jata hai (at temperature T).

Look at the above diagram. Call the length of the rod at T_0 as L_0. When the rod is heated to T, then the length increases to $L = L_0\left[1+\alpha\left(T-T_0\right)\right]$. The difference of the lengths is, $L - L_0 = \alpha L_0\left(T - T_0\right)$, or $\Delta L = \alpha L_0 \Delta T$. Here α is just a dimensionless number that tells you how a particular material expands. It is called the coefficient of linear expansion. If α was zero, then the material would not expand at all. You can also write it as $\boxed{\alpha = \dfrac{\Delta L / L_0}{\Delta T}}$.

α hota hai coefficient of linear expansion, jo batata hai material kitna expand karta

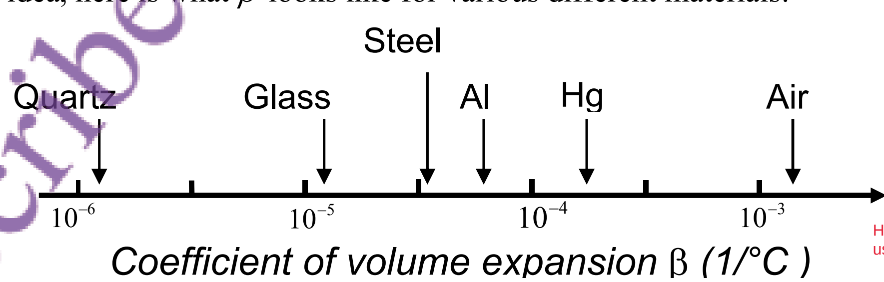
Similarly, define a coefficient of volume expansion – call it β – as $\boxed{\beta \equiv \dfrac{\Delta V / V_0}{\Delta T}}$.

Agar α = 0 ho to koi expansion nahi hoti.

12. There is a relation between α and β, the linear and volume coefficients. Let's look at the change in volume due to expansion:

$$\begin{aligned} V' &= (L+\Delta L)^3 = (L+\alpha L \Delta T)^3 \\ &= L^3 + 3\alpha L^3 \Delta T + 3\alpha^2 L^3 \Delta T^2 + \alpha^3 L^3 \Delta T^3 \\ &\approx L^3 + 3\alpha L^3 \Delta T = V + 3\alpha V \Delta T \end{aligned}$$

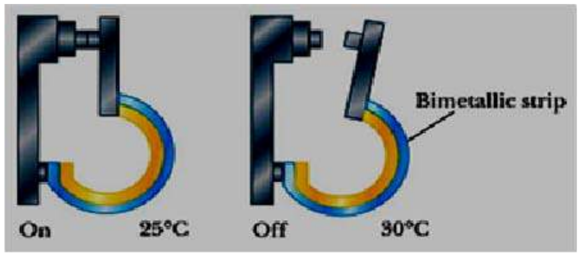
We are only looking for small changes, so the higher terms in ΔT can be safely dropped. Hence $\boxed{\Delta V = 3\alpha V \Delta T}$ From the definition, this we immediately see that $\boxed{\beta = 3\alpha}$. Just to get an idea, here is what β looks like for various different materials:



Har metal ka expansion rate alag hota hai is property ka use thermostats banane mein hota hai.

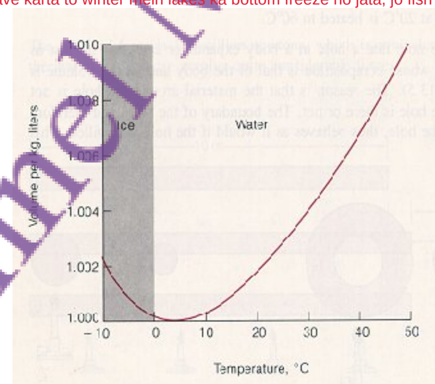
13. We can use the fact that different metals expand at different rates to make thermostats. For example, you need a thermostat to prevent an electric iron from getting too hot or a

refrigerator from getting too cold. In the diagram below, one metal is bonded to another. When they expand together, one expands less than the other and the shape is distorted. This breaks off a circuit, as shown here.



Normally heating se materials expand karte hain. Lekin paani 0°C se 4°C tak heat karne par contract karta hai (chhota hota hai). Isay anomalous behaviour kehte hain. Agar paani normal behave karta to winter mein lakes ka bottom freeze ho jata, jo fish ke liye dangerous hota.

14. Water behaves strangely - for certain temperatures, it contracts when heated (instead of expanding)! Look at the graph: between 0^0C and 4^0C, it exhibits strange or *anomalous* behaviour. The reason is complicated and has to do with the molecular structure of water. If water behaved normally, it would be very bad for fish in the winter because they rely upon the bottom of the lake or sea to remain liquid even though there the surface is frozen.



15. We define the "heat capacity" of a body as $\boxed{C = \frac{Q}{\Delta T}}$, where ΔT is the increase in temperature when an amount of heat Q is added to the body. Heat capacity is always positive; Q and ΔT have the same sign. The larger the heat capacity, the smaller is the change in the body's temperature when a fixed amount of heat is added. In general, $\boxed{Q = mc\Delta T}$ where Q = heat added , m = mass , c = specific heat , and ΔT = change in temperature. Water has a very large specific heat c, $c = 1.0\ cal\ /(^oC\ g)$; this means it takes one calorie to raise the temperature of 1 gm of water by 1 degree Celsius. In joules per kilogram this is the same as 4186. In the table are the specific heats of various common materials. You can see that metals have small c, which means that it is relatively easy to raise or lower their temperatures. The opposite is true of water. Note also that steam and ice have smaller c's than water. This shows that knowing the chemical composition is not enough.

Specific Heats at Atmospheric Pressure

Substance	Specific heat, c [J / (kg·K)]
Water	4186
Ice	2090
Steam	2010
Beryllium	1820
Air	1004
Aluminum	900
Glass	837
Silicon	703
Iron (steel)	448
Copper	387
Silver	234
Gold	129
Lead	128

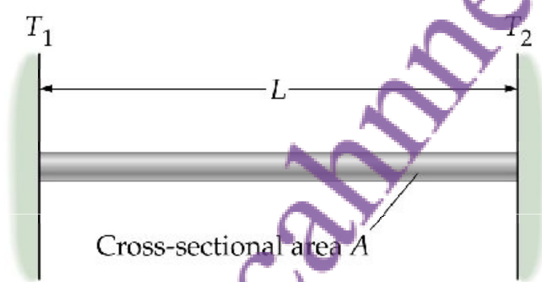
Metals ka c chhota hota hai to unka temperature jaldi change hota hai.

16. A 0.5-kg block of metal with an initial temperature of $30.0^{\circ}C$ is dropped into a container holding 1.12 kg of water at $20.0^{\circ}C$. If the final temperature of the block-water system is $20.4^{\circ}C$, what is the specific heat of the metal?

SOLUTION: Write an expression for the heat flow out of the block $Q_{block} = m_b c_b (T_b - T)$. Do the same for water, $Q_{water} = m_w c_w (T - T_w)$. Now use the fact that all the energy that is lost by the block is gained by the water: $Q_{block} = Q_{water} \Rightarrow T m_b c_b (T_b - T) - m_w c_w (T - T_w) = 0$

From this,
$$\boxed{c_b = \frac{m_w c_w (T - T_w)}{m_b (T_b - T)} = \frac{(1.12kg)[4186J/(kg\cdot K)](20.4^{\circ}C - 20.0^{\circ}C)}{(0.500kg)(30.0^{\circ}C - 20.4^{\circ}C)} = 391J/(kg\cdot K)}$$

17. When lifting a "daigchee" from a stove, you would be wise to use a cloth. Why? Because metals transfer, or conduct, heat easily whereas cloth does not. Scientifically we define conductivity using experiments and apparatus similar to the following:



Jab daigchee uthate ho, kapra use karo. Kyun?
Metals heat easily conduct karte hain, kapra nahi karta.

Heat flows from the hotter to the colder plate. Let us use the following symbols:

k = thermal conductivity	Q = heat transferred
A = cross sectional area	t = duration of heat transfer
L = length	ΔT = temperature difference

Then the heat transferred in time t is, $\boxed{Q = kA\left(\frac{\Delta T}{L}\right)t}$. This formula allows us to measure k if all the other quantities in it are measured.

Agar sab values maloom hoon to k find kiya ja sakta hai

18. Conduction is one possible way by which heat is transferred from one portion of a system to another. It does not involve physical transport of particles. However, there is another way by which heat can be transferred - convection. In convection, heat is carried by a moving fluid. So when you heat a pot of water, molecules at the bottom move up, and the ones at the top come down - the water has currents inside it that transfer heat. Another mechanism for transferring heat is through radiation. We have already talked about this while discussing blackbody radiation and the Stefan-Boltzman Law.

Conduction: solid ke particles touch mein hotay hain, direct heat pass hoti hai.
Convection: liquid/gas mein heat moving fluid se transfer hoti hai (e.g. garam paani upar jaata hai).
Radiation: bina kisi medium ke heat transfer hoti hai (e.g. sunlight).

Jab hum kisi cheez ko study kar rahe hote hain jaise ke ice + water ka mixture ya garam gas, to usay system kehte hain.

Summary of Lecture 37 – THERMAL PHYSICS II

1. Let us agree to call whatever we are studying the "system" (a mixture of ice and water, a hot gas, etc). The state of this system is specified by giving its pressure, volume, temperature, etc. These are called "thermodynamic variables". The relation between these variables is called the "equation of state". This equation relates P,V,T. So, for example:

$P = f(V,T)$ (knowing V and T gives you P) (volume aur temperature se pressure nikaal sakte ho)

$V = g(P,T)$ (knowing P and T gives you V) (Pressure aur temperature se volume nikaal sakte ho)

$T = h(P,V)$ (knowing P and V gives you T) (Pressure aur volume se temperature nikaal sakte ho)

For an ideal gas, the equation of state is $\boxed{PV = Nk_BT}$ Here N is the number of molecules in the gas, T is the temperature in degrees Kelvin, and k_B is called the Boltzman cosntant. This EOS is easy to derive, although I shall not do it here. In principle it is possible to mathematically derive the EOS from the underlying properties of the atoms and molecules in the system. In practice, however, this is a very difficult task (except for an ideal gas). However the EOS can be discovered experimentally.

N = molecules ki number
T = temperature (Kelvin mein)
kB = Boltzmann constant

2. Thermodynamics is the study of heat and how it flows. The First Law of Thermodynamics is actually just an acknowledgment that heat is a form of energy (and, of course, energy is always conserved!). Mathematically, the First Law states that:

$$\boxed{\Delta E = \Delta q + \Delta w}$$

where:
- E = internal energy of the system
- q = heat transferred to the system from the surroundings
- w = work done on the system by the surroundings.

Agar aap kisi system ko heat do (Q positive) aur uspe kaam bhi karo (W positive), to system ki energy barhti hai.

In words, what the above formula says is this: if you do an amount of work Δw and also transfer an amount of heat Δq, then the sum of these two quantities will be the additional amount of energy ΔE that is stored in the system. There could be nothing simpler! Note: Δq and Δw are positive if heat is added to or work is done on a system, and negative if heat is removed from the system, or if the system does work on the environment.

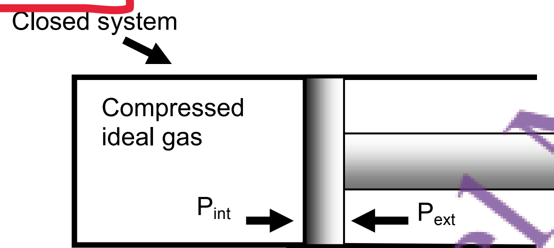
Agar system se heat nikal lo ya system khud kaam kare (W negative), to energy kam hoti hai.

3. Work and heat are called *path variables* - their values depend on the steps leading from one state to another. To give an obvious example: suppose you drag a heavy box on the floor in a straight line from point A to point B, and then in a very roundabout way from A to B again. Obviously the work done by you will be very different. On the other hand, the internal energy does not depend upon the path. Example: a gas has internal energy proportional to its temperature. It makes no difference whether the gas had been slowly heated or rapidly; it will have the same internal energy. Since the internal energy does not depend on the path we can write $\boxed{\int_{State1}^{State2} dE = E_2 - E_1 = \Delta E.}$

Agar aap ek box ko sidha uthake point A se B tak le jao ya ghoom ke le jao, dono mein work alag hoga.
Lekin internal energy sirf temperature pe depend karti hai, path pe nahi.

Jab gas piston ko dhakelti hai aur piston move karta hai, to gas work karti hai.

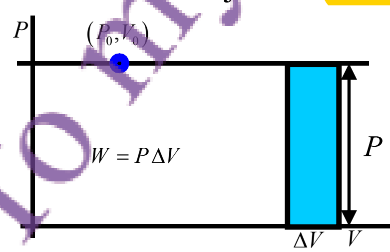
4. Let us calculate the work done by an expanding gas. In the diagram below, suppose the piston moves out by distance dx, then the work done is $dw = Fdx$ where F is the force exerted on the gas. But $F = P \times A = -P_{ext} \times A$. (The minus sign is important to understand; remember that the force exerted by the gas on the piston is the negative of the force that the piston exerts on the gas). So $dw = Fdx = -P_{ext} \times A \times dx = -P_{ext}dV$. The work done by the expanding gas is $\boxed{w = -\int_{V_1}^{V_2} P_{ext}dV}$ where V_1 and V_2 are the initial and final volumes.



Note that w depends on the path. Let's consider 3 different paths.

a)Suppose that the gas expands into the vacuum, $\boxed{P_{ext} = 0}$. Then $w = 0$. So, if in the figure below, the valve is opened, then gas will flow into the vacuum without doing work.

b)Suppose the pressure outside has some constant value. Then, $\boxed{w = P_{ext} \times (V_2 - V_1)}$ or, $\boxed{w = P_{ext}\Delta V}$. You can see that this is just the area of the rectangle below.

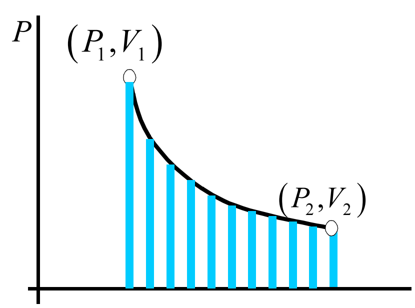


Agar hum gas ko aise expand karne dein ke internal aur external pressure almost barabar ho jaayein, to isay reversible expansion kehte hain.
Is halat mein system pe jo kaam hota hai (work done on the system), wo hota ha

c)We can also let the gas expand so that the internal and external pressures are almost the same (this is called reversible expansion). Then the work done *on* the system is: $dw = -P_{ext}dV$. Using $PV = Nk_BT$ gives,

$$w = -\int_{V_1}^{V_2} \frac{Nk_BT}{V} dV = -Nk_BT(\ln V_2 - \ln V_1) = -Nk_BT \ln \frac{V_2}{V_1}$$

Of course, the work done *by* the gas as it expands is positive since V_2 is larger than V_1. Remember that the log function is positive if its argument is bigger than 1.



Gas jab expand karti hai to V2 > V1 hota hai, is liye ln(V2 / V1) positive hota hai.
Iska matlab hai gas positive work karti hai (apne aap kaam karti hai jab expand karti hai).

Gas ki internal energy sirf temperature aur molecules ki tadaad par depend karti hai.

5. The internal energy of a gas depends only on the number of molecules it contains and on the temperature, $\boxed{E = \frac{3}{2} N k_B T}$. In a free expansion of gas (it doesn't do any work in this case), the initial and final internal energies are equal, $E_f = E_i$.

Agar gas khud se expand kare (free expansion), to koi kaam nahi hota, is liye energy same rehti hai

Agar aap kisi substance ko heat dein aur uska volume constant rakhein, to zyada temperature barhne ke liye zyada heat chahiye.

6. Imagine that a substance is heated while keeping its volume constant. Obviously, you have to supply more heat as you raise the temperature higher, $\boxed{\Delta Q_V = C_V \Delta T}$ where C_V is given the name "specific heat at constant volume". No work is done since there is no expansion, $\Delta W = 0$. Hence, using the First Law, $\Delta Q_V = \Delta E + \Delta W = \Delta E$, and so $\Delta E = C_V \Delta T$.

Agar gas ko expand karne dein constant pressure par, to aur zyada heat chahiye hoti hai.

7. Now take the same system as above, but allow it to expand at constant pressure as it is heated. Then you will have to supply an amount of heat, $\Delta Q_P = C_P \Delta T$. Here, C_P is called the specific heat at constant pressure, $\Delta Q_P = \Delta E + \Delta w = \Delta E + P\Delta V$. Hence, $C_P \Delta T = \Delta U + P\Delta V$. If we supply only a small amount of heat then, $C_P dT = dU + PdV$. This gives, $C_P dT = C_V dT + PdV$. If we consider the special case of an ideal gas, then $PV = Nk_B T$. Hence, $PdV = Nk_B dT$ and $C_P dT = C_V dT + Nk_B dT$. Hence, $C_P = C_V + Nk_B$.

Since the internal energy is $E = \frac{3}{2} Nk_B T$, it follows that $\boxed{C_V = \frac{dU}{dT} = \frac{3}{2} Nk_B}$. From this, the specific heat at constant pressure is: $\boxed{C_P = \frac{3}{2} Nk_B + Nk_B = \frac{5}{2} Nk_B}$. From the equation $C_P = C_V + Nk_B$, it follows that $\frac{Nk_B}{C_V} = \frac{C_P - C_V}{C_V}$. Let us define a new symbol, $\boxed{\frac{C_P}{C_V} \equiv \gamma}$.

Agar gas adiabatically expand ho (yaani koi heat exchange nahi),

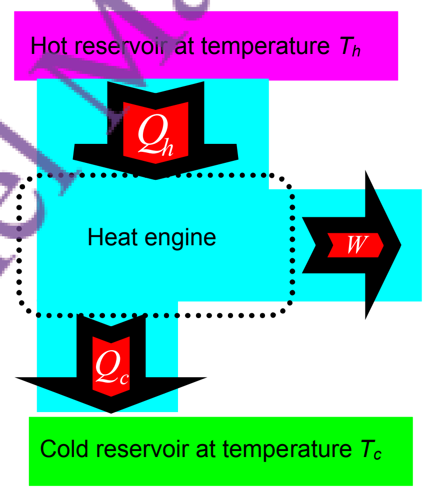
8. We can find the EOS of an ideal gas when it expands without losing any heat (this is called adiabiatic expansion). In this case, $\boxed{dQ = dE + dW = 0}$. Hence, $C_V dT + PdV = 0$, or $C_V dT + Nk_B T \frac{dV}{V} = 0$. Dividing by dT gives, $\frac{dT}{T} + \frac{Nk_B}{C_V}\frac{dV}{V} = 0$ or $\frac{dT}{T} + (\gamma - 1)\frac{dV}{V} = 0$. Integrating this gives, $\ln T + (\gamma - 1)\ln V =$ Constant. Equivalently, $\ln\left(TV^{\gamma-1}\right) =$ Constant. A more convenient form is: $TV^{\gamma-1} =$ Constant. This holds for the entire adiabatic expansion.

9. Almost nothing beats the importance of the Second Law Of Thermodynamics. There are many equivalent ways of stating it. The one I like is: "There can be no process whose only final result is to transfer thermal energy from a cooler object to a hotter object". This seems extremely simple (and almost useless), but in fact it tells you, among other things, that no perpetual motion machine (such as that which generates electricity without any fuel input) can ever be built. If someone could build such a machine, then it could also be used to transfer heat from a cold object to a hot object - a contradiction with the Second Law.

Second Law of Thermodynamics kehta hai ke heat kabhi bhi khud se thandi cheez se garam cheez ko transfer nahi hoti. Ye perpetual motion machines ko impossible banata hai jo bina kisi fuel ke kaam karein.

Thermal reservoir ek aisi cheez hoti hai jo apna temperature constant rakhti hai.
Jaise thermometer jab muh mein hota hai, aapka body ek thermal reservoir ki tarah kaam karta hai. Heat engine (jaise car engine, fridge) 2 reservoirs ke darmiyan kaam karta hai ek hot aur ek cold.

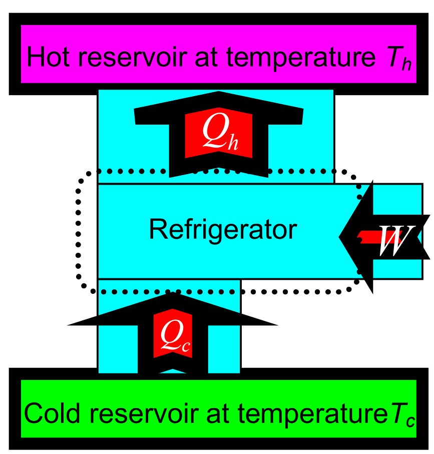
10. Before we consider some implications of the Second Law, I want to introduce the concept of a *thermal reservoir*. Suppose you put a thermometer in your mouth. You will transfer a small amount of heat to the thermometer, but because your body is so big this will make no measureable difference to your body temperature. Your body therefore is a thermal reservoir as far as the thermometer is concerned. More generally, any large mass of material will act as a thermal reservoir if it is at constant temperature. A *heat engine* (e.g. a car engine, refrigerator, etc) operates between reservoirs at two different temperatures.



11. A heat engine works between a high temperature T_H and a low temperature T_C. It aborbs heat $Q_{in,h}$ from the hot reservoir and rejects heat $Q_{out,c}$ into the cold reservoir. Since the internal energy of the engine does not change, $\Delta U = 0$ and the First Law gives simply, $Q = \Delta U + W = W$. The work done by the engine is $\boxed{W = Q_{in,h} - Q_{out,c}}$. Now define the efficiency of the machine as $\varepsilon = \dfrac{\text{work done by engine}}{\text{heat put in}} = \dfrac{W}{Q_{in,h}}$. Then $\varepsilon = \dfrac{Q_{in,h} - Q_{out,c}}{Q_{in,h}} = 1 - \dfrac{Q_{out,c}}{Q_{in,h}}$. Now, the heat that is transferred out of or into a reservoir is proportional to its absolute temperature. Hence we arrive at the important result that $\boxed{\varepsilon = 1 - \dfrac{T_C}{T_H}}$. This number is always less than one, showing that no machine can convert all the input heat into useful work. As an example, a nuclear reactor has temperature $300^0 C$ at the core and rejects heat into a river at $30^0 C$. The maximum efficiency it can have is $\boxed{\varepsilon = 1 - \dfrac{30+273}{300+273} = 0.471}$

Work (motor se) kiya jata hai Taake heat cold reservoir se hot reservoir mein pump ki ja sake (jaise frige mein heat bahar nikalna)

12. A refrigerator is a heat engine working in reverse. Work is done (by an electric motor) to pump heat from a cold reservoir (inside of refrigerator) to the hot exterior. The Second Law can be shown to imply the following: it is impossible for a refrigerator to produce no other effect than the transfer of thermal energy from a cold object to a hot object. Again, the efficiency of a refrigerator is always less than one.



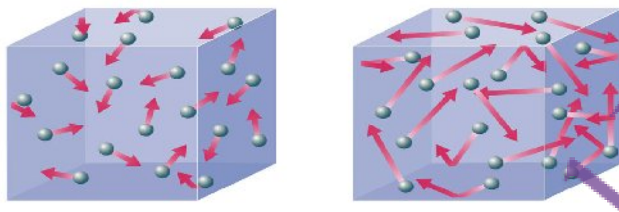
Second Law kehta ha Koi bhi refrigerator kabhi 100% efficient nahi ho sakta kyun ke ye sirf thermal energy ko cold se hot object mein transfer karta h
Efficiency hamesha 1 se kam hoti hai

Heat engine: hot => cold + work
Refrigerator: work + cold => hot

Heat hamesha garam cheez se thandi cheez ki taraf jaati hai.
Heat asal mein atoms ki random (bila tareet) movement hai.
Garam gas ke atoms tez move karte hain, aur thandi gas ke dheere.

Summary of Lecture 38 – THERMAL PHYSICS III

1. In the two previous lectures, I concentrated exclusively on heat as a form of energy that flows from a hot to a cold body. I made no mention of the fact that all matter is made of atoms and molecules; the notion of heat would exist even if we did not know this. But, the modern understanding of heat is that it is the random kinetic energy of atoms. So, for example, the difference between a cold and hot gas is illustrated below.



cold gas **hot gas**

Longer arrows denote atoms that are moving faster.
The hotter gas has, on the average, faster moving atoms.

Temperature : atoms average kitni speed se move kar rahe hain.
Heat : energy ka transfer hona ek object se doosre mein.

Statistical mechanics ek physics ka hissa hai jo atoms ke random motion ko samajhta hai.

2. The study of heat, considered as arising from the random motion of the basic constituents of matter, is an area of physics called statistical mechanics. Its goal is the understanding, and prediction of macroscopic phenomena, and the calculation of macroscopic properties from the properties of individual molecules. Temperature is the average energy related to the speed of atoms in an object, and heat is the amount of energy transferred from one object to another.

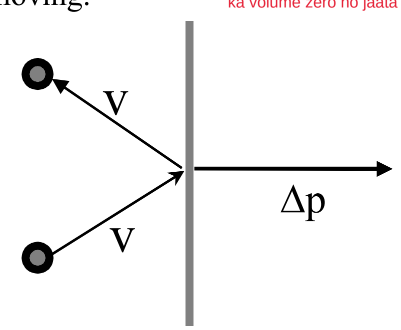
Jab temperature barhta hai, atoms zyada speed se takrate hain Pressure barhta hai. Agar pressure constant ho, to volume barhta hai jab temperature barhta hai. Isi wajah se balloon garam karne par phoolta hai.

3. Imagine a gas so dilute that atoms rarely collide with each other (this is also called an ideal gas). We can readily understand why the pressure is directly proportional to the temperature for a gas confined to a box: increasing the temperature T makes gas molecules move faster, striking the walls of the container harder and more often, thus giving an increase in pressure P, i.e. $P \propto T$. On the other hand, for an ideal gas at constant pressure, the volume is directly proportional to the temperature : $V \propto T$. So, as the air in a balloon is heated up, its volume will increase in direct proportion because of the impact of the atoms on the balloon walls. Note that this is where the "absolute" or Kelvin scale comes from: at 0° K an ideal gas would have zero volume because the atoms would not be moving.

0 Kelvin par atoms bilkul move nahi karte, aur ideal gas ka volume zero ho jaata hai.

Gas ke atoms jab container ke wall se takrate hain, to pressure create hota hai.

4. Pressure is related to the outward force per unit area exerted by the gas on the container wall. Newton's 2nd Law is: $\boxed{F = \frac{\Delta p}{\Delta t}}$. Here Δp is to be understood as the momentum destroyed in time Δt. Using this, we can do a little calculation to find the pressure in an ideal gas.



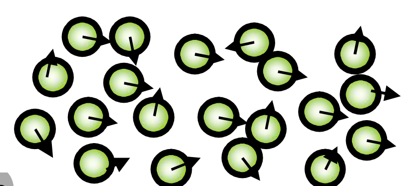
Box mein N atoms hain.
Half +x direction mein move karte hain, half -x mein.
Jo atoms wall se takrate hain, woh pressure banate hain.

Suppose there are N atoms in a box of volume, and the average speed of an atom in the x direction is v_x. By symmetry, half are moving in the $+x$ and half in the $-x$ directions. So the number of atoms that hit the wall in time Δt is $\frac{N}{2}\frac{1}{V}v_x\Delta tA$. Hence the change in momentum is $\Delta p = (2mv_x)\times\left(\frac{1}{2}\frac{N}{V}v_x\Delta tA\right) = \frac{N}{V}mv_x^2A\Delta t$. From this, we can calculate the pressure $\boxed{P = \frac{F}{A} = \frac{1}{A}\frac{\Delta p}{\Delta t} = \frac{N}{V}mv_x^2. \text{ Hence } PV = Nmv_x^2 = 2N\left(\frac{1}{2}mv_x^2\right)_{av}}$.

Jab koi choti particle move karti hai, uske paas energy hoti hai: (1/2 mv²)
Agar temperature zyada ho to particle zyada tez move karegi, matlab zyada energy hogi.
kB (Boltzmann constant) ka formula use hota hai: (1/2 mv²) = (1/2) kB T
Pura gas ke liye formula ban jaata hai: PV = NkBT
Total energy of 1 atom hoti hai: (3/2)kBT
Aur agar gas mein N atoms hain to total energy hoti hai: (3/2)NkBT

5. Let us return to our intuitive understanding of heat as the random energy of small particles. Now, $\boxed{\left(\frac{1}{2}mv_x^2\right)_{av}}$ is the average kinetic energy of a particle on account of its motion in the x direction. This will be greater for higher temperatures, so $\left(\frac{1}{2}mv_x^2\right)_{av} \propto T$, where T is the absolute temperature. So we write $\left(\frac{1}{2}mv_x^2\right)_{av} = k_BT$, which you can think of as the definition of the Boltzmann constant, k_B. So we immediately get $\boxed{PV = Nk_BT}$, the equation obeyed by an ideal gas (dilute, no collisions). Now, there is nothing special about any particular direction, so $(v_x^2)_{av} = (v_y^2)_{av} = (v_z^2)_{av}$ and therefore the average total velocity is $(v^2)_{av} = (v_x^2)_{av} + (v_y^2)_{av} + (v_z^2)_{av} = 3(v_x^2)_{av}$. So the total average kinetic energy of one atom is, $K_{av} = \left(\frac{1}{2}mv^2\right)_{av} = \frac{3}{2}k_BT$, and the energy of the entire gas is $\boxed{K = N\left(\frac{1}{2}mv^2\right)_{av} = \frac{3}{2}Nk_BT.}$ From this, the mean squared speed of atoms in a gas at temperature is $\boxed{(v^2)_{av} = \frac{3k_BT}{m}}$.

6. Heat, which is random kinetic energy, causes changes of phase in matter:



a) Water molecules attract each other, but if water is heated then they can escape and water becomes steam.
b) If steam is heated, the water molecules break up and water becomes separated hydrogen and oxygen atoms.
c) If a gas of atoms gets hot enough, then the atoms collide so violently that they lose electrons (i.e. get ionized).
d) More heat (such as inside the core of a star) will break up atomic nuclei into protons and neutrons.

a) Agar paani garam ho jaaye to molecules escape kar ke gas ban jaate hain.
b) Steam banne par molecules aur zyada door ho jaate hain (hydrogen aur oxygen alag ho jaate hain).
c) Agar gas bahut zyada garam ho jaaye to electrons toot jaate hain (ionization).
d) Aur agar aur bhi zyada heat mile (jaise star ke core mein), to protons aur neutrons bhi alag ho jaate hain.

Jab paani glass mein chhod do, kuch der baad gaayab ho jaata hai Kyun? Kyunke kuch molecules zyada energy le ke escape kar jaate hain.Ye energy random collisions se milti hai do molecules takrate hain, ek ruk jaata hai, doosra tez ho kar surface se nikal jaata hai.

7. Evaporation: if you leave water in a glass, it is no longer there after some time. Why?

Water molecules with high enough KE can break free from the water and escape. But how does a water molecule get enough energy to escape? This comes from random inter-molecular collisions, such as illustrated below. Two molecules collide, and one slows down while the other speeds up. The faster molecule might escape the water's surface.

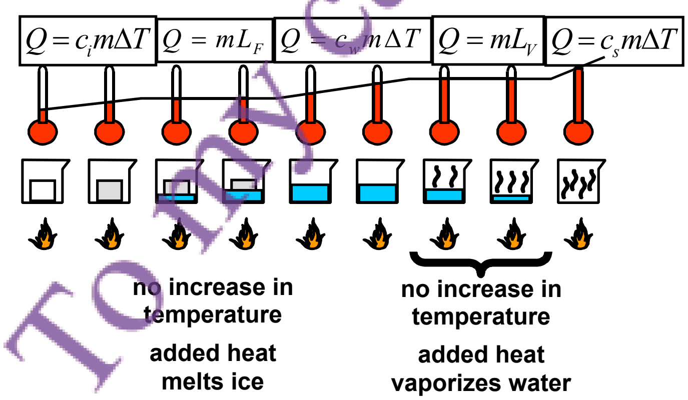
8. Heat needs to be supplied to cause a change of phase:

a) To melt a solid (ice, wax, iron) we must supply the "latent heat of fusion" L_F for unit mass of the solid. For mass m we must supply an amount of heat $\boxed{Q = mL_F}$ to overcome the attraction between molecules. No temperature change occurs in the melting.

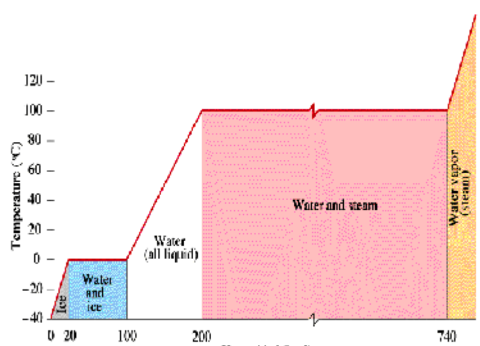
b) To vapourize a substance (convert to gas or vapour), we must supply the "latent heat of vapourization" L_V for unit mass of the substance. For mass m we must supply an amount of heat $\boxed{Q = mL_V}$. No temperature change occurs in the vapourization .

c) To raise the temperature of a substance we must supply the "specific heat" C for unit mass of the substance by 1 degree Kelvin. To change the temperature of mass m by ΔT, we must supply heat $\boxed{Q = mC\Delta T}$.

a) Melting (ice to water): Heat deni padti hai Q = mLf
b) Vapourization (water to steam): Heat deni padti hai Q = mLv
c) Temperature change: Jab sirf garam karna ho Q = mcΔT



In the above diagram you see what happens as ice is heated. The amount of heat needed is indicated as well. Note that no change of temperature happens until the phase change is completed. The same physical situation is represented to the right as well where the temperature is plotted against Q.



Jab ice melt hoti hai ya water steam banta hai, temperature nahi badhta sirf state change hoti hai. Jab states change nahi ho rahi hoti, tab temperature badhta hai.

Entropy ka matlab hota hai: disorder ya randomness
Jab kisi system mein heat add ki jati hai, toh uski entropy barhti hai

9. A very important concept is that of *entropy*. It was first introduced in the context of thermodynamics. Then, when the statistical nature of heat became clear a century later, it was understood in very different terms. Let us begin with thermodynamics: suppose a system is at temperature T and a small amount of heat dQ is added to it. Then, the small increase in the entropy of the system is $\boxed{dS = \frac{dQ}{T}}$. What if we keep adding little bits of heat? Then the temperature of the system will change by a finite amount, and the change in entropy is got by adding up all the small changes: $\boxed{\Delta S = \int_{T_1}^{T_2} \frac{dQ}{T}}$.

Now let's work out how much the entropy changes when we add heat dQ to an ideal gas. From the First Law, $\boxed{dQ = dU + dW = dU + PdV = C_V dT + Nk_B T \frac{dV}{V}}$. The entropy change is $dS = \frac{dQ}{T} = C_V \frac{dT}{T} + Nk_B \frac{dV}{V}$, and for a finite change we simply integrate,

$$\Delta S = \int \frac{dQ}{T} = C_V \ln \frac{T_2}{T_1} + Nk_B \ln \frac{V_2}{V_1} = \frac{3}{2} Nk_B \ln \frac{T_2}{T_1} + Nk_B \ln \frac{V_2}{V_1}$$

. Note that the following:

a)The entropy increases if we heat a gas $(T_2 > T_1)$.

b)The entropy increases if the volume increases $(V_2 > V_1)$.

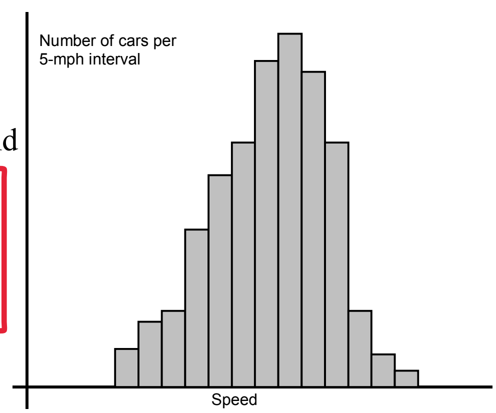
Statistical mechanics mein Entropy ko disorder ke tor par samjha jata hai. Jab gas ke atoms randomly move karte hain, toh unka disorder barhta hai, isliye entropy bhi barhti hai.

10. In statistical mechanics, we interpret entropy as the degree of disorder. A gas with all atoms at rest is considered ordered, while a hot gas having atoms buzzing around in all directions is more disordered and has greater entropy. Similarly, as in the above example, when a gas expands and occupies greater volume, it becomes even more disordered and the entropy increases.

11. We always talk about averages, but how do you define them mathematically? Take the example of cars moving along a road at different speeds. Call $n(\mathrm{v}_i)$ the number of cars with speed v_i , then $N = \sum_i n(\mathrm{v}_i)$ is the total number of cars and

Agar cars different speeds se move kar rahi hain, toh average speed nikalne ka tareeqa

the average speed is defined as $\boxed{\bar{\mathrm{v}} = \frac{\sum_i \mathrm{v}_i n(\mathrm{v}_i)}{\sum_i n(\mathrm{v}_i)}}$.

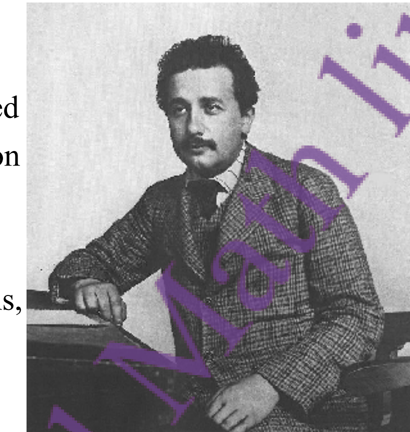


We define the probability of finding a car with speed v_i as $P(\mathrm{v}_i) \equiv \frac{n(\mathrm{v}_i)}{\sum_i n(\mathrm{v}_i)} = \frac{n(\mathrm{v}_i)}{N}$. We can

also write the average speed as $\boxed{\bar{\mathrm{v}} = \sum_i \mathrm{v}_i P(\mathrm{v}_i)}$.

Ye formula batata hai ke aik random car ka average speed kya ho sakta hai.

Summary of Lecture 39 – SPECIAL RELATIVITY I



1. The Special Theory of Relativity was created by Albert Einstein, when he was a very young man in 1905. It is one of the most solid pillars of physics and has been tested thousands of times. Special Relativity was a big revolution in understanding the nature of space and time, as well as mass and energy. It is absolutely necessary for a proper understanding of fast-moving particles (electrons, photons, neutrinos..). Einstein's General Theory of Relativity - which we shall not even touch here - goes beyond this and also deals with gravity.

Yah physics ka aik important pillar hai jo space aur time ka naya nazariya deta hai, especially jab koi object light ke qareeb speed se move karta hai

Is theory ne bataya ke time aur space absolute nahi hain, balki reference frame par depend karte hain.Fast moving particles (like photons, electrons) ke behavior ko samajhne ke liye yeh theory zaroori hai. General Relativity is se bhi aage jaati hai aur gravity ko bhi explain karti hai.

2. Relativity deals with time and space. So let us first get some understanding of how we measure these two fundamental quantities:

   a) *Time* **:** we measure time by looking at some phenomenon that repeats itself. There are endless examples: your heart beat, a pendulum, a vibrating quartz crystal, rotation of the earth around its axis, the revolution of the earth about the sun,... These can all be used as clocks. Of course, an atomic clock is far more accurate than using your heart beat and is accurate to one part in a trillion. Although different systems of measurement have different units it is fortunate that time is always measured in seconds.

   b) *Distance* **:** intuitively we know the difference between short and long. But to do a measurement, we first have to agree on what should be the unit of length. If you use a metre as the unit, then you can use a metre rod and measure any length you want. Of course, sometimes we may use more sophisticated means (such as finding how high a satellite is) but the basic idea is the same: the distance between point A and point B is the number of metre rods (or fractions thereof) that can be made to fit in between the two points.

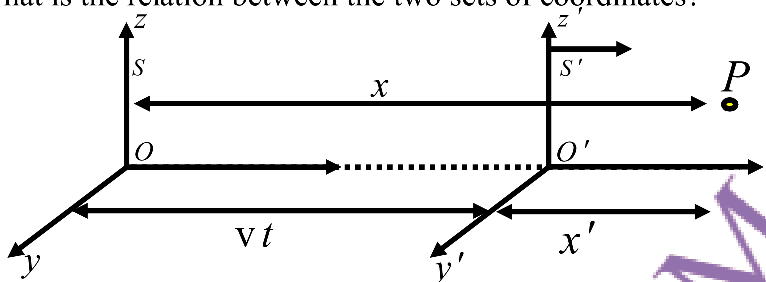
Time: Kisi repeating event se measure hota hai (jaise pendulum, heartbeat, clock).
Distance: Point A se B tak jitne meter rods fit hote hain, wohi distance hai.

3. Newton had believed that there was one single time for the entire universe. In other words, time was absolute and could be measured by one clock held somewhere in the centre of the universe. Similarly, he believed that space was absolute and that the true laws of physics could be seen in that particular frame which was fixed to the centre of the universe. (As we shall see, Einstein shocked the world by showing that time and space are not absolute quantities, but depend on the speed of your reference frame. Even more shocking was his proof that time and space are not entirely separate quantities!)

Newton ke mutabiq time aur space absolute the (har jagah ek jese). Einstein ne yeh concept wrong prove kiya aur dikhaya ke speed ke hisaab se time aur space change karte hain.

Ek event kisi space aur time point par hota hai (x,y,z,t). Agar ek frame se dusre moving frame mein dekhen, toh coordinates change ho jaate hain.Newton ke hisaab se time har frame mein same hota hai (Galilean Transformation).Einstein ne bataya ke yeh galat hai, aur time bhi reference frame ke mutabiq badalta hai.

4. An *event* is something that happens at some point in space at some time. With respect to the frame S below, the event P happened at (x,y,z,t). Now imagine a girl running to the right at fixed speed v in frame S′. According to her, the same event P happened at (x′,y′,z′,t′). What is the relation between the two sets of coordinates?



If you were Newton, then you would look at the above figure and say that obviously it is the following: $\boxed{x' = x - vt,\ y' = y,\ z' = z,\ t' = t.}$ These are called *Galilean coordinate transformations*. Note that it is assumed here that the time is the same in both frames because of the Newtonian belief that there is only one true time in the world.

5. There are certain obvious consequences of using the Galilean transformations. So, for example a rod is at rest in S-frame. The length in S-frame $= x_B - x_A$, while the length in the S′-frame $= x'_B - x'_A \;=\; x_B - x_A - \mathrm{v}(t_B - t_A)$. Since $t_B = t_A$, the length is the same in both frames: $x'_B - x'_A \;=\; x_B - x_A$. (As we shall see later this will not be true in Einstein's Special Relativity).

6. Let us now see what the Galilean transformation of coordinates implies for transformations of velocities. Start with $x' = x - \mathrm{v}t$ and differentiate both sides. Then $\frac{dx'}{dt} = \frac{dx}{dt} - \mathrm{v}$. Since $t = t'$, it follows that $\frac{dx'}{dt} = \frac{dx'}{dt'}$ and so $\frac{dx'}{dt'} = \frac{dx}{dt} - \mathrm{v}$. Similarly, $\frac{dy'}{dt'} = \frac{dy}{dt}$ and $\frac{dz'}{dt'} = \frac{dz}{dt}$.

Now, $\frac{dx'}{dt'} = u'_x$ is the x-component of the velocity measured in S′-frame, and similarly $\frac{dx}{dt} = u_x$ is the x-component of the velocity measured in S-frame. So we have found that:

$$u'_x = u_x - \mathrm{v},\quad u'_y = u_y,\quad u'_z = u_z \text{ (or, in vector form, } \vec{u}' = \vec{u} - \vec{\mathrm{v}})$$

Taking one further derivative, $\frac{du'_x}{dt'} = \frac{d}{dt}(u_x - \mathrm{v}) = \frac{du_x}{dt}$ (remember that $\mathrm{v} = \text{constant}$), we find that the components of acceleration are the same in S and S′:

$$\boxed{\frac{du'_x}{dt'} = \frac{du_x}{dt},\quad \frac{du'_y}{dt'} = \frac{du_y}{dt},\quad \frac{du'_z}{dt'} = \frac{du_z}{dt}.}$$

Einstein ne socha: agar mein torch le ke bhaag raha hoon to kya uski light aur tez chalegi?Us waqt yeh maana jata tha ke light aether mein travel karti hai (ek nazar na aane wali medium). Lekin koi evidence nahi mila ke aether exist karta hai. Light ki speed hamesha 2.9979 x 10⁸ m/s hoti hai, har frame mein.

7. We had learned in a previous chapter that light is electromagnetic wave that travels at a speed measured to be $\boxed{c = 2.907925 \times 10^8 m/\sec}$. Einstein, when he was 16 years old, asked himself the question: in which frame does this light travel at such a speed? If I run holding a torch, will the light coming from the torch also move faster? Yes, if we use the formula for addition of velocities derived in the previous section! So if light actually travels at $2.907925 \times 10^8 m/\sec$ then it is with respect to the frame in which the "aether" (a massless fluid which we cannot feel) is at rest. But there is no evidence for the aether!

1.Physics ke tamam laws sabhi inertial frames mein same hote hain.
2. Light ki speed vacuum mein sab inertial frames mein barabar hoti hai yeh source ya observer ki motion par depend nahi karti.

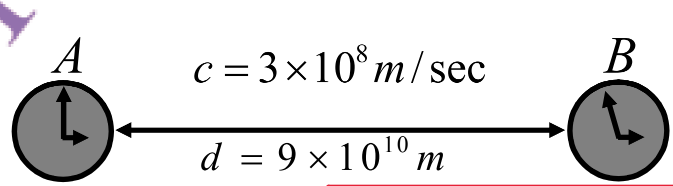
8. Einstein made the two following postulates (or assumptions).
   1) The laws of physics have the same form in all inertial frames. (In other words, there is no constant-velocity frame which is preferred, or better, than any other).
   2) The speed of light in vacuum has the same value in all inertial systems, independent of the relative motion of source and observer.

   [Note: as stressed in the lecture, no postulate of physics can ever be mathematically proved. You have to work out the consequences that follow from the postulates to know whether the postulates are good ones or not.]

Agar 2 clocks A aur B par hain, aur light ko B se A aane mein 5 min lagta hai, to observer time difference ko samajh ke clocks ko synchronize kar sakta hai.

9. Let's first get one thing clear: in any one inertial frame, we can imagine that there are rulers and clocks to measure distances and times. We can synchonize all the clocks to read one time, which will be called the time in that frame S. But how do we do this? If are two clocks, then we can set them to read the same time (i.e. synchronize them) by taking account of the time light takes to travel between them.

   Example: The observer with clock A sees the time on clock B as 2:55pm. But he knows that light took 5 minutes to travel from B to A, and therefore A and B are actually reading exactly the same time.

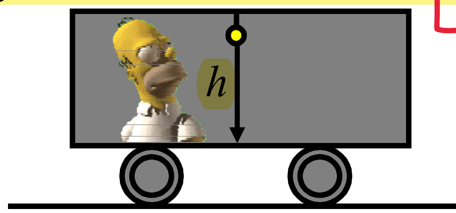


(Time taken by light in going from B to A is equal to $\boxed{\dfrac{d}{c} = \dfrac{9 \times 10^{10} m}{3 \times 10^8 m/\sec} = 300\sec = 5\min.}$)

10. Now I shall derive the famous formula which shows that a moving clock runs slow. This will be something completely different from the older Newtonian conception of time. Einstein derived this formula using a "gedanken" experiment, meaning an experiment which can imagine but not necessarily do. So imagine the following: a rail carriage has a

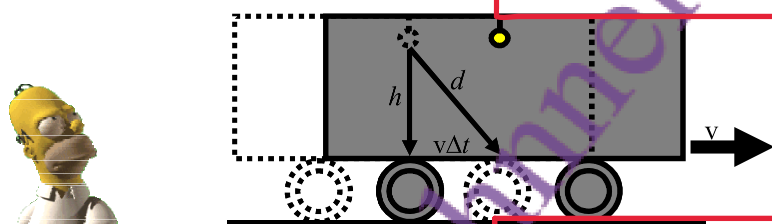
Socho ek rail carriage hai jisme ek bulb chhat se laga hua hai. Bulb flash karta hai aur roshni neeche floor tak jati hai. Train ke andar jo observer hai uske liye time = Δt' = h / c

a bulb that is fixed to the ceiling. The bulb suddenly flashes, and the light reaches the floor. Time taken according to the observer inside the train is $\boxed{\Delta t' = h/c}$.



Now suppose that the same flash is observed by an observer S standing on the ground. According to S, the train is moving with speed v. Let's consider the same light ray. Clearly, the train has moved forward between the time when the light left the ceiling and when it reached the floor. According to S, the time taken is $\boxed{\Delta t = \frac{d}{c} = \frac{\sqrt{h^2 + (\mathrm{v}\Delta t)^2}}{c}}$. Now

Lekin jo observer train ke bahar zameen par hai, uske liye scene thoda different hai Train speed v se move kar rahi hoti hai. Roshni neeche jate waqt train aage bhi move karti hai. Is observer ke liye light ka path diagonally hota hai (triangle ka hypotenuse).



square both sides: $(c\Delta t)^2 = h^2 + (\mathrm{v}\Delta t)^2$, which gives $\boxed{\Delta t = \frac{1}{\sqrt{1 - \frac{\mathrm{v}^2}{c^2}}} \frac{h}{c} = \gamma \Delta t'}$. Here γ is the relativistic factor $\boxed{\gamma = \frac{1}{\sqrt{1 - \frac{\mathrm{v}^2}{c^2}}}}$ and is a number that is always bigger than one. As v gets closer and closer to c, the value of γ gets larger and larger. For v=4c/5, γ=5/3. So, if 1 sec elapses between the ticks of a clock in S′ (i.e. $\Delta t' = 1$), the observer in S will see 5/3 seconds between the ticks. In other words, he will think that the moving clock is slow!

γ ka value hamesha 1 se zyada hota hai, jab v c ke gareeb ho.

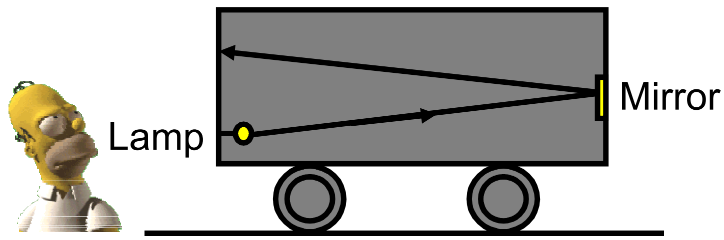
agar moving clock ka ek second guzarta hai, to bahar wala observer usay 5/3 seconds me dekhega matlab moving clock slow lagti hai.

11. The muon is an unstable particle. If at rest, it decays in just 10^{-18} seconds. But if traveling at 3/5 the speed of the light, it will last 25% longer because $\boxed{\gamma = \frac{1}{\sqrt{1 - (3/5)^2}} = \frac{5}{4}}$. If it is traveling at v=0.999999c, it will last 707 times longer. We can observe these shifts due to time dilation quite easily, and they are an important confirmation of Relativity.

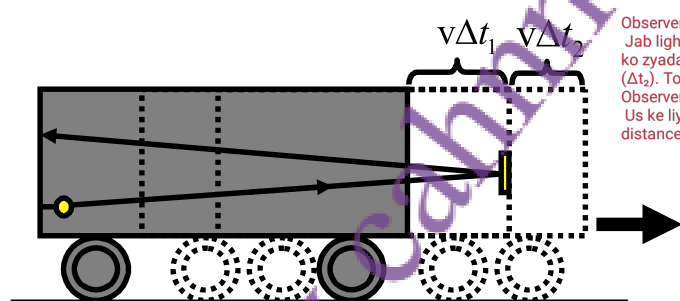
Agar woh light ki speed ke 3/5 se travel kare, to γ = 5/4, aur uski life 25% zyada ho jati hai.Agar woh 0.9999999c speed se chale, to woh 707 times zyada zinda rehta hai.

12. Another amazing prediction of Relativity is that objects are shortened (or contracted) along the direction of their motion. Einstein reached this astonishing conclusion on the basis of yet another gedanken experiment. Again, consider a moving railway carriage with a bulb

Relativity yeh bhi kehta hai ke moving objects ka length unki motion ke direction me chhota ho jata hai. Isay bhi Einstein ne ek thought experiment ke zariye derive kiya.



at one end that suddenly flashes. Let Δx be the length of the carriage according to ground observer S, and $\Delta x'$ be the length according to the observer S′ inside the carriage. So, $\Delta t' = \dfrac{2\Delta x'}{c}$ is the time taken for the light to go from one end to the other, and then return after being reflected by a mirror. Now let's look at this from the point of view of S, who is fixed to the ground. Let Δt_1 be the time for the signal to reach the front end. Then, because the mirror is moving forward, $\Delta t_1 = \dfrac{\Delta x + \mathrm{v}\Delta t_1}{c}$. Call Δt_2 the return time. Then,



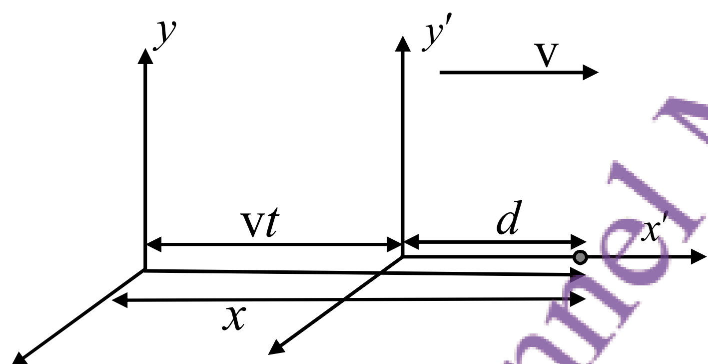
Observer S (jo zameen par hai):
Jab light mirror tak ja rahi hoti hai, mirror bhi move kar raha hota hai. Is liye light ko zyada waqt lagta hai aagay jaane mein (Δt_1) aur kam waqt wapas aane mein (Δt_2). Total waqt (Δt) dono ka sum hota hai.
Observer S' (jo train ke andar hai):
Us ke liye mirror aur lamp dono stationary hain.Is liye usay sirf ek straight line ka distance nazar aata hai, aur total time $\Delta t'$ hota hai.

$\Delta t_2 = \dfrac{\Delta x - \mathrm{v}\Delta t_2}{c}$. Solving for Δt_1 and Δt_2: $\Delta t_1 = \dfrac{\Delta x}{c - \mathrm{v}}$ and $\Delta t_2 = \dfrac{\Delta x}{c + \mathrm{v}}$. The total time is therefore $\Delta t = \Delta t_1 + \Delta t_2 = \dfrac{2\Delta x / c}{\left(1 - \mathrm{v}^2/c^2\right)}$. Now, from the time dilation result derived earlier $\boxed{\Delta t' = \sqrt{1 - \mathrm{v}^2/c^2}\,\Delta t}$ and so $\Delta t' = \sqrt{1 - \mathrm{v}^2/c^2}\left(\dfrac{2\Delta x / c}{\left(1 - \mathrm{v}^2/c^2\right)}\right) = \dfrac{2\Delta x / c}{\sqrt{1 - \mathrm{v}^2/c^2}}$. This gives, $\boxed{\Delta x' = \dfrac{\Delta x}{\sqrt{1 - \mathrm{v}^2/c^2}}, \text{ or } \Delta x = \Delta x'/\gamma}$. This an astonishing result! Suppose that there is a metre rod. Then the observer riding with it has $\Delta x' = 1$. But according to someone who sees the metre rod moving towards/away from him, the length is $1/\gamma$. This is less than 1 metre!

13. Although an object shrinks in the direction of motion (both while approaching and receding), the dimensions perpendicular to the velocity are not contracted. It is easy to conceive of a gedanken experiment that will demonstrate this. One of the exercises will guide you in this direction.

Agar ek event kisi jagah aur waqt par hota hai (x, t) S frame mein, to S' frame mein wo event (x', t') par hoga, jise Lorentz Transformation ke zariye calculate kiya jata hai.

14. We shall now derive the "Lorentz Transformation", which is the relativistic version of the Galilean transformation discussed earlier. Consider an event that occurs at position x (as measured in S) at a distance d (again, as seen in S) from the origin of S'. Then, $x = d + \mathrm{v}t$. From the Lorentz contraction formula derived earlier, $d = \frac{x'}{\gamma}$ where x' is the distance measured in S'. This gives $x' = \gamma(x - \mathrm{v}t)$. Now, by the same logic, $x' = d' - \mathrm{v}t'$.



Here $d' = \frac{x}{\gamma}$. This gives $x = \gamma(x' + \mathrm{v}t') = \gamma(\gamma(x - \mathrm{v}t) + \mathrm{v}t')$. We find that the time in S' is related to the time in S by $\boxed{t' = \gamma\left(t - \frac{\mathrm{v}}{\mathrm{c}^2}x\right)}$. Note that if we make c very large, then $t' = t$.

To summarize:

LORENTZ TRANSFORMATION

$x' = \gamma(x - \mathrm{v}t)$

$y' = y$

$z' = z$

$\boxed{t' = \gamma\left(t - \frac{\mathrm{v}}{c^2}x\right).}$

(Note: in various books you will find slightly different derivations of the above Lorentz transformation. You should look at one of your choice and understand that as well.)

Summary of Lecture 40 – SPECIAL RELATIVITY II

1. Recall the Lorentz Transformation: $\boxed{x' = \gamma\left(x - \mathrm{v}t\right)}$ and $\boxed{t' = \gamma\left(t - \frac{\mathrm{v}}{c^2}x\right)}$ Suppose we take the space interval between two events $\Delta x = x_1 - x_2$, and the time interval $\Delta t = t_1 - t_2$.

Then, these intervals will be seen in S′ as $\boxed{\Delta t' = \gamma\left(\Delta t - \frac{\mathrm{v}}{c^2}\Delta x\right)}$ and $\boxed{\Delta x' = \gamma\left(\Delta x - \mathrm{v}\Delta t\right)}$.

Now consider two particular cases:

a) Suppose the two events occur at the same place (so $\Delta x = 0$) but at different times (so $\Delta t \neq 0$). Note that in S′ they do not occur at the same point: $\boxed{\Delta x' = \gamma\left(0 - \mathrm{v}\Delta t\right)}$!

b) Suppose the two events occur at the same time (so $\Delta t = 0$) but at different places (so $\Delta x \neq 0$). Note that in S′ they are not simultaneous $\boxed{\Delta t' = \gamma\left(0 - \frac{\mathrm{v}}{c^2}\Delta x\right).}$

Frame S mein dekha jaye to ek particle distance dx move karta hai time dt mein. Uski velocity u = dx/dt hoti hai. Frame S' mein wohi particle distance dx' move karta hai.

2. As seen in the frame S, suppose a particle moves a distance dx in time dt. Its velocity u is then $u = \frac{dx}{dt}$ (in S-frame). As seen in the S′-frame, meanwhile, it has moved a distance dx' where $dx' = \gamma\left(dx - \mathrm{v}dt\right)$ and the time that has elapsed is $dt' = \gamma\left(dt - \frac{\mathrm{v}}{c^2}dx\right)$. The velocity in S′-frame is therefore $\boxed{u' = \frac{dx'}{dt'} = \frac{\gamma\left(dx - \mathrm{v}dt\right)}{\gamma\left(dt - \frac{\mathrm{v}}{c^2}dx\right)} = \frac{dx/dt - \mathrm{v}}{1 - \frac{\mathrm{v}}{c^2}dx/dt} = \frac{u - \mathrm{v}}{1 - \frac{u\mathrm{v}}{c^2}}}$. This is the Einstein velocity addition rule. It is an easy exercise to solve this for u in terms of u',

$$\boxed{u = \frac{u' + \mathrm{v}}{1 + \frac{u'\mathrm{v}}{c^2}}.}$$

agar ek car speed v se chal rahi ho aur uske headlights on ho jayein, to ground pe observer ke liye light ki speed kya hogi?

3. Note one very interesting result of the above: suppose that a car is moving at speed v and it turns on its headlight. What will the speed of the light be according to the observer on the ground? If we use the Galilean transformation result, the answer is v+c (wrong!). But using the relativistic result we have $u' = c$ and $u = \frac{c + \mathrm{v}}{1 + \frac{c\mathrm{v}}{c^2}} = \frac{c + \mathrm{v}}{1 + \frac{\mathrm{v}}{c}} = c\frac{c + \mathrm{v}}{c + \mathrm{v}} = c$. In other words, the speed of the source makes no difference to the speed of light in your frame. Note that if either u or v is much less than c, then u' reduces to the familiar result: $\boxed{u' = \frac{u - \mathrm{v}}{1 - \frac{u\mathrm{v}}{c^2}} \to u - \mathrm{v}}$, which is the Galilean velocity addition rule.

Lorentz transformations ka aik khaas property ha jis m Agar 2 events ke darmiyan time aur space intervals ko frame S mein aur S' mein compare karein to result ma I milta h

4. The Lorentz transformations have an interesting property that we shall now explore. Take the time and space intervals between two events as observed in frame S, and the corresponding quantities as observed in S'. We will now prove that the quantities defined respectively as $I=(c\Delta t)^2-(\Delta x)^2$ and $I'=(c\Delta t')^2-(\Delta x')^2$ are equal. Let's start with I':

$$I'=(c\Delta t')^2-(\Delta x')^2=\gamma^2\left((c\Delta t)^2+(\Delta x)^2 \mathrm{v}^2/c^2-\cancel{2\mathrm{v}\Delta t\Delta x}\ -(\Delta x)^2-(\mathrm{v}\Delta t)^2+\cancel{2\mathrm{v}\Delta t\Delta x}\right)$$

$$=\frac{1}{\left(1-\mathrm{v}^2/c^2\right)}\left(\left(c^2-\mathrm{v}^2\right)(\Delta t)^2-(\Delta x)^2\left(1-\mathrm{v}^2/c^2\right)\right)$$

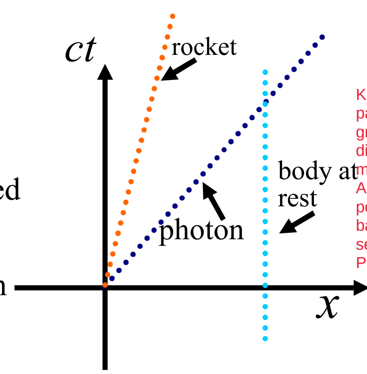
$$=(c\Delta t)^2-(\Delta x)^2=\boxed{I}$$

This guarantees that all inertial observers measure the same speed of light !!

I > 0, interval ko timelike kehte hain.
I < 0, interval ko spacelike kehte hain.
I = 0 ho to interval lightlike kehlata hai.

5. If the time separation is large, then $I>0$ and we call the interval *timelike*.
If the space separation is large, then $I<0$ and we call the interval *spacelike*.
If $I=0$ and we call the interval *lightlike*.
Note: a) If an interval is timelike in one frame, it is timelike in all other frames as well.
b) If interval between two events is timelike, their time ordering is absolute.
c) If the interval is spacelike the ordering of two events depends on the frame from which they are observed.

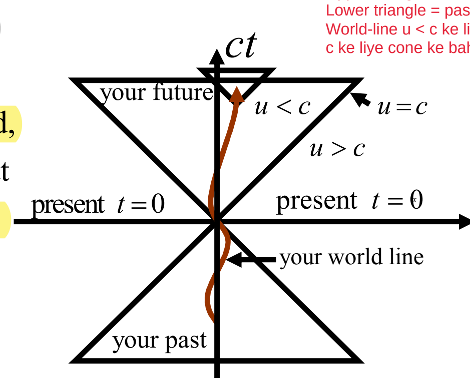
6. It is sometimes nice to look at things graphically. Here is a graph of position versus time for objects that move with different speeds along a fixed direction. First look at an object at rest. Its position is fixed even though time (plotted on the vertical axis) keeps increasing. Then look at the rocket moving at constant speed (which has to be less than c), and finally a photon (which can only move at c).



Kabhi kabhi cheezon ko graphical tor par dekhna achha hota hai. Yeh aik graph hai jo position aur time ka taluq dikhata hai un objects ke liye jo mukhtalif speeds se move karte hain. Agar aik object rest mein hai, uski position constant rahegi jabke time barhta rahega. Rocket constant speed se move karta hai (jo c se kam hoti hai). Photon sirf speed c se move karta hai.

Upper triangle = Future light cone
Lower triangle = past light cone
World-line u < c ke liye cone ke andar hoti hai, jabke u > c ke liye cone ke bahar.

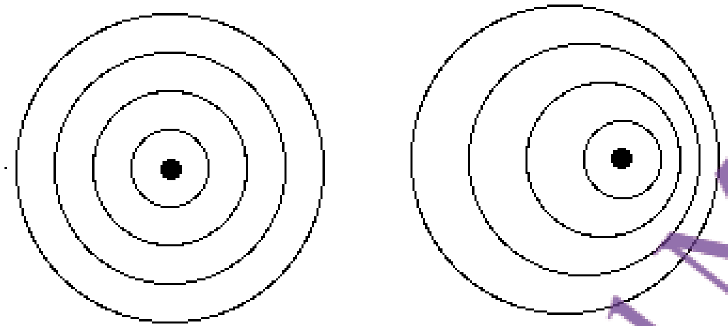
7. The trajectory of a body as it moves through space-time is called its world-line. Let's take a rocket that is at $x=0$ at $t=0$. It moves with non-constant speed, and that is why its world-line is wavy. A photon that moves to the right will have a world line with slope equal to +1, and that to the right with slope -1. The upper triangle (with t positive) is called the future light cone (don't forget we also have y and x). The lower light cone consists of past events.



Agar rocket x = 0, t = 0 par hai, aur non-constant speed se move karta hai, to uski world-line wavy hoti hai.
Photon ki world-line ka slope hota hai +1 ya -1 (right ya left movement ke liye).

Pehle Doppler effect ko sound ke context mein discuss kiya gaya tha. Ab light ke liye Light hammesha fixed speed se travel karti hai, is liye observer ke liye source ki motion wavelength aur frequency ko effect karti hai.

8. Earlier on we had discussed the Doppler effect in the context of sound. Now let us do so for light. Why are they different? Because light always moves with a fixed speed while the speed of sound is different according to a moving and a fixed observer. Consider the case of a source of light at rest, and one that is moving to the right as shown below:



waves symmetrically spread hoti hain. waves ahead compress ho jati hain, aur peeche expand.

(a) stationary source (b) moving source

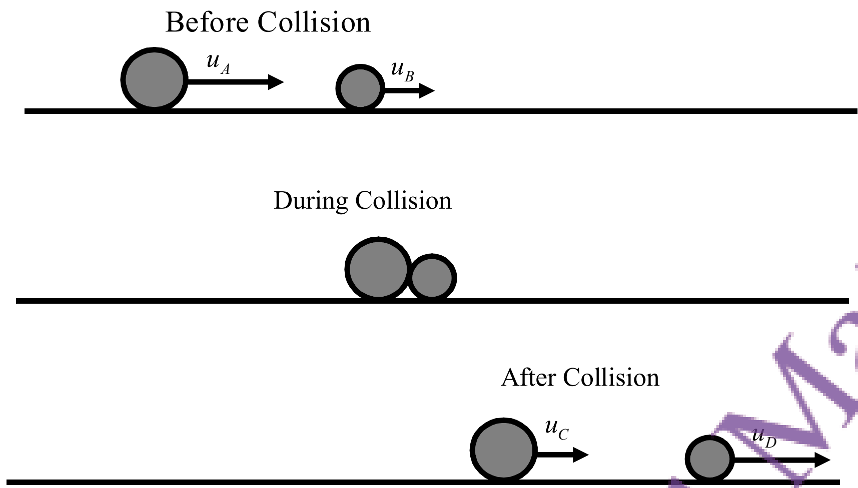
Let $\nu_0 = \frac{1}{T_0}$ be the frequency measured in the source's rest frame S, where T_0 is the time for one complete cycle in S. We want to calculate ν, the frequency as seen by the observer in S′ moving to the right at speed v. Call λ the distance between two successive wave crests (i.e. the wavelength according to the observer). In time T the crests ahead of the source move a distance cT, even as the source moves a shorter distance vT in the same direction. Hence $\lambda = (c - \text{v})T$ and so $\nu = \frac{c}{\lambda} = \frac{c}{(c-\text{v})T}$. Now, as discussed earlier, the time measured by observer will not be T_0 because of time dilation. Instead,

$$T = \frac{T_0}{\sqrt{1-\text{v}^2/c^2}} = \frac{cT_0}{\sqrt{c^2-\text{v}^2}}.\text{ Hence, } \nu = \frac{c}{(c-\text{v})T} = \left(\frac{c}{c-\text{v}}\right)\frac{\sqrt{c^2-\text{v}^2}}{c}\nu_0 = \sqrt{\frac{c+\text{v}}{c-\text{v}}}\nu_0.$$

(If source moves away from the observer just change the sign of v: $\boxed{\nu = \sqrt{\frac{c-\text{v}}{c+\text{v}}}\nu_0}$)

This is the famous Doppler effect formula. In the lecture I discussed some applications such as finding the speed at which stars move away from the earth, or finding the speed of cars or aircraft.

9. We now must decide how to generalize the concept of momentum in Relativity theory. The Newtonian definition of momentum is $\vec{p} = m\vec{u}$. The problem with this definition is that we are used to having momentum conserved when particles collide with each other, and this old definition will simply not work when particles move very fast. Consider the collision of two particles as in the diagrams below:



In the frame fixed to the lab (S-frame) conservation of momentum implies: $m_A u_A + m_B u_B = m_C u_C + m_D u_D$. Mass is also conserved: $m_A + m_B = m_C + m_D$. Now suppose we wish to observed the collision from a frame S′ moving at speed v. Then, from the relativistic addition of velocities formula, $u' = \dfrac{u - \mathrm{v}}{1 - \dfrac{u\mathrm{v}}{c^2}}$ and, from it, $u = \dfrac{u' + \mathrm{v}}{1 + \dfrac{u'\mathrm{v}}{c^2}}$. Insert this into the equation of conservation of momentum (using $p = m\mathrm{v}$ as the definition):

$$\boxed{m_A\left(\frac{u'_A + \mathrm{v}}{1 + u'_A \mathrm{v}/c^2}\right) + m_B\left(\frac{u'_B + \mathrm{v}}{1 + u'_B \mathrm{v}/c^2}\right) = m_C\left(\frac{u'_C + \mathrm{v}}{1 + u'_C \mathrm{v}/c^2}\right) + m_D\left(\frac{u'_D + \mathrm{v}}{1 + u'_D \mathrm{v}/c^2}\right)}$$

This is clearly not the equation $m_A u'_A + m_B u'_B = m_C u'_C + m_D u'_D$ So momentum will not be conserved relativistically if we insist on using the old definition!!

10. Can we save the situation and make the conservation of momentum hold by finding some suitable new definition of momentum? The new definition must have two properties:

1) At low speeds it must reduce to the old one.

2) At all speeds momentum must be conserved.

Let's see if the definition $\vec{p} = \dfrac{m\vec{u}}{\sqrt{1 - u^2/c^2}} = \gamma m\vec{u}$ will do the job. Obviously if u<<c we get $\vec{p} = m\vec{u}$, so requirement 1 is clearly satisfied. . Let's now see if the conservation of momentum equation will hold if the new definition of momentum is used:

$$m_A\gamma_A u_A + m_B\gamma_B u_B = m_C\gamma_C u_C + m_D\gamma_D u_D.$$

After doing some algebra you find,

$$m_A\left(\frac{1}{\gamma}\gamma'_A u'_A + \gamma_A \mathrm{v}\right) + m_B\left(\frac{1}{\gamma}\gamma'_B u'_B + \gamma_B \mathrm{v}\right) = m_C\left(\frac{1}{\gamma}\gamma'_C u'_C + \gamma_C \mathrm{v}\right) + m_D\left(\frac{1}{\gamma}\gamma'_D u'_D + \gamma_D \mathrm{v}\right)$$

This gives $\boxed{m_A\gamma'_A u'_A + m_B\gamma'_B u'_B = m_C\gamma'_C u'_C + m_D\gamma'_D u'_D}$ which is just what we want.

11. We shall now consider how energy must be redefined relativistically. The usual expression for the kinetic energy $K = \frac{1}{2}mu^2$ is not consistent with relativistic mechanics (it does not satisfy the law of conservation of energy in relativity). To discover a new definition, let us start from the basics: the work done by a force F when it moves through distance dx adds up to an increase in knetic energy, $K = \int F dx = \int \frac{dp}{dt} dx = \int \frac{dx}{dt} dp = \int u dp$. Now use:

$$dp = md\left(\frac{u}{\sqrt{1-u^2/c^2}}\right) = \frac{mdu}{\sqrt{1-u^2/c^2}} + \frac{m\left(u^2/c^2\right)du}{\left(1-u^2/c^2\right)^{3/2}} = \frac{mdu}{\left(1-u^2/c^2\right)^{3/2}}$$

This give $K = \int_0^u u dp = m\int_0^u \frac{udu}{\left(1-u^2/c^2\right)^{3/2}} = \frac{mc^2}{\sqrt{1-u^2/c^2}} - mc^2$

Note that K is zero if there is no motion, or $K = E - E_0$ where $E = \frac{mc^2}{\sqrt{1-u^2/c^2}}$ and $E_0 = mc^2$. Now expand $\frac{1}{\sqrt{1-u^2/c^2}} = \left(1-u^2/c^2\right)^{-1/2} = 1 + \frac{u^2}{2c^2} + \cdots$

$$\boxed{\text{Hence, } K = mc^2\left(1 + \frac{u^2}{2c^2} + \cdots\right) - mc^2 \to \frac{1}{2}mu^2 \text{ as } u/c \to 0.}$$

12. Now that we have done all the real work, let us derive some alternative expressions using our two main formulae: $\boxed{\vec{p} = \frac{m\vec{u}}{\sqrt{1-u^2/c^2}} = \gamma m\vec{u} \text{ and } E = \frac{mc^2}{\sqrt{1-u^2/c^2}} = \gamma mc^2.}$

a) $\vec{u} = \frac{\vec{p}}{\gamma m} = \frac{\vec{p}c^2}{E}$ or $pc = E\left(\frac{u}{c}\right)$

b) Clearly $(pc)^2 = E^2\left(\frac{u}{c}\right)^2 = \gamma^2 m^2 c^4\left(1 - 1/\gamma^2\right) = \gamma^2 m^2 c^4 - m^2 c^4 = E^2 - m^2 c^4$. Hence,

$E^2 = p^2c^2 + m^2c^4$.

c) For a massless particle $(m = 0)$, $E = pc$.

13. A particle with mass m has energy mc^2 even though it is at rest. This is called its rest energy. Since c is a very large quantity, even a small m corresponds to a very large energy. We interpret this as follows: suppose all the mass could somehow be converted into energy. Then an amount of energy equal to mc^2 would be released.

Hum particles ko aise matter ke tor par sochtay hain jo kisi space ke chhote hisse mein bohat ziada hota hai, aur waves ko aise sochtay hain jo har taraf phaili hoti hain. Socho aik cricket ball ka, aur samandar mein lehron ka dono bilkul mukhtalif hain. Matter kabhi wave ki tarah aur kabhi particle ki tarah behave karta hai. Light bhi dono jaisi sifat rakhti hai.

Summary of Lecture 41 – WAVES AND PARTICLES

1. We think of particles as matter highly concentrated in some volume of space, and of waves as being highly spread out. Think of a cricket ball, and of waves in the ocean. The two are completely different! And yet today we are convinced that matter takes the form of waves in some situations and behaves as particles in other situations. This is called wave-particle duality. But do not be afraid - there is no logical contradiction here! In this lecture we shall first consider the evidence that shows the particle nature of light.

2. The photoelectric effect, noted nearly 100 years ago, was crucial for understanding the nature of light. In the diagram shown, when light falls upon a metal plate connected to the cathode of a battery, electrons are knocked out of the plate. They reach a collecting plate that is connected to the battery's anode, and a current is observed. A vacuum is created in the apparatus so that the electrons can travel without hindrance. According to classical (meaning old!) physics we expect the following:

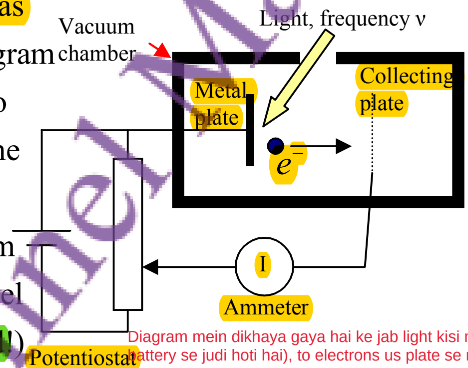


Diagram mein dikhaya gaya hai ke jab light kisi metal plate par girti hai (jo battery se judi hoti hai), to electrons us plate se nikal kar dusri plate tak pohanchtay hain aur current observe hota hai. Vacuum banaya jata hai taake koi rukawat na ho.

a)As intensity of light increases, the kinetic energy of the ejected electrons should increase. Jaise jaise light ki intensity barhti hai, ejected electrons ki kinetic energy bhi barhni chahiye.

b)Electrons should be emitted for any frequency of light ν, so long as the intensity of the light is sufficiently large. Har frequency ki light pe electrons emit hone chahiyein agar intensity zyada ho.

But the actual observation was completely different and showed the following:

a)The maximum kinetic energy of the emitted electrons was completely independent of the light intensity, but it did depend on ν. Electrons ki maximum kinetic energy light ki intensity par depend nahi karti, balkay frequency (v) par karti hai.

b)For $\nu < \nu_0$ (i.e. for frequencies below a cut-off frequency) no electrons are emitted no matter how large the light intensity was. Agar $\nu < \nu_0$ (cut-off frequency se kam), to koi electron emit nahi hota, chaahe intensity kitni bhi zyada ho.

1905 mein Einstein ne bataya ke photoelectric effect tabhi samjha ja sakta hai agar hum light ko chhoti packets (quanta) mein samjhein, jinki energy hoti hai E = hv.

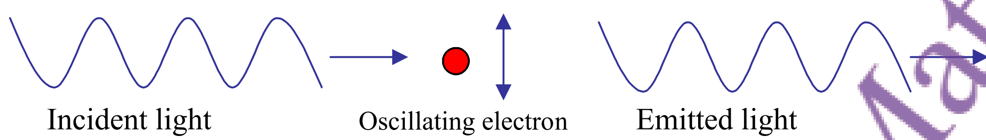
3. In 1905, Einstein realized that the photoelectric effect could be explained if light actually comes in little packets (or quanta) of energy with the energy of each quantum $E = h\nu$. Here h is a universal constant of nature with value $h = 6.63\times10^{-34}$ Joule-seconds, and is known as the Planck Constant. If an electron absorbs a single photon, it would be able to leave the material if the energy of that photon is larger than a certain amount W. W is called the work function and differs from material to material, with a value varying from 2-5 electron volts. The maximum KE of an emitted electron is then $\boxed{K_{\max} = h\nu - W}$. We visualize the photon as a particle for the purposes of this experiment. Note that this is completely different from our earlier understanding that light is a wave!

Agar photon ki energy W (material se related quantity) se zyada hai, to electron emit hoga.

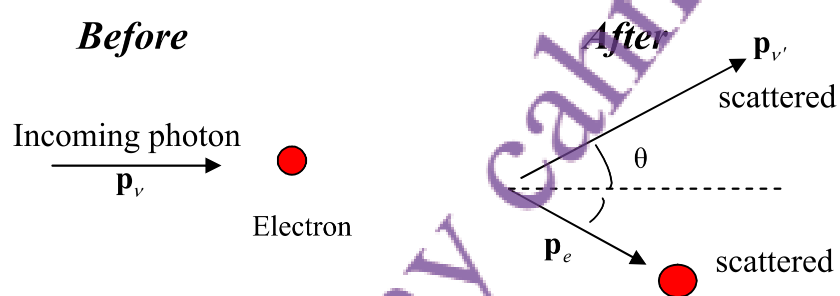
Jab ek electron ko light beam ke raste mein rakha gaya, to dekha gaya ke scattered light aur incident light ki frequency mein farq hota hai ye classical theory ke khilaaf tha.

4. That light is made of photons was confirmed by yet another experiment, carried out by Arthur Compton in 1922. Suppose an electron is placed in the path of a light beam. What will happen? Because light is electromagnetic waves, we expect the electron to oscillate with the same frequency as the frequency of the incident light ν. But because a charged particle radiates em waves, we expect that the electron will also radiate light at frequency ν. So the scattered and incident light have the same ν. But this is not what is observed!

Light photons ka bana hota hai, aur jab photon electron se takrata hai, to energy aur momentum transfer hota hai photon ki wavelength badal jaati hai.



To explain the fact that the scattered light has a different frequency (or wavelength), Compton said that the scattering is a collision between particles of light and electrons. But we know that momentum and energy is conserved in scattering between particles. Specifically, from conservation of energy $h\nu + m_e c^2 = h\nu' + \left(p_e^2 c^2 + m_e^2 c^4\right)^{1/2}$. The last term is the energy of the scattered electron with mass m_e. Next, use the conservation



λ = original wavelength
λ' = scattered wavelength
h = Planck's constant
m_e = electron mass
c = speed of light
θ = scattering angle

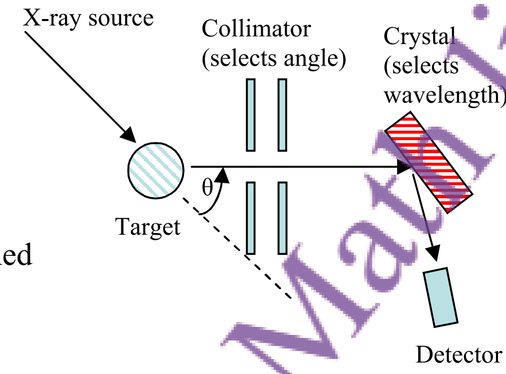
of momentum. The initial momentum of the photon is entirely along the $\hat{z}$ direction, $\mathbf{p}_\nu = \frac{h}{\lambda}\hat{z} = \mathbf{p}_{\nu'} + \mathbf{p}_e$. By resolving the components and doing a bit of algebra, you can get the change in wavelength $\boxed{\lambda' - \lambda = \frac{h}{m_e c}(1-\cos\theta) = \lambda_c(1-\cos\theta)}$, where the Compton wavelength $\lambda_c = \frac{h}{m_e c} = 2.4\times10^{-12}$ m. Note that $\lambda' - \lambda$ is always positive because $\cos\theta$ has magnitude less than 1. In other words, the frequency of the scattered photon is always less than the frequency of the incoming one. We can understand this result because the incoming photon gives a kick to the (stationary) electron and so it loses energy. Since $E = h\nu$, it follows that the outgoing frequency is decreased. As remarked earlier, it is impossible to understand this from a classical point of view. We shall now see how the Compton effect is actually observed experimentally.

Wavelength hamesha barhti hai (λ' > λ) Kyunkay cosθ ≤ 1 hota hai, is liye (1 − cosθ) ≥ 0, aur result hamesha positive hota hai.

Photon aur electron ka collision elastic nahi hota, photon energy lose karta hai aur wavelength barhti hai.

X-rays jab kisi target par girti hain jahan electrons hote hain, to wo scatter hoti hain. Collimator ek specific angle θ par scatter hone wali X-rays ko choose karta hai, jo baad mein ek crystal par girti hain.
Crystal diffraction ke zariye wavelength ko measure kiya jata hai.

5. In the apparatus shown, X-rays are incident upon a target which contains electrons. Those X-rays which are scattered in a particular angle θ are then selected by the collimator and are incident upon a crystal. The crystal diffracts the X-rays and, as you will recall from the lecture on diffraction, $2d\sin\theta_D = n\lambda$ is the necessary condition. It was thus determined that the changed wavelength of the scattered follows $\boxed{\lambda' - \lambda = \lambda_c(1-\cos\theta)}$.



6. Here is a summary of photon facts:
   a)The relation between the energy and frequency of a photon is $\boxed{E = h\nu}$.
   b)The relativistic formula relating energy and momentum for photons is $\boxed{E = pc}$. Note that in general $E^2 = p^2c^2 + m^2c^4$ for a massive particle. The photon has $m = 0$.
   c)The relation between frequency, wavelength, and photon speed is $\boxed{\lambda\nu = c}$.
   d)From the abbove, the momentum-wavelength relation for photons is $\boxed{p = \frac{h\nu}{c} = \frac{h}{\lambda}}$.
   e)An alternative way of expressing the above is: $E = \hbar\omega$ and $p = \hbar k$. Here $\omega = 2\pi\nu$ and $k = \frac{2\pi}{\lambda}$, $\hbar \equiv \frac{h}{2\pi}$ ($\hbar$ is pronounced h-bar).
   f)Light is always detected as packets (photons); if we look, we never observe half a photon. The number of photons is proportional to the energy density (i.e. to square of the electromagnetic field strength).

7. So light behaves as if made of particles. But do all particles of matter behave as if they are waves? In 1923 a Frenchman, Louis de Broglie, postulated that ordinary matter can have wave-like properties with the wavelength λ related to the particle's momentum p in the same way as for light, $\boxed{\lambda = \frac{h}{p}}$. We shall call λ the de Broglie (pronounced as Deebrolee!) wavelength.

Debroglie Wavelength formula

1923 mein Louis de Broglie ne suggest kiya ke har particle jese electron, cricket ball waghera ki bhi wave-like properties hoti hain.

8. Let us estimate some typical De Broglie wavelengths:
   a) The wavelength of 0.5 kg cricket ball moving at 2 m/sec is:

$$\boxed{\lambda = \frac{h}{p} = \frac{6.63\times10^{-34}}{0.5\times 2} = 6.63\times10^{-34}\ \text{m}}$$

Agar electron ki kinetic energy 50 eV ho, to uski wavelength ka formula hai
λ = h / √(2mₑK)
h = Planck's constant
mₑ = electron ka mass
K = kinetic energy

This is extremely small even in comparison to an atom, 10^{-10}m.

b) The wavelength of an electron with 50eV kinetic energy is calculated from:

$$\boxed{K=\frac{p^2}{2m_e}=\frac{h^2}{2m_e\lambda^2}\Rightarrow\lambda=\frac{h}{\sqrt{2m_eK}}=1.7\times10^{-10}\,\text{m}}$$

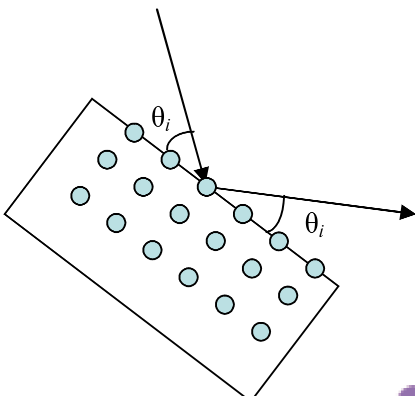
Now you see that we are close to atomic dimensions.

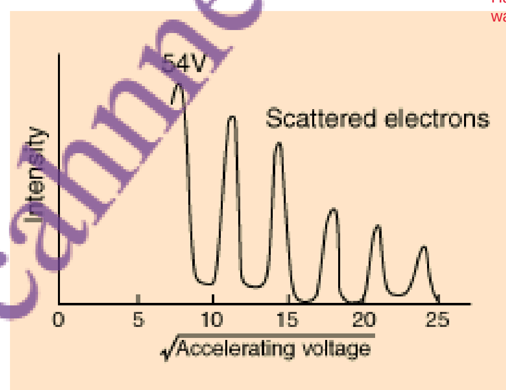
Mean electron ki wavelength atomic scale ke barabar hai.

Agar De Broglie ka idea sahi hai to electrons bhi light ki tarah interference dikhayenge.
Is experiment mein electrons ko crystal surface pe bheja gaya aur scattering angles ko record kiya gaya.

9. If De Broglie's hypothesis is correct, then we can expect that electron waves will undergo interference just like light waves. Indeed, the Davisson-Germer experiment (1927) showed that this was true. At fixed angle, one find sharp peaks in intensity as a function of electron energy. The electron waves hitting the atoms are re-emitted and reflected, and waves from different atoms interfere with each other. One therefore sees the peaks and valleys that are typical of interference (or diffraction) patterns in optical experiments.

He observe Electron waves ne interference patterns dikhaye (jaise peaks and valleys). Har angle par intensity change hoti thi, jo wave nature ka saboot hai.

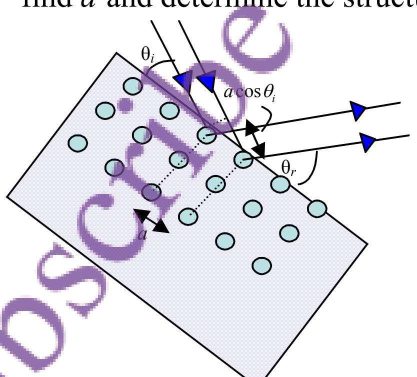


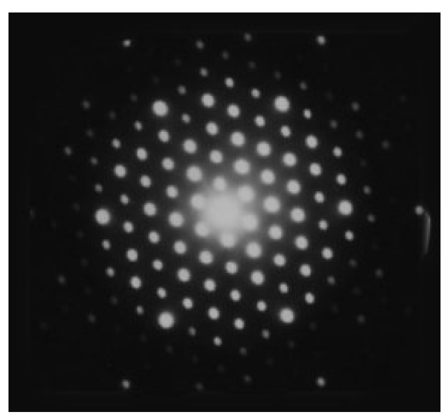


Jab electrons crystal surface se takraye, to scattering surface layers se hoti hai.

10. Let us look at the interference in some detail. When electrons fall on a crystalline surface, the electron scattering is dominated by surface layers. Note that the identical scattering planes are oriented perpendicular to the surface. Looking at the diagram, we can see that constructive interference happens when $\boxed{a(\cos\theta_r-\cos\theta_i)=n\lambda}$. When this condition is satified, there is a maximum intensity spot. This is actually how we find a and determine the structure of crystals.

Is condition se hum crystal structure ka size aur shape maloom karte hain.

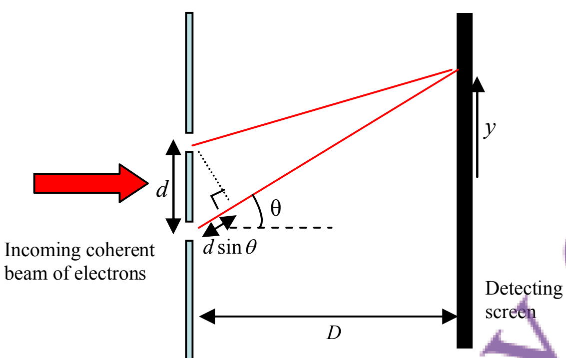


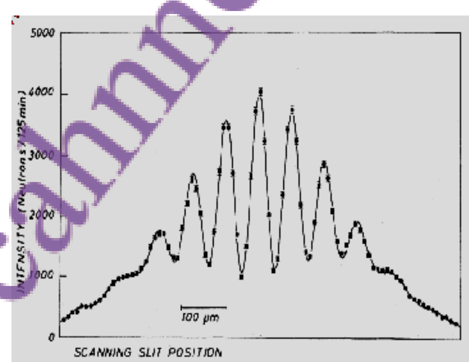


Electron bhi wave ki tarah behave karte hain interference aur diffraction patterns show karte hain

Electrons ko ek metal plate par bheja jata hai jismein do chhote holes (slits) hotay hain. Yeh slits aik doosre ke qareeb hotay hain aur unka distance d hota hai.
Peechay aik screen hoti hai jo electron se takra kar flash karti hai. Jab electrons guzarte hain, to interference pattern (ujalay aur andheray daag) nazar aata hai - just like light waves.

11. Let's take a still simpler situation: electrons are incident upon a metal plate with two tiny holes punched into it. The holes - spearated by distance d - are very close together. A screen behind, at distance D, is made of material that flashes whenever it is hit by an electron. A clear interference pattern with peaks and valleys is observed. Let us analyze: there will be a maximum when $\boxed{d\sin\theta = n\lambda}$. If the screen is very far away, i.e. $D >> d$, then θ will be very small and $\sin\theta \approx \theta$. So we then have $\boxed{\theta \approx \frac{n\lambda}{d}}$ and the angular separation between two adjacent minima is $\Delta\theta \approx \frac{\lambda}{d}$. The position on the screen is y, and $y = D\tan\theta \approx D\theta$. So the separation between adjacent maxima is $\Delta y \approx D\Delta\theta$ and hence $\boxed{\Delta y = \frac{\lambda D}{d}}$. This is the separation between two adjacent bright spots. You can see from the experimental data that this is exactly what is observed.





12. The double-slit experiment is so important that we need to discuss it further. Note the following:

Chahe light ho, electrons ya atoms, sab wave jaise behave karte hain.

a) It doesn't matter whether we use light, electrons, or atoms - they all behave as waves. in this experiment. The wavelength of a matter wave is unconnected to any internal size of particle. Instead it is determined by the momentum, $\lambda = \frac{h}{p}$.

b) If one slit is closed, the interference disappears. So, in fact, each particle goes through both slits at once.

matlab yeh hai ke har particle dono slits se aik hi waqt mein guzarta hai

c) The flux of particles arriving at the slits can be reduced so that only one particle arrives at a time. Interference fringes are still observed !Wave-behaviour can be shown by a single atom. In other words, a matter wave can interfere with itself.

Matlab yeh ke aik particle khud apne aap se interfere karta ha

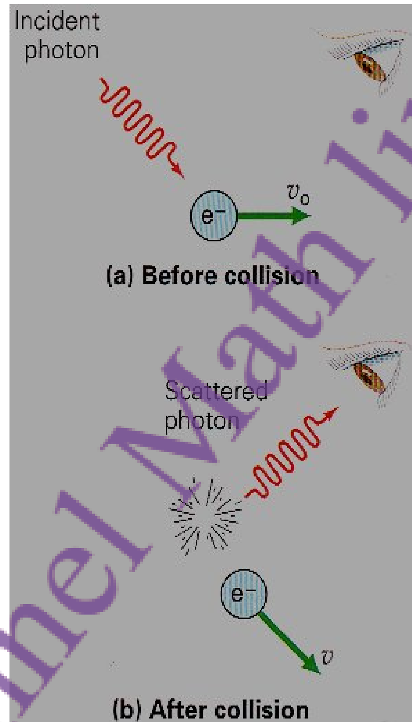
d) If we try to find out which slit the particle goes through the interference pattern vanishes! We cannot see the wave/particle nature at the same time.

Wave aur particle ki nature ko ek saath nahi dekh sakte.

All this is so mysterious and against all our expectations. But that's how Nature is!

Heisenberg (Germany ka mashhoor physicist) ne bataya ke agar hum kisi electron ko dekhna chahte hain to humein usay kisi particle (jaise photon) se takrana parta hai. Jab photon electron se takraata hai, to wo uski position aur velocity ka kuch data to de deta hai, lekin us takraar ki wajah se electron ki state badal jaati hai

13. *Heisenberg Uncertainty Principle.* In real life we are perfectly familiar with seeing a cricket ball at rest - it has both a fixed position and fixed (zero) momentum. But in the microscopic world (atoms, nuclei, quarks, and still smaller distance scales) this is impossible. Heisenberg, the great German physicist, pointed out that if we want to see an electron, then we have to hit it with some other particle. So let's say that a photon hits an electron and then enters a detector. It will carry information to the detector of the position and velocity of the electron, but in doing so it will have changed the momentum of the electron. So if the electron was initially at rest, it will no longer be so afterwards. The act of measurement changes the state of the system! This is true no matter how you do the experiment. The statement of the principle is:



Agar hum kisi particle ki position ko Δx accuracy ke saath jaan lein, to momentum ki maximum accuracy Δp hogi, jahan Δx·Δp ≥ ħ / 2.

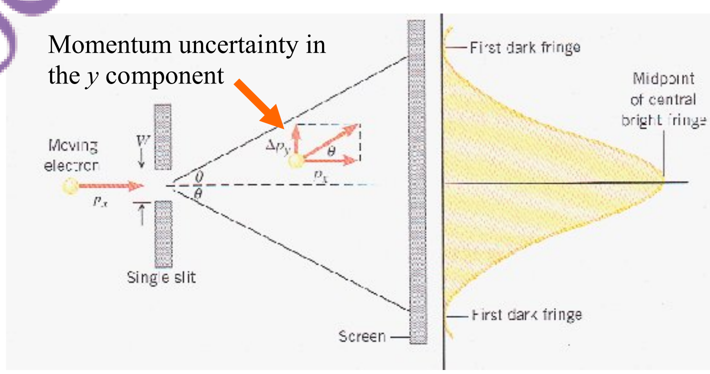
"If the position of a particle can be fixed with accuracy Δx, then the maximum accuracy with which the momentum can be fixed is Δp, where $\Delta x \Delta p \geq \hbar/2$." We call Δx the position uncertainty, and Δp the momentum uncertainty. Note that their product is fixed. Therefore, if we fix the position of the particle (make Δx very small), then the particle will move about randomly very fast (and have Δp very large).

Δx = position uncertainty
Δp = momentum uncertainty
ħ = reduced Planck's constant (h/2π)

Agar hum position ko bohat exact set kar dein (Δx choti), to momentum ka uncertainty bohat bara ho jata hai (Δp bara), aur particle randomly fast move karega.

14. By using the De Broglie hypothesis in a simple gedanken experiment, we can see how the uncertainty principle emerges. Electrons are incident upon a single slit and strike a screen far away. The first dark fringe will be when $W\sin\theta = \lambda$. Since θ is small, we can use $\sin\theta \approx \theta$, and so $\theta \approx \dfrac{\lambda}{W}$. But, on the other hand, $\tan\theta = \dfrac{\Delta p_y}{p_x}$ and $\theta \approx \dfrac{\Delta p_y}{p_x}$. So,

$\dfrac{\Delta p_y}{h/\lambda} = \dfrac{\lambda}{W}$. But W is really the uncertainty in the y position and we should call it Δy.



Agar hum electron ki position y mein narrow kar dete hain (slit se guzarte waqt), to uske momentum py mein uncertainty barh jati hai. Matlab Jitni position zyada exact hogi, utna zyada momentum uncertain hoga.

Thus we have found that $\Delta p_y \Delta y \approx h$. The important point here is that by localizing the y position of the electron to the width of the slit, we have forced the electron to acquire a momentum in the y direction whose uncertainty is Δp_y.

15. In a proper course in quantum mechanics, one can give a definite mathematical meaning to Δx and Δp_x etc, and derive the uncertainty relations:

$$\boxed{\Delta x \Delta p_x \geq \hbar / 2, \quad \Delta y \Delta p_y \geq \hbar / 2, \quad \Delta z \Delta p_z \geq \hbar / 2}$$

Note the following:

Agar x direction mein position aur px momentum ka relation apply hota hai, to y direction ke saath koi uncertainty nahi hoti.

a) There no uncertainty principle for the product $\Delta x \Delta p_y$. In other words, we can know in principle the position in one direction precisely together with the momentum in another direction.

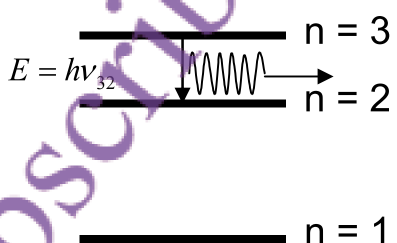
Jab tak measurement nahi ki jati, particle ki position ya momentum define nahi hoti. Measurement hi uncertainty laata hai. Aaj kal maana jata hai ke yeh uncertainty nature ka hissa hai yeh kisi measurement ki kami nahi, balke quantum duniya ki asliat hai.

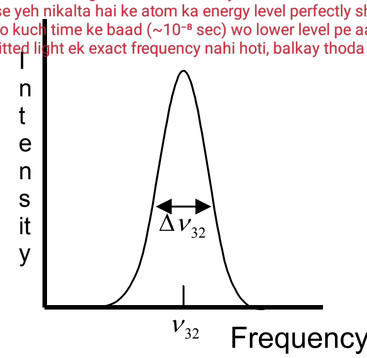
b) The thought experiment I discussed seems to imply that, while prior to experiment we have well defined values, it is the act of measurement which introduces the uncertainty by disturbing the particle's position and momentum. Nowadays it is more widely accepted that quantum uncertainty (lack of determinism) is intrinsic to the theory and does not come about just because of the act of measurement.

16. There is also an Energy-Time Uncertainty Principle which states that $\Delta E \Delta t \geq \hbar / 2$. This says that the principle of energy conservation can be violated by amount ΔE, but only for a short time given by Δt. The quantity ΔE is called the uncertainty in the energy of a system.

17. One consequence of $\Delta E \Delta t \geq \hbar / 2$ is that the level of an atom does not have an exact value. So, transitions between energy levels of atoms are never perfectly sharp in frequency. So, for example, as shown below an electron in the $n = 3$ state will decay to a lower level after a lifetime of order $t \approx 10^{-8}$s. There is a corresponding "spread" in the emitted frequency.

Jitni zyada certainty kisi cheez ke position mein hogi, utni hi uncertainty uske momentum mein hogi aur yeh Nature ka fundamental rule hai. ΔE Δt principle se yeh nikalta hai ke atom ka energy level perfectly sharp nahi hota.
Jaise agar electron n = 3 level par ho, to kuch time ke baad (~10⁻⁸ sec) wo lower level pe aa jata hai (n = 2), aur is transition mein frequency spread hota hai matlab emitted light ek exact frequency nahi hoti, balkay thoda range hoti hai.





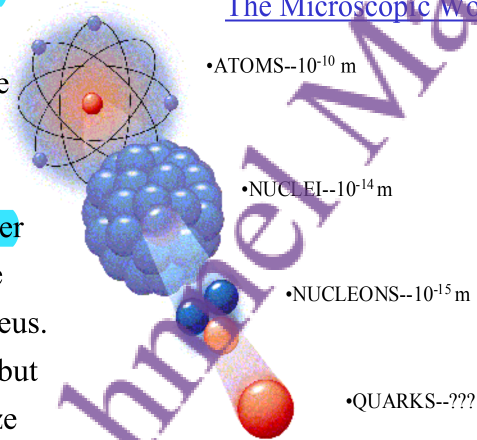
Summary of Lecture 42 – QUANTUM MECHANICS

1. The word "quantum" means packet or bundle. We have already encountered the quantum of light - called photon - in an earlier lecture. Quanta (plural of quantum) are discrete steps. Walking up a flight of stairs, you can increase your height one step at a time and not, for example, by 0.371 steps. In other words your height above ground (and potential energy can take discrete values only).

Is duniya ke sizes ka andaza lagane ke liye hum atom se shuru karte hain, jo bohot chhoti cheez hoti hai. Magar atomic nucleus atom se 100,000 guna chhota hota hai. Neutron aur proton nucleus se bhi 10 guna chhote hote hain. Aur nucleus quarks se milke banta hai, lekin quarks ka size humein nahi pata shayad woh sirf point-like particles hain.

2. Quantum Mechanics is the true physics of the microscopic world. To get an idea of the sizes in that world, let us start from the atom which is normally considered to be a very small object. But, as you can see, the atomic nucleus is 100,000 times smaller than the atom. The neutron and proton are yet another 10 times smaller than the nucleus. We know that nuclei are made of quarks, but as yet we do know if the quarks have a size or if they are just point-like particles.

The Microscopic World



3. It is impossible to cover quantum mechanics in a few lectures, much less in this single lecture. But here are some main ideas:

Classical (Newtonian) Mechanics sirf bade objects ke liye kaam karti hai. Atomic level par yeh fail ho jati hai. Iski wajah uncertainty principle hai kisi particle ki position aur momentum ek sath exact nahi pata kiya ja sakta.

a) Classical (Newtonian) Mechanics is extremely good for dealing with large objects (a grain of salt is to be considered large). But on the atomic level, it fails. The reason for failure is the uncertainty principle - the position and momenta of a particle cannot be determined simultaneously (this is just one example; the uncertainty principle is actually more general). Quantum Mechanics properly describes the microscopic - as well as macroscopic - world and has always been found to hold if applied correctly.

b) Atoms or molecules can only exist in certain energy states. These are also called "allowed levels" or quantum states. Each state is described by certain "quantum numbers" that give information about that state's energy, momentum, etc.

Atoms ya molecules sirf certain energy states mein hote hain jise quantum states kehte hain. In states ko specific quantum numbers se describe kiya jata hai.

c) Atoms or molecules emit or absorb energy when they change their energy state. The amount of energy released or absorbed equals the difference of energies between the two quantum states.

Atoms energy absorb ya emit karte hain jab woh energy states change karte hain.

d) Quantum Mechanics always deals with probabilities. So, for example, in considering the outcome of two particles colliding with each other, we calculate probabilities to scatter in a certain direction, etc.

Quantum Mechanics probabilities ke basis par kaam karti hai har outcome ka chance hota hai, certainty nahi.

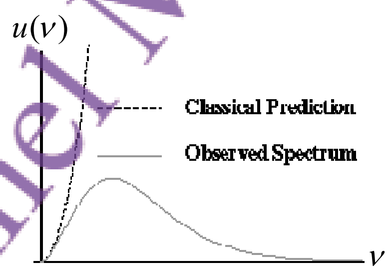
19th century ke aakhir tak kai experimental results classical theory se match nahi kar rahe thay.

4. What brought about the Quantum Revolution? By the end of the 19th century a number of serious discrepancies had been found between experimental results and classical theory. The most serious ones were:
   A) The blackbody radiation law
   B) The photo-electric effect
   C) The stability of the atom, and puzzles of atomic spectra

   In the following, we shall briefly consider these discrepancies and the manner in which quantum mechanics resolved them.

Classical physics kehta tha ke hot body se nikalne wali radiation ka energy har frequency par infinite ho jata hai (jo ghalat hai). Max Planck ne suggest kiya ke energy quantized hoti hai E = hv yani sirf specific amounts mein energy radiate hoti hai.

A) Classical physics gives the wrong behaviour for radiation emitted from a hot body. Although in this lecture it is not possible to do the classical calculation, it is not difficult to show that the electromagnetic energy $u(\nu)$ radiated at frequency ν increases as ν^3 (see graph). So the energy radiated over all frequencies is infinite. This is clearly wrong. The correct calculation was done by Max Planck.



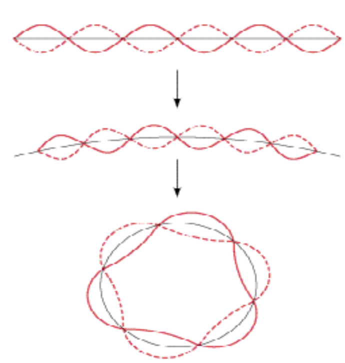
Black Body Spectrum

Planck's result is shown above, and it leads to the sensible result that $u(\nu)$ goes to zero at large ν. He assumed that radiation of a given frequency ν could only be emitted and absorbed in quanta of energy $\varepsilon = h\nu$. If the electromagnetic field is thought of as harmonic oscillators, Planck assumed that the total energy of this large number of oscillators is made of finite energy elements $h\nu$. With this assumption, he came up with a formula that fitted well with the data. But he called his theory "an act of desperation" because he did not understand the deeper reasons.

Einstein (1905) ne yeh theory di ke light photon (quantum of light) ki form mein hoti hai jisme energy aur momentum hota hai. Photon tabhi electron ko metal se nikal sakta hai jab uske paas kafi energy ho.

B) I have already discussed the photoelectric effect in the previous lecture. Briefly, Einstein (1905) postulated a quantum of light called photon, which had particle properties like energy and momentum. The photon is responsible for knocking electrons out of the metal - but only if it has enough energy.

C) Classical physics cannot explain the fact that atoms are stable. An accelerating charge always radiates energy if classical electromagnetism is correct. So why does the hydrogen atom not collapse? In 1921 Niels Bohr, a great Danish physicist, made the following hypothesis: if an electron moves around a nucleus so that its angular momentum is $\hbar, 2\hbar, 3\hbar, \cdots$ then it will not radiate energy. In the next lecture we explore the consequences of this hypothesis. Bohr's hypothesis called for the quantization



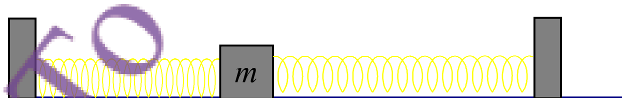
Classical physics yeh explain nahi kar sakti ke atoms stable kyun hote hain. Accelerating electron ko radiation emit karna chahiye (jo atom ko collapse kar deta), lekin aisa hota nahi.
Niels Bohr (1921) ne yeh hypothesis diya ke electron sirf specific orbits mein ghoomta hai jisme angular momentum h, 2h, 3h... hota hai — aur aise electron energy radiate nahi karta.

of angular momentum. If the electron is regarded as a De Broglie wave, then there must be an integer number of waves that go around the centre. Of course, this is not proper quantum mechanics and cannot be taken too seriously, but it definitely was a major step in the ultimate development of the subject. Even if one's understanding was imperfect, it was now possible to understand why atoms had certain energies only, and why the light radiated by atoms was of discrete frequencies only. In contrast, classical physics predicted that atoms could radiate at any frequency.

5. One consequence of quantum mechanics is that it explains through the uncertainty principle the stability of the atom. But before talking of that, let us consider a particle moving between two walls. Each time it hits a wall, a force pushes it in the opposite direction. There is no friction, so classically the particle just keeps moving forever between the two potential walls. In QM the uncertainty of the particle's position is $\Delta x = a$ and so, from $\Delta x \Delta p \geq \hbar/2$, we have $\Delta p\, a \approx \dfrac{\hbar}{2}$. From this we learn that the kinetic energy $\dfrac{(\Delta p)^2}{2m} \approx \dfrac{\hbar^2}{8ma^2}$. This is telling us that as we squeeze the particle into a tighter and tighter space, the kinetic energy goes up and up!

Quantum mechanics ka aik nateeja yeh hai ke yeh uncertainty principle ke zariye atom ki stability ko samjhata hai. Maslan agar ek particle do walls ke darmiyan move karta hai, toh classical physics ke mutabiq yeh particle hamesha move karta rahega. Lekin quantum mechanics mein, position ki uncertainty Δx ≈ a hai, toh momentum ki uncertainty Δp ≈ h/2a ho jaati hai. Isse humein yeh pata chalta hai ke kinetic energy (Δp)²/2m ≈ h²/8ma² hoti hai. Matlab yeh ke jitni tight space mein particle ko band karenge, uski energy utni zyada ho jaayegi.

6. The story repeats for a harmonic oscillator. So imagine that a mass moves in a potential of the type shown below, and that its frequency of oscillation is ω. Classically, the lowest energy would be that in which the mass is at rest (no kinetic energy) and it is at the position where the potential is minimum. But this means that the body has both a



well-defined momentum and position. This is forbidden by the Heisenberg uncertainty principle. A proper quantum mechanical calculation shows that the minimum energy is actually $\frac{1}{2}\hbar\omega$. This is called the zero-point energy and comes about because it is not possible for the mass to be a rest. The oscillator's other energy states have energies $\frac{1}{2}\hbar\omega, \frac{3}{2}\hbar\omega, \frac{5}{2}\hbar\omega, \frac{7}{2}\hbar\omega, \cdots$ We say that the energies are quantized. This is completely impossible to understand from classical mechanics where we know that we can excite an oscillator to have any energy we want. Quantized energy levels are indeed what we observe in so many physical systems: atoms, rotating or vibrating molecules, nuclei, etc.

Classical physics ke mutabiq, agar ek mass rest mein ho aur potential energy minimum ho, toh uski energy zero honi chahiye. Lekin quantum mechanics ke mutabiq, Heisenberg uncertainty principle ki wajah se aisa nahi ho sakta. Mass kabhi bilkul rest mein nahi ho sakta. Minimum energy hoti hai ½ħω. Isay hum zero-point energy kehte hain. Baaki energy levels quantized hote hain, jaise: ½ħω, 3/2ħω, 5/2ħω... Matlab yeh ke energy sirf kuch specific levels par hi ho sakti hai.

Hum pichlay examples se yeh samajh saktay hain ke hydrogen atom collapse kyun nahi karta, chahe electron ka orbital angular momentum na bhi ho. Uncertainty principle electron ko proton ke kareeb aanay se rokti hai, kyunke agar Δx choti ho jaaye to Δp barh jaata hai, jiska matlab hai ke kinetic energy bohot zyada ho jaayegi.

7. We can understand from both the previous examples why the hydrogen atom does not collapse even if the electron does not have any orbital angular momentum around the proton. The uncertainty principle essentially forces the electron to stay away from the proton - if it tried to get too close, the kinetic energy would rise enormously because $\Delta x \Delta p \geq \hbar / 2$ says that if Δx becomes small then Δp must become big to compensate.

8. Quantum mechanics predicts what is called "tunneling" of particles through a potential barrier. Again, we shall use the uncertainty principle but this time the energy-time one. Imagine a particle that moves in a potential of the shape shown below. Suppose it does not have enough energy to go over the peak and on to the other side. In that case, we

Quantum mechanics aik aur ajeeb cheez predict karta hai jise kehte hain tunneling yani particles ka potential barrier ko paar karna, jabke unke paas itni energy bhi nahi hoti. Yeh uncertainty principle ka energy-time version use karta hai. Agar aik particle hill ke neeche phans gaya hai aur uske paas energy nahi ke woh hill ke upar jaaye, toh bhi quantum mechanics ke mutabiq woh kuch energy steal karke thodi dair ke liye barrier cross kar sakta hai. Isay tunneling kehte hain.

know from our experience that it will just keep oscillating - moving first towards the hill and then down again, etc. But quantum mechanically, it can "steal" energy ΔE for a time Δt and this may be enough to surmount the hill. Of course, the particle must respect $\Delta E \Delta t \geq \hbar / 2$ so the time is small if it needs a large amount of energy to cross over. Again, I have given only a rough argument here, but in quantum mechanics we can do proper calculations to find tunneling probabilities.

Agar yeh tunneling na hoti to hamara sooraj thanda ho jaata. Sooraj hydrogen fusion ke zariye chal raha hai, jisme protons ko ek dosray ke kareeb aana padta hai. Lekin normal temperature mein energy kaafi nahi hoti. Sirf tunneling ki wajah se yeh protons nuclear force ko mehsoos karte hain aur fusion hota hai.

9. Without tunneling our sun would go cold. As you may know, it is powered by hydrogen fusion. The protons must somehow overcome electrostatic repulsion to get close enough so that they can feel the nuclear force and fuse with each other. But the thermal energy at the core of the sun is not high enough. It is only because the tunneling effect allows protons to sometimes get close enough that fusion happens.

Quantum mechanics probabilities par based hai. Probability ka matlab hota hai kisi cheez ke honay ka chance. Agar hum koi experiment N dafa karein, aur koi khaas natija n dafa aaye, toh uski probability hoti hai P = n/N.

10. Quantum mechanics is all about probabilities.What is probability? Probability is a measure of the likelihood that an event will occur. Probability values are assigned on a scale of zero (the event can never occur) to one (the event definitely occurs). More precisely, suppose we do an experiment (like rolling dice or flipping a coin) N times where N is large. Then if a certain outcome occurs n times, the probability is $P = n / N$.

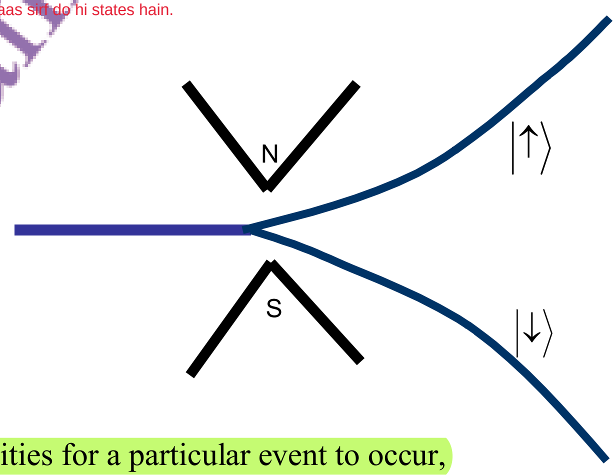
Quantum mechanics mein sab se simple system wo hota hai jisme sirf do states hoti hain up aur down Yeh kisi electron ke spin up/down ko represent kar sakti hain.

11. The simplest system for discussing quantum mechanics is one that has only two states. Let us call these two states "up" and "down" states, $|\uparrow\rangle$ and $|\downarrow\rangle$. They could denote an electron with spin up/down, or a switch which is up/down, or an atom which can be only in one of two energy states, etc. Things like $|\uparrow\rangle$ were called kets by their inventor, Paul Dirac.

For definiteness let us take the electron example. If the state of the electron is known to be $|\Psi\rangle = \sqrt{\frac{2}{3}}|\uparrow\rangle + \sqrt{\frac{1}{3}}|\downarrow\rangle$, then the probability of finding the electron with spin up is $P(\uparrow) = \frac{2}{3}$, and with spin down is $P(\downarrow) = \frac{1}{3}$. More generally: $|\Psi\rangle = c_1|\uparrow\rangle + c_2|\downarrow\rangle$ denotes an electron with $P(\uparrow) = |c_1|^2$ and $P(\downarrow) = |c_2|^2$. This means that if we look at look at a large number of electrons N all of which are in state $|\Psi\rangle$, then the number with spin up is $N|c_1|^2$ and with spin down is $N|c_2|^2$. We sometimes call $|\Psi\rangle = c_1|\uparrow\rangle + c_2|\downarrow\rangle$ a *quantum state*, and c_1 and c_2 *quantum amplitudes*.

Stern-Gerlach experiment mein electron ko magnetic field se guzara jata hai. Field electron ko majboor karti hai ke woh sirf do states mein se kisi ek ko chune ya to up ya down. Beam sirf do hisso mein split hoti hai, is ka matlab hai electron ke paas sirf do hi states hain.

12. The Stern-Gerlach experiment illustrated here shows an electron beam entering a magnetic field. The electrons can be pointing either up or down relative to any chosen axis. The field forces them to choose one of the two states. That the beam splits into only two parts shows that the electron has only two states. Other particle beams might split into $3, 4, \cdots$

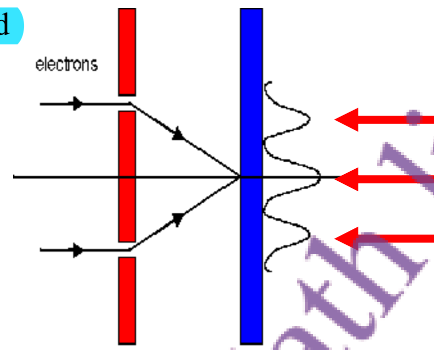


13. If a_1 and a_2 are the amplitudes of the two possibilities for a particular event to occur, then the amplitude for the total event is $A = a_1 + a_2$. Here a_1 and a_2 are complex numbers in general. But the probability for the event to occur is given by $P = |A|^2 = |a_1 + a_2|^2$. In daily experience we add probabilities, $P = P_1 + P_2$ but in quantum mechanics we add amplitudes: $P = |a_1 + a_2|^2 = a_1^* a_1 + a_2^* a_2 + a_1^* a_2 + a_2^* a_1 = P_1 + P_2 + a_1^* a_2 + a_2^* a_1$. The cross terms $a_1^* a_2 + a_2^* a_1$ are called interference terms. They are familiar to us from the lecture on light where we add amplitudes first, and then square the sum to find the intensity. Of course, if we add all possible outcomes then we will get 1. So, for example, in the electron case $P(\uparrow) + P(\downarrow) = |c_1|^2 + |c_2|^2 = 1$. Note that amplitudes can be complex but probabilities are always real.

amplitudes complex ho sakte hain, lekin probabilities hamesha real hoti hain.

Yahan electron wave jo ek slit se aati hai, doosri slit se aane wali wave ke sath interfere karti hai. Yeh interference kabhi destructive hota hai (jahaan electron bilkul nahi milte) aur kabhi constructive (jahaan bohot sare electron milte hain). Yeh pattern wave nature ki wajah se hota hai. Is liye humein matter waves ko mathematically samajhna hoga.

14. Let us return to the double slit experiment discussed earlier. Here the amplitude for an electron wave coming from one slit interferes with the amplitude for an electron wave coming from the other slit. This is what causes a pattern to emerge in which electron are completely absent in certain places (destructive interference) and are present in large numbers where there is constructive interference. So what we must deal with are matter waves. But how to treat this mathematically?



1926 mein Schrödinger ne ek quantity propose ki jo electron ya matter waves ko describe karti hai.

15. The above brings us to the concept of a "wave function". In 1926 Schrödinger proposed a quantity that would describe electron waves (or, more generally, matter waves).

- The wavefunction $\Psi(x,t)$ of a particle is the amplitude to be at position x at time t.

Wave function Ψ(x, t) yeh batata hai ke particle kisi waqt t par position x par honay ki kya amplitude rakhta hai.

- The probability of finding the particle at position x between x and $x+dx$ (at time t) is $|\Psi(x,t)|^2\,dx$. Since the particle has to be somewhere, if we add up all possibilities then we must get one, i.e. $\int_{-\infty}^{+\infty}|\Psi(x,t)|^2\,dx=1$.

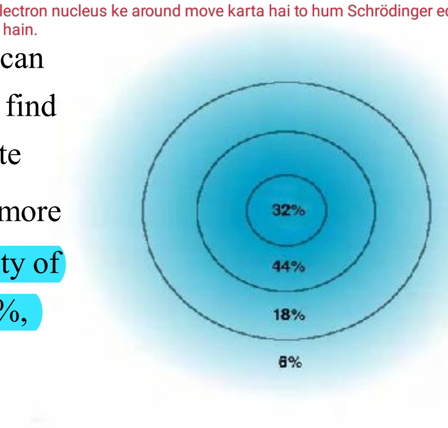
particle kahin na kahin zarur hota hai, is liye agar hum sab possibilities ko add karein to total 1 hona chahiye.

Ψ(x, t) ko Schrödinger equation solve karke hasil kiya jata hai. Yeh physics ki sab se ahem equations mein se ek hai. Agar atom ke liye isay solve kar liya jaye to sab kuch pata lag sakta hai: energy levels, electron kis jagah mil sakta hai, uska momentum kya hoga, waghera.

- $\Psi(x,t)$ is determined by solving the "Schrodinger equation" which, unfortunately, I shall not be able to discuss here. This is one of the most important equations of physics. If it is solved for the atom then it tells you all that is possible to know: energies, the probability of finding an electron here or there, the momenta with which they move, etc. Of course, one usually cannot solve this equation in complicated situations (like a large molecule, for example) and this is what makes the subject both difficult and interesting.

Jab electron nucleus ke around move karta hai to hum Schrödinger equation ko solve kar ke Ψ(x,t) nikaal sakte hain.

16. For an electron moving around a nucleus, one can easily solve the Schrodinger equation and thus find the wavefunction $\Psi(x,t)$. From this we compute $|\Psi(x,t)|^2$, which is large where the electron is more likely to be found. In this picture, the probability of finding the electron inside the first circle is 32%, between the second and first is 44%, etc.



Taqreeban 2500 saal pehle aik Greek philosopher Democritus ne yeh sawal uthaya: Duniya kis cheez se bani hai? Us ne kaha ke sab kuch choti choti particles yani atoms se bana hai. 300 saal pehle, French chemist Lavoisier ne dekha ke kisi bhi chemical reaction mein reactants ka total mass reaction ke baad bhi utna hi rehta hai. Us ne yeh demonstrate kiya ke lakri ko jalaane ke bawajood mass mein koi farq nahi aata. Isay kehte hain: Law of Conservation of Matter

Summary of Lecture 43 – INTRODUCTION TO ATOMIC PHYSICS

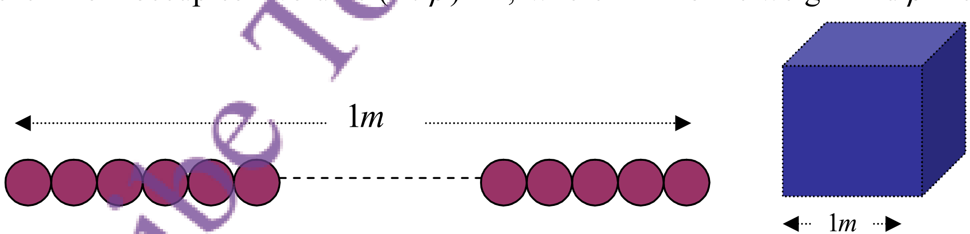
1. • About 2500 years ago, the ancient Greek philosopher Democritus asked the question: what is the world made of? He conjectured that it is mostly empty, and that the remainder is made of tiny "atoms". By definition these atoms are indivisible.
   • Then 300 years ago, it was noted by the French chemist Lavoisier that in all chemical reactions the total mass of reactants before and after a chemical reaction is the same. He demonstrated that burning wood caused no change in mass. This is the Law of Conservation of Matter.
   • A major increase in understanding came with Dalton (1803) who showed that:
      1) Atoms are building blocks of the elements. Atoms elements ke building blocks hain.
      2) All atoms of the same element have the same mass. Har element ke atoms ka mass aik jaisa hota hai.
      3) Atoms of different elements are different. Har element ke atoms doosre elements se farq hotay hain.
      4) Two or more different atoms bond in simple ratios to form compounds.

Do ya zyada atoms simple ratio mein mil kar compounds banate hain.

2. Avogadro made the following hypothesis : "Equal volumes of all gases, under the same conditions of temperature and pressure, contain equal numbers of molecules". Why? Because we know that pressure is caused by molecules hitting the sides of the containing vessel. If the temperature of two gases is the same, then their molecules move with the same speeds, and so Avogadro's hypothesis follows for ideal gases. The famous number $N_0 = 6.023\times 10^{26}$ per kilogram-mole is called Avogadro's Number.

"Har gas ka equal volume, agar temperature aur pressure same ho, to us mein molecules ki tadaad bhi equal hogi."
Agar do gases ka temperature barabar ho, to unke molecules ki speed bhi barabar hoti hai. Isi liye har ideal gas ka volume barabar hone par usmein molecules ki tadaad bhi barabar hoti hai.

3. Let's get an idea of the size of atoms. Amazingly, we do not need high-powered particle accelerators to do so. Consider a cube of $1m\times 1m\times 1m$. If the radius of an atom is r, then we have $(1/2r)^3$ atoms in the cube. Now in 1kg.atom we have $N_0 = 6\times 10^{26}$ atoms and each atom occupies a volume (A/ρ) m^3, where $A =$ atomic weight and $\rho =$ density.

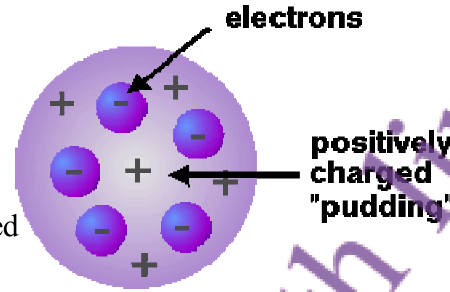


Hence $N_0 = (1/2r)^3 \times A/\rho$. This give $r = \dfrac{1}{2}\left(\dfrac{A}{\rho N_0}\right)^{1/3}$. Putting in some typical densities, we find that $r_{Ag} \approx r_{Be} \approx 10^{-10} m$. This shows that atoms are mostly of the same size. This is quite amazing because one expects a Be atom to be much smaller than an Ag atom.

Atoms ka size estimate karne ke liye heavy instruments ki zarurat nahi.

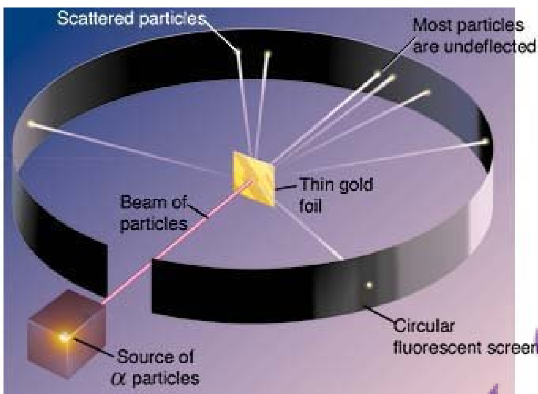
Agar aapko yeh pata bhi ho ke atom kitna bara hota hai, iska yeh matlab nahi ke uska internal structure bhi maloom hai. 1895 mein J.J. Thomson ne plum pudding model propose kiya. Is model ke mutabiq atom ek positively charged material ka gola tha jisme negatively charged electrons idhar udhar scattered hote hain.

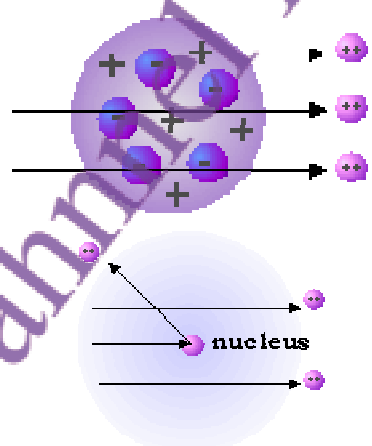
4. Even if you know how big an atom is, this does not mean that its internal structure is known. In 1895 J.J.Thomson proposed the "plum pudding" model of an atom. Here the atom is considered as made of a positively charged material with the negatively charged electrons scattered through it.



5. But the plum-pudding model was wrong. In 1911, Rutherford carried out his famous experiment that showed the existence of a small but very heavy core of the atom. He arranged for a beam of α particles to strike gold atoms in a thin foil of gold.

1911 mein Rutherford ne ek mashhoor experiment kiya jisme usne gold foil par alpha particles ka beam maara. Usne dekha ke zyada tar alpha particles seedha guzar gaye, lekin kuch deflect hue aur kuch wapas bhi chale gaye. Iska matlab tha ke atom ke andar koi bohot chhoti aur bohot heavy cheez hai jise hum nucleus kehte hain.

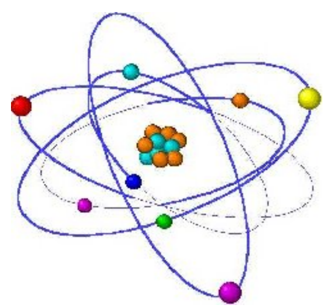




If the positive and negative charges in the atom were randomly distributed, all α's would go through without any deflection. But a lot of backscattering was seen, and some α's were even deflected back in the direction of the incident beam. This was possible only if they were colliding with a very heavy object inside the atom. Rutherford had discovered the atomic nucleus.

atom mein zyada space khaali hai, aur beech mein chhota sa positive nucleus hota hai jisme protons hote hain. Electrons uske gird orbits mein ghoomte hain, bilkul planets ki tarah jo sun ke gird ghoomte hain. Unhein coulomb force nucleus ki taraf khenchti hai.

6. The picture that emerged after Rutherford's discovery was like that of the solar system - the atom was now thought of as mostly empty space with a small, positive nucleus that contained protons. Negative electrons moved around the outside in orbits hat resembled those of planets, attracted towards the centre by a coulomb force.

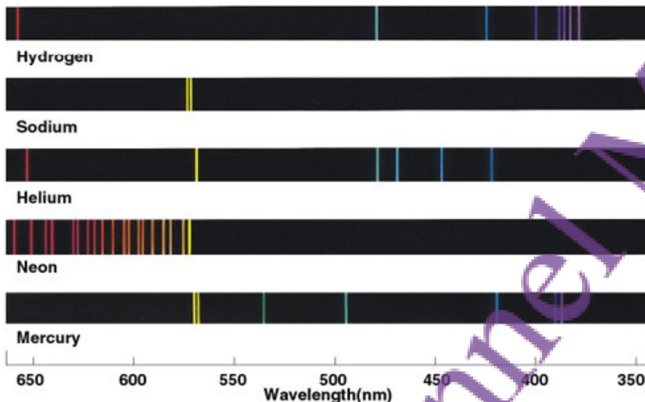


7. This sounds fine, but there is a serious problem: we know that a charge that accelerates radiates energy. In fact the power radiated is $P \propto e^2 a^2$, where e is the charge and a is the acceleration. Now, a particle moving in a circular orbit has an acceleration even if it is moving at constant speed because it is changing its direction all the time. So this means

Yeh model to theek lagta hai, lekin ismein ek badi problem hai: jab koi charge accelerate karta hai to wo energy radiate karta hai. Circular motion mein bhi particle constantly direction change karta hai, is liye wo accelerate karta hai aur energy lose karta hai.

that the electron will be constantly radiating power and thus will slow down, collapsing eventually into the nucleus.

8. This is not the only thing wrong with the solar system model. If you look at the light emitted by any atom, you do not see a continuous distribution of colours (frequencies). Instead, a spectroscope will easily show that light is emitted at only certain discrete frequencies.

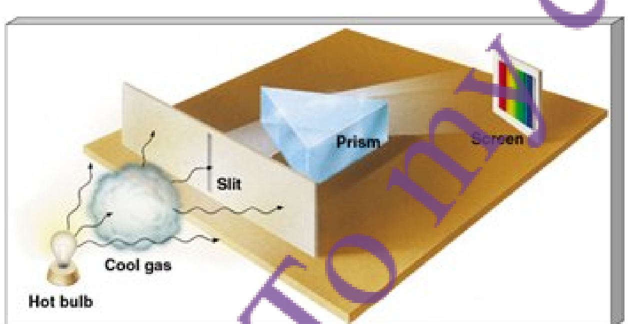


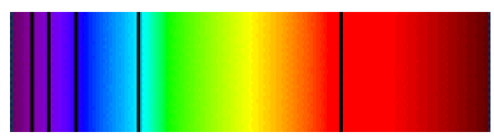
Sirf yeh nahi ke solar system model mein electron nucleus mein gir jata, balkay ek aur masla bhi hai. Agar aap kisi atom se emit hone wali roshni dekhein, to woh ek continuous range mein nahi hoti. Balkay spectroscope yeh dikhata hai ke roshni sirf kuch specific frequencies par hi emit hoti hai.
image mein mukhtalif atoms (Hydrogen, Sodium, Helium, Neon, Mercury) ka emission spectrum dikhaya gaya hai har ek atom sirf kuch khas wavelengths par hi roshni nikalta hai.

In the above you see the emission spectrum of different atoms.

9. Similarly, if white light is passed through a gas of atoms of a certain type, only certain colours are absorbed, and the others pass through without a hindrance.

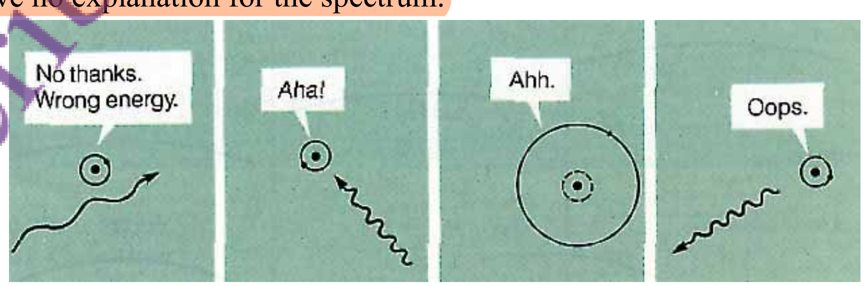
Isi tarah agar aap white light ko kisi specific gas (atoms ke gas) mein se guzarte hain, to sirf kuch rang absorb hote hain aur baqi as it is guzarte hain.





Upar wali image mein absorption spectrum dikhaya gaya hai. Emission aur absorption lines bilkul barabar hoti hain. Classical physics aur Rutherford model is cheez ko explain nahi kar sakte.

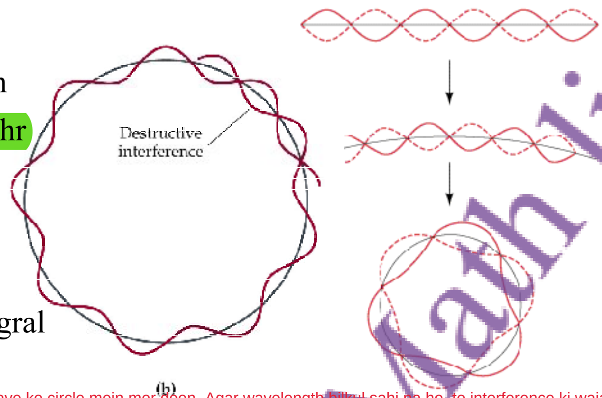
The above shows the absorption spectrum of a certain atom. The wavelengths for both emission and absorption lines are exactly equal. Classical physics and the Rutherford model have no explanation for the spectrum.



atom sirf tabhi energy absorb karta hai jab wo exact sahi amount ho warna wo photon reject kar deta hai

De Broglie relation aur Davisson-Germer experiment se maloom ho gya k electrons ka dual character hai yani woh wave bhi hain

10. Then came Niels Bohr. By this time it was known that electrons had a dual character as waves (De Broglie relation and Davisson-Germer experiment). Bohr said: suppose I bend a standing wave into a circle. If the wavelength is not exactly correct, wave interference will make the wave disappear. So only integral numbers of wavelengths can interfere constructively.



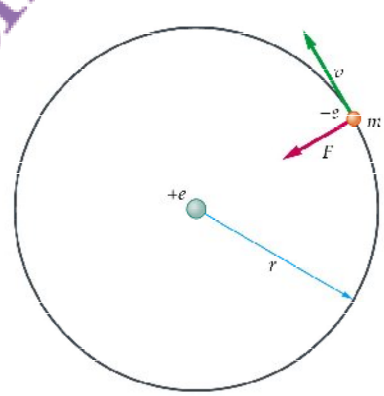
socho agar main ek standing wave ko circle mein mor doon. Agar wavelength bilkul sahi na ho, to interference ki wajah se wave khatam ho jaayegi. Sirf wohi waves constructively interfere karengi jinki wavelength integral multiples ho.

11. Let us pursue this idea further. The electron has a wavelength and forms standing waves in its orbit around the nucleus. An integral number of electron wavelengths must fit into

Electron ki ek wavelength hoti hai aur woh nucleus ke gird standing wave banaata hai. Ek integral number of wavelengths circular orbit ke circumference mein fit honi chahiye.

the circumference of the circular orbit. Hence $n\lambda = 2\pi r$ with $n = 1,2,3\cdots$ The momentum is $\boxed{p = m\text{v} = \dfrac{h}{\lambda} = \dfrac{h}{(2\pi r/n)} = \dfrac{n\hbar}{r}}$. The angular momentum $L = rm\text{v} = n\hbar$ is therefore quantized in units of $\hbar$.

12. Now let us suppose that the electron moves in an orbit of radius r when it has $L_n = n\hbar$. Equilibrium demands that the centrifugal force be equal to the coulomb attraction: $\dfrac{m\text{v}_n^2}{r} = k\dfrac{e^2}{r^2}$. From $\text{v}_n = \dfrac{n\hbar}{mr_n}$ we find that the radius $r_n = \left(\dfrac{\hbar^2}{mke^2}\right)n^2$.



- For $n = 1$ the electron orbit which is closest to the nucleus, $r_n = a_0 \equiv 0.53\times 10^{-8}\,cm$ (this is called the Bohr radius).
- For higher n, $r_n = a_0 n^2$. The atom becomes huge for n ≈ 100, the so-called Rydberg atom. Such atoms are of experimental interest these days.
- Note that the speed of the electron is smaller in orbits farther from the nucleus, $\text{v}_n = \dfrac{ke^2}{n\hbar}$. As n becomes very large, the electron is very far out and very slow.

Agar n=> ∞ ho, to electron nucleus se bohat door hota hai.

- In the above $n = 0$ is strictly not allowed. As you can see, none of the formulae make any sense for this case. The minimum angular momentum that the electron can have is $\hbar$. (In proper quantum mechanics the minimum is $0\hbar$ and this is a big difference with the Bohr model of the atom).

n = 0 allowed nahi hota, warna koi bhi formula kaam nahi karega. Minimum angular momentum h ho sakta hai, zero nahi.

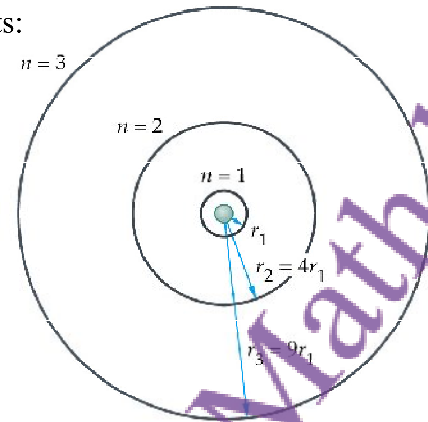
13. We can compute the energies of the various orbits:

$$E = K + U = \tfrac{1}{2}mv^2 + U$$

$$= \tfrac{1}{2}\left(\frac{ke^2}{r}\right) - \frac{ke^2}{r} = -\frac{ke^2}{2r}$$

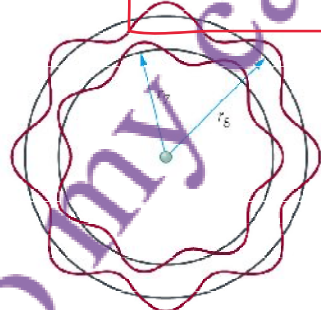
$$\text{Hence, } E_n = -\left(\frac{ke^2}{2}\right)\left(\frac{mke^2}{\hbar^2}\right)\frac{1}{n^2}$$

$$= -\left(\frac{mk^2e^4}{2\hbar^2}\right)\frac{1}{n^2} = \boxed{-\frac{13.6\ \text{eV}}{n^2}}$$



Bohr model ke mutabiq electron ek orbit se doosri orbit mein jump kar sakta hai, agar photon absorb ya emit kare.
Absorption spectra mein dark lines tab hoti hain jab photons absorb hote hain, aur emission spectra mein bright lines tab hoti hain jab photons emit hote hain.
Photon ki energy hoti

14. In the Bohr model, electrons can jump between different orbits due to the absorption or emission of photons. Dark lines in the absorption spectra are due to photons being absorbed, and bright lines in the emission spectra are due to photons being emitted. The energy of the emitted or absorbed photon is equal to the difference of the initial and final energy levels, $h\nu = E_f - E_i$. The picture below shows the electron in the $n = 7$ and $n = 8$ levels. The photon emitted has $\boxed{h\nu = E_8 - E_7 = -13.6\left(\frac{1}{8^2} - \frac{1}{7^2}\right) \text{ ev.}}$



(a)

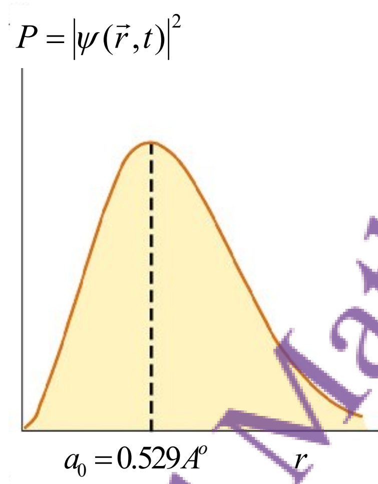
Bohr model hydrogen ke spectrum ke liye achhe results deta tha, aur yeh pehla indication tha ke atomic level par koi new physics zaroori hai.

15. The Bohr model gave wonderful results when compared against the hydrogen spectrum. It was the among the first indications that some "new physics" was needed at the atomic level. But this model is not to be taken too seriously - it fails to explain many atomic properties, and fails to explain why the H atom can exist even when the electron has no orbital angular momentum (and hence no centrifugal force to balance against the Coulomb attraction). It cannot predict all the lines observed for H, much less for multi-electron atoms such as Oxygen. The real value of this model was that it showed the way forward towards developing quantum mechanics, which is the true physics of the world, both microscopic and macroscopic. I have discussed some elements of QM in the last lecture, in particular the wavefunction $\psi(\vec{r},t)$ of the electron.

Lekin is model ko zyada seriously nahi lena chahiye Qk
Yeh bahut si atomic properties ko explain nahi karta
Yeh nahi samjha sakta ke H atom bina orbital angular momentum ke kaise stable rehta hai
Multielectron atoms (jaise Oxygen) ke lines predict nahi kar sakta

Quantum mechanics ek mukhtalif tasveer deta hai mukable mein solar system model ke. Hum Schrödinger Equation ko solve karte hain taake wavefunction ψ(r,t) mil jaye, jiska square hume yeh batata hai ke electron kisi r point par milne ka kitna chance hai.
Lowest energy state mein, electron ek gol cloud ki shakal mein hota hai jo nucleus ke ird gird hota hai. Jahan cloud ghana hota hai, wahan electron milne ka chance zyada hota hai.

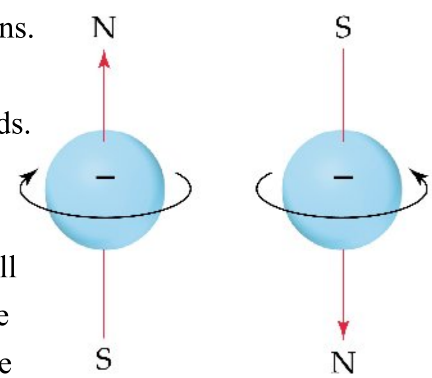
16. Quantum mechanics gives a picture that is quite different from the solar-system model. We solve the "Schrodinger Equation" to find the wavefunction $\psi(\vec{r},t)$, whose square gives the probability of finding the electron at the point $\vec{r}$. In the lowest energy state, the electron can be viewed as a spherical cloud surrounding the nucleus. The densest regions of the cloud represent the highest probability for finding the electron.



- The principal quantum number $n = 1,2,3\cdots$ determines the allowed energy levels. An electron can only have energy $E_n = -\frac{13.6}{n^2}$ eV. Miraculously this is the same result as in the Bohr model.
- The orbital angular momentum is determined by the number l, and $L = \sqrt{l(l+1)}\hbar$. Allowed values of l are, $l = 0,1,2,3\cdots,n-1$.
- The magnetic quantum number m_l determines the projection (or component) of the vector angular momentum $\vec{L}$ on to any fixed axis, $L_z = m_l\hbar$. Allowed values are, $m_l = -l,\cdots -2,-1,0,1,2,\cdots,l$.

Electrons ko chhoti ghoomnay wali charge wali ballon ki tarah samjha ja sakta hai. Har electron same angular momentum ke sath ghoomta hai (value ħ/2 hoti hai).
Electron sirf 2 directions mein spin kar sakta hai. Ghoomta hua charge current create karta hai, aur current magnetic field deta hai. Isi wajah se electron chhoti magnet ki tarah behave karta hai.

17. Electrons can be thought of as little spinning balls of charge. All electrons spin at the same speed (more accurately, their spin angular momentum is the same and equals $\hbar/2$). An electron can spin in only one of two possible directions. When charges move around in a circle, that constitutes a current. As you know, currents give rise to magnetic fields. This is why electrons are also little magnets that interact with other magnets. Now go back to the Stern-Gerlach experiment described in the previous lecture, and you will understand better why the electrons were deflected by the applied magnetic field. Now that we have learned that the electron has spin, we can describe the two spin states by giving the "magnetic" quantum number m_s where $m_s = -\frac{1}{2},\frac{1}{2}$. These two states have the same energy except when there is some magnetic field present.



18. States (or orbitals) having $l = 0,1,2,\cdots$ are called s,p,d,$\cdots$. The s-states are spherical. As n increases, the s-orbitals get larger and the wavefunction is larger away from the nucleus.

hum aasanai se ek jese particles ko pehchan lete hain (jaise ke same mass, charge, spin waghera). Lekin Quantum Mechanics (QM) mein, unki pehchan confuse ho sakti hai aur identical particles ko differentiate karna mushkil hota hai.Sochiye ke A aur B do particles hain jo har tarah se bilkul same hain, aur hum unki jagah badal dete hain. Probability of finding A or B koi farq nahi padta.

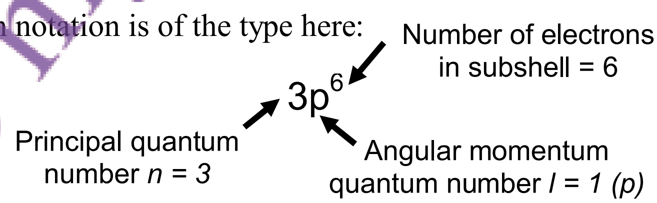
19. In the world we are used to, we can always tell apart identical particles (same mass, charge, spin,...) by simply watching them. But in QM, their idetities can get confused and identical particles are indistinguishable. Suppose that A and B are particles that are identical in every possible way, and we exchange them. Of course, the probability of finding one or the other must remain unchanged. In other words, $|\Psi(1,2)|^2 = |\Psi(2,1)|^2$, where 1 and 2 denote the positions of the first and second particles. But something very interesting happens now because either one of two possibilities can be true: $\Psi(1,2) = +\Psi(2,1)$ or $\Psi(1,2) = -\Psi(2,1)$. Particles obeying the first are called *bosons*, while those obeying the second are called *fermions*. What if we bring two fermions to same point in space? Then: $\Psi(1,1) = -\Psi(1,1)$. This means that $\Psi(1,1) = 0$! In other words, two identical fermions will never be at the same point or in the same quantum state. This is the famous Pauli Exclusion Principle.

matlab do fermions kabhi ek hi point ya quantum state mein nahi ho sakte. Yeh famous Pauli Exclusion Principle hai.

20. Let us apply the Pauli Exclusion Principle to the multi-electron atom where each electron has the quantum numbers $\{n, l, m_l, m_s\}$. Only one electron at a time may have a particular set of quantum numbers. Now for some definitions:

- Shell - electrons with the same value of n
- Subshell - electrons with the same values of n and l
- Orbital - electrons with the same values of n, l, and m_l

Once a particular state is occupied, other electrons are excluded from that state. The electron configuration is how the electrons are distributed among the various atomic orbitals in an atom. A common notation is of the type here:

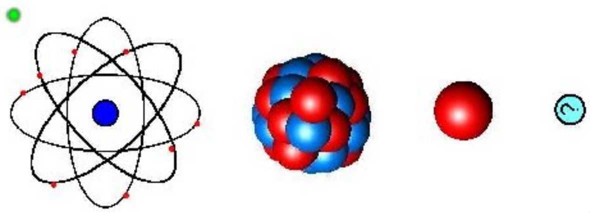


21. Building the shell structure of multi-electron atoms through $n = 4$ using the Pauli Principle.

n	Possible Values of l	Subshell Designation	Possible Values of m_l	Number of Orbitals in Subshell	Total Number of Orbitals in Shell
1	0	1s	0	1	1
2	0	2s	0	1	
	1	2p	1, 0, −1	3	4
3	0	3s	0	1	
	1	3p	1, 0, −1	3	
	2	3d	2, 1, 0, −1, −2	5	9
4	0	4s	0	1	
	1	4p	1, 0, −1	3	
	2	4d	2, 1, 0, −1, −2	5	
	3	4f	3, 2, 1, 0, −1, −2, −3	7	16

Summary of Lecture 44 – INTRODUCTION TO NUCLEAR PHYSICS

Q.1 In the previous lecture you learned how it was discovered that the atom is mostly empty space with a cloud of electrons. At the centre is a small but very heavy nucleus that has protons and neutrons. The word "nucleon" refers to both of these. So you can think of



the neutron or proton as being two different varieties of the nucleon. The masses of the two are very similar, and they are roughly 2000 times heavier than the electron.

$$\text{proton mass} = M_p = 1.672\times10^{-27}\,kg$$

$$\text{neutron mass} = M_n = 1.675\times10^{-27}\,kg$$

$$\text{electron mass} = M_e = 9.109\times10^{-31}\,kg$$

The neutron is neutral, of course, but the charge on the proton is $1.6\times10^{-19}\,C$ while the charge on the electron is the negative of this, $-1.6\times10^{-19}\,C$.

2. Using kilograms is very awkward if you are dealing with such small particles. Instead we we use $E = mc^2$ to write the mass of a particle in terms of its rest energy, $m = E/c^2$. So mass is measured in units of MeV/c^2.

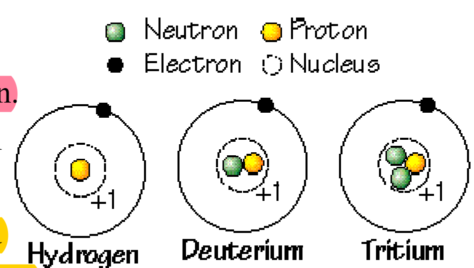
$$\text{proton mass} = M_p = 938\ MeV/c^2$$

$$\text{neutron mass} = M_n = 940\ MeV/c^2$$

$$\text{electron mass} = M_e = 0.5\ MeV/c^2$$

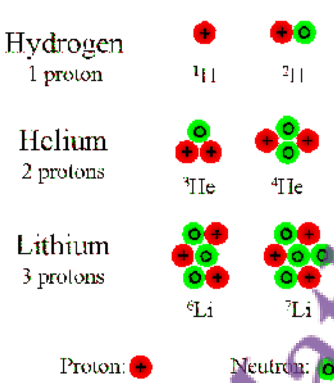
3. a) A hydrogen nucleus is just one proton.
   b) A deuteron has a proton plus one neutron.
   c) A triton (or tritium nucleus) has a proton plus two neutrons.

   All three atoms have one electron only, and thus completely identical chemical properties.

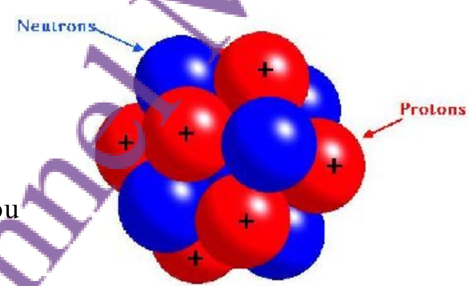


4. Hydrogen, deuterium, and tritium are called isotopes. If a nucleus with Z protons has N neutrons then its isotopes will have fewer, or more, neutrons. Since the number of electrons is also Z, the chemical properties of all isotopes are exactly the same. But for any given element, at most there is only one stable isotope.

4. A commonly used notation is ${}^{A}_{Z}X$ where X is the element, and $A = Z + N$. Of the elements that you see on the right, the most stable ones are ^{1}H, ^{3}He, and ^{6}Li. Now consider oxygen. The most stable isotope is ^{16}O. When you breathe in oxygen from the atmosphere, 99.8% is ^{16}O, 0.037% is ^{17}O, and 0.163% is ^{18}O. We shall see later what unstable means and how nuclei decay.



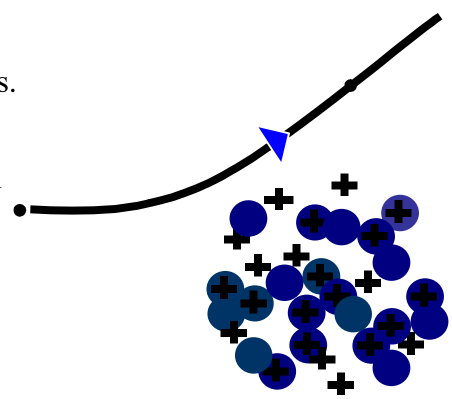
5. The diameter of the nucleus is about 10 million times smaller that the overall diameter of the atom. Nuclei follow an approximate rule for the radius, $\boxed{r \approx r_0 A^{1/3}}$ where $r_0 = 1.2\ fm$ (remember, 1 fermi $= 10^{-13}$ cm) and $A = Z + N$. Now, $A^{1/3}$ increases very slowly with A. You can check that $16^{1/3} = 2.52$ while $208^{1/3} = 5.93$. This means that a very heavy lead nucleus ${}^{208}Pb$ is only about 2.4 times the size of the much lighter ${}^{16}O$ nucleus.



matlab yeh hai ke bohot heavy nucleus bhi chhoti increase ke saath size mein badhta hai.

Agar humein nucleus ke andar protons ki distribution dekhni ho to electron beam nucleus pe fire ki jati hai. Electrons positive protons se scatter hote hain, aur detector mein unka angle dekha jata hai. Is tarah hum andaza laga sakte hain ke nucleus ke andar charge distribution kaisa hai.

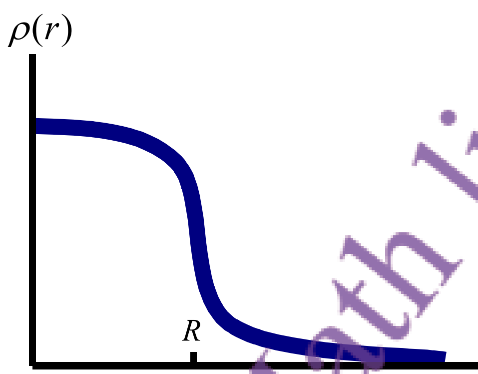
6. To learn about how protons are distributed inside a nucleus, we send a beam of electrons at a nucleus and observe how they scatter in different directions. The negatively charged electrons interact with the positively charged protons, but they obviously will not see the neutrons. The scattered electrons are captured in a detector which can be moved around to different angles. In this way one can reconstruct the charge distribution which caused the electrons to be scattered in that particular way.



7. What energy should electrons have in order to see a nucleus? We know that electrons are waves with $\lambda = h/p$ (the De Broglie relation). To see something as small as $1\,fm$ requires a wave with wavelength at least $\lambda \approx 1\,fm$. A wave with longer wavelength would simply pass over the nucleus without being disturbed. So the minimum electron energy is, $\boxed{E = \dfrac{p^2}{2m} = \dfrac{h^2}{2m\lambda^2}}$. Evaluation gives this to be a few MeV, requiring an electron accelerator of more than this minimum energy.

Electron scattering se pata chalta hai ke nucleus ke andar proton density lagbhag constant hoti hai aur sirf surface par jaake sharply girti hai. Is wajah se nucleus ko fuzzy ball samjha ja sakta hai. Distance R ≈ r_0A^(1/3) ko nuclear radius kehte hain.

8. From electron scattering we see that the proton density is almost constant throughout the nucleus and falls sharply at the surface. So nuclei should be thought of as rather fuzzy balls. A typical plot of density versus distance from the centre looks like this. Here the distance $R \approx r_0 A^{1/3}$ can be called the nuclear radius. The distribution of neutrons is very similar to this plot, but requires other techniques.



Nuclear radius ka formula use karte hue volume nikaal ke yeh estimate hota hai ke nucleus ke andar nucleon density ≈ 0.14 nucleons per cubic fermi hoti hai. Yeh value almost har nucleus ke liye same hoti hai.

9. Let us consider the implication of the approximate formula for the nuclear radius, $r \approx r_0 A^{1/3}$ where $r_0 = 1.2\ fm$. The volume of the nucleus is: $V = \frac{4}{3}\pi r^3 = \frac{4}{3}\pi r_0^{\ 3} A$. From this $\frac{A}{V} = \frac{3}{4\pi r_0^{\ 3}} \approx 0.14\ \text{nucleons/fm}^3$. This is the number of nucleons per cubic fermi, and is independent of the nucleus considered. This is the density you would find at the centre of any nucleus. Of course, this is approximately true only but it is quite remarkable.

Protons apas mein repulsion karte hain lekin phir bhi nucleus nahi phatta. Iska matlab hai ke ek strong attractive force bhi hoti hai isy strong nuclear force kehte hain.

10. Protons repel protons through the electrostatic force. So why does the nucleus not blow apart. Obviously there must be some attractive force that is stronger than this repulsion. It is, in fact, called the strong force. From what we have learned so far, we can guess some of its important features:

N-P (neutron-proton) force alag hoti hai N-N ya P-P force se.

a) Since neutron and proton distributions are almost the same, the N-P force cannot be very different from the N-N or P-P force.

Nucleons sirf nearby particles ke sath interact karte hain strong force ka range sirf 1-2 fermi hota hai.

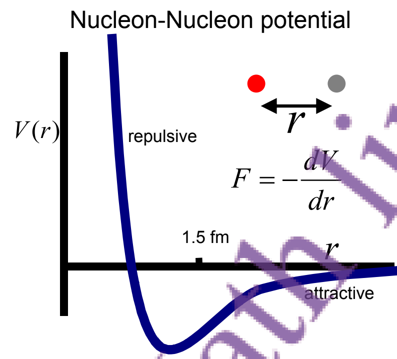
b) Since the density of nucleons in large nuclei is the same as in lighter nuclei, this means that a given nucleon feels only the force due to its immediate neighbours, and does not interact much with nucleons on the other side of the nucleus. In other words, the range of the nucleon-nucleon force is very short and of the order of 1-2 fm only.

11. The force between two charges is always of one sign - repulsive if the signs are the same, and attractive if they are opposite. In the early years of quantum theory, people realized that this force comes about because of the exchange of photons between charges. The nucleon-nucleon force is different. It has to be attractive to keep the nucleus together, and has to short range (as discussed above). But, to prevent nucleons from sticking to each other, it must be repulsive at short distances. Now here, "short" and long means distances on the scale of fermis. Typically the distances between nucleons is on this scale as well.

Charge ke darmiyan force hamesha ya to attractive hoti hai (agar opposite charge ho) ya repulsive (same charge). Lekin nucleon-nucleon force photons se nahi, balkay kisi aur mechanism se hoti hai. Yeh force short-range aur attractive hoti hai nucleus ko bind rakhne ke liye, lekin close range par repulsive bhi hoti hai takay nucleons collapse na kar jayein.

Yeh graph Nucleon-Nucleon potential ko dikhata hai. Jab r (distance) zyada hoti hai, force attractive hoti hai (kyunki slope negative hota hai).
- Jab r ≈ 1.4 fm, potential apne minimum (most attractive point) par hota hai.
- Agar r is se kam ho jaaye, to force repulsive ho jaati hai.
- Iska matlab: Nucleons ek dosray ke kareeb aane se 0.5 fm se zyada nahi aa sakte.

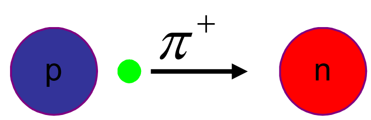
Here is roughly what the N-N potential looks like. Since the force is the negative of the slope, you can see that for large value of r, the force is attractive (negative means directed towards smaller values of r). At roughly 1.4 fm the potential reaches its most attractive point. For values of r smaller than this, the force is repulsive. In fact there is a very strong core that almost completely forbids the nucleons from getting closer than about 0.5 fm.



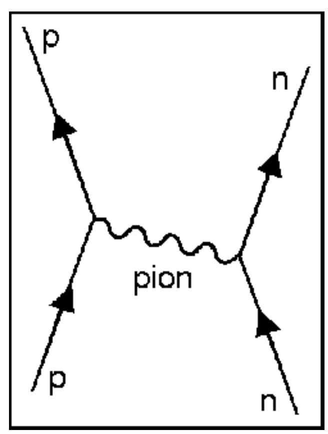
Japanese physicist Hideki Yukawa ne yeh idea diya ke strong force (N-N force) aik particle ke exchange se hoti hai jaise EM force mein photons hote hain. Usne is particle ko pi-meson (pion) kaha.

12. In 1935, the Japanese physicist Hideki Yukawa made an astonishing breakthrough in understanding the basis for the attractive N-N force. He assumed that, just as between charges, there must be a particle that is emitted by one nucleon and then captured by the other. He called this a "pi-meson" or "pion", and made a good guess for what its mass should be. His argument uses the time-energy uncertainty principle discussed earlier:

- Let Δt be the time that the pion takes between emission and capture. In this time it will have travelled a distance approximately equal to $c\Delta t$ because light particles can travel no faster than light.
- Creating a pion from "nothing" means that energy conservation has been violated. The amount of violation is ΔE = minimum energy of pion $\approx mc^2$ and this must obey $\Delta E\,\Delta t \approx \hbar/2$. Hence, $\Delta t \approx \frac{\hbar}{2mc^2}$ and the distance travelled by the pion $\approx$ c $\Delta t \approx \frac{\hbar}{2mc}$.



- Put $\frac{\hbar}{2mc} \approx 1.2\,fm$ (range of nuclear force). This gives the mass of the pion as close to $mc^2 \approx 124\,MeV$. This was an amazing prediction - the first time a particle had been predicted to exist on the basis of a theoretical argument. When experimentalists searched for it in 1947, they indeed found a particle of mass rather close to it, with $mc^2 \approx 138\,MeV$. It was a very dramatic confirmation for which Yukawa got the Nobel prize.



- Pions can rightfully be called the carriers of the strong nuclear force. They have 3 possible charge states: π^+, π^0, π^-. They belong to a larger family of particles called mesons. Other family members are rho-mesons, omega-mesons, K-mesons,... Today we can produce mesons in huge amounts by smashing nucleons against each other in an accelerator.

Pions strong nuclear force ke Carriers hote hain.
Inke 3 types hain: π⁺, π⁰, π⁻ Ye particles baryons aur mesons family ka hissa hain.
Aaj ke zamane mein pion ko accelerator se nucleons ko collide karke banaya jaata hai.

Har nucleus stable nahi hota zyada nuclei time ke sath decay karte hain.

13. A few nuclei are stable, most decay. The decay law is simply derived: if the number of nuclei decreases by dn in time dt, then dn must be proportional to both the the number of nuclei n that are decaying and dt, so $dn \propto -ndt$ (minus sign for decrease). with the proportionality constant λ, we have $dn = -\lambda ndt$, or $\frac{dn}{dt} = -\lambda n$. We have encountered the solution of this type of equation before, $n = n_0 e^{-\lambda t}$. You can see that at $t = 0$, $n = n_0$. Taking the log, we have $\ln\frac{n}{n_0} = -\lambda t$. We define the half-life $T_{\frac{1}{2}}$ as the time it takes for half the original sample to decay. If $n = n_0/2$ then $\log\frac{n_0}{2n_0} = -\lambda t$, from which the half life is related to λ by, $T_{\frac{1}{2}} = \frac{\log 2}{\lambda} = \frac{0.693}{\lambda}$. The larger λ, the more radioactively unstable a nucleus is. Some typical half-lives are:

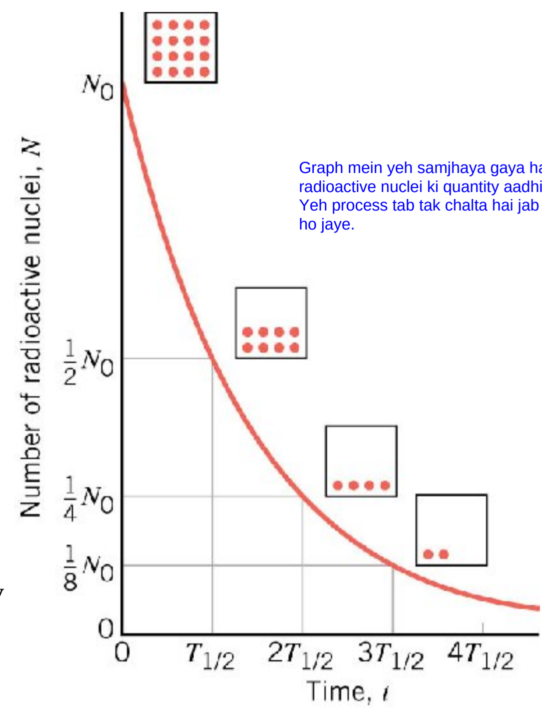
Agar ek waqt mein n nuclei hain, to unka decay rate dn/dt = -λn hota hai. Yahaan λ decay constant hai.

Zyada λ = zyada radioactive decay speed = kam half-life

Polonium	${}^{214}_{84}P$	$1.64\times10^{-4}\,s$
Krypton	${}^{89}_{36}\mathrm{K}$	3.16 minutes
Strontium	${}^{90}_{38}\mathrm{Sr}$	28.5 years
Radium	${}^{226}_{88}\mathrm{Ra}$	1600 years
Carbon	${}^{14}_{6}\mathrm{C}$	5730 years
Uranium	${}^{238}_{92}\mathrm{U}$	4.5×109 years

You can see how hugely different the lifetimes of different nuclei are!

14. Here is a plot of the number of unstable nuclei left as a function of time. After each half-life, the number of nuclei decreases in number by half of the previous. Eventually there is only one nucleus left, and that too will eventually decay. So how can the derivation for the decay law be correct? Strictly speaking, we are not allowed to write down, or solve, a differential equation like $\frac{dn}{dt} = -\lambda n$ because this assumes that $n(t)$ is a continuous function. But this is almost true because in real life we deal with very large numbers of nuclei and so it makes a lot of sense to think of $n(t)$ as continuous.



Graph mein yeh samjhaya gaya hai ke har half-life ke baad radioactive nuclei ki quantity aadhi hoti jaati hai.
Yeh process tab tak chalta hai jab tak last nucleus bhi decay na ho jaye.

Radon (^{222}Rn) aik dangerous gas hai jo zameen ke neeche milti hai. Yeh ^{218}Po (bohot poisonous) aur helium nucleus (^4He, harmless) mein decay hoti hai. Iski half-life 3.8 din hai.

15. Just to get an idea, consider the decay of $^{222}_{86}$Rn (Radon, a very dangerous gas that is found underground) into $^{218}_{84}$Po (Polonium, another terrible poison) and $^{4}_{2}$He (harmless, fortunately!). The half life is 3.8 days. So, if we started with 20,000 atoms of $^{222}_{86}$Rn, then in 3.8 days we would have 10,000 atoms of $^{222}_{86}$Rn and 10,000 atoms of $^{218}_{84}$Po In 7.6 days we would have 5000 atoms of $^{222}_{86}$Rn, in 11.4 days, 2500 , $^{222}_{86}$Rn etc.

Radioactive dating ka use cheezon ki age maloom karne ke liye hota hai.

16. The decay law can be used to see how old things are. This is called radioactive dating. Carbon dating is widely used for living things that died a few hundred or few thousand years ago. How does it work?This uses the decay of the unstable isotope, $^{14}_{6}C$. Of course, the stable isotope of carbon is $^{12}_{6}C$.

- When a living organism dies, CO_2 is no longer absorbed. Thus the ratio of carbon 14:12 decreases by half every 5730 years. We can measure the rate of decrease through $N = N_o e^{-\lambda t}$ or the "activity" A = $A_o \; e^{-\lambda t}$ with $A_o = 0.23\ Bq/g$. (The becquerel Bq is the unit of radioactivity, defined as the activity of a quantity of radioactive material in which one nucleus decays per second.)
- The amount of isotopes in the atmosphere is approximately constant, despite a half-life of 5730 y because there is a constant replenishment of $^{14}_{6}C$ through the reaction, $^{14}_{7}N + n \rightarrow {}^{14}_{6}C + p$

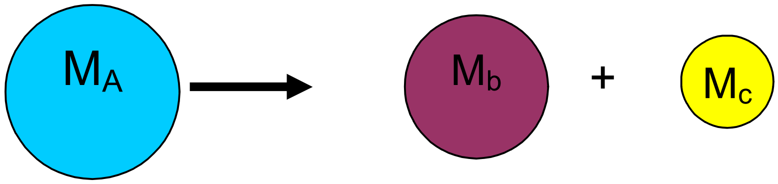
17. Let us use the above idea to find the time when this man died. His body was found a few years ago buried under deep snow in a mountain pass, so it it did not decay as usual. By looking at the radioactivity in his body, it was found that that the activity of $^{14}_{6}C$ was 0.121 Bq/g of body tissue. This is less than the normal activity 0.23 Bq/g. because $^{14}_{6}C$ has been decaying away. First find λ, $\lambda = \dfrac{0.693}{T_{1/2}} = \dfrac{0.693}{5730} = 1.21\times10^{-4} \quad y^{-1}$

Then use, $0.121 = 0.23 \;\; e^{-1.21\times10^{-4}\times t}$ which gives,

$\ln\dfrac{0.121}{0.23} = -1.21\times10^{-4}\times t$ and so $t = 5300$ years is when this poor man was killed (or died somehow)!

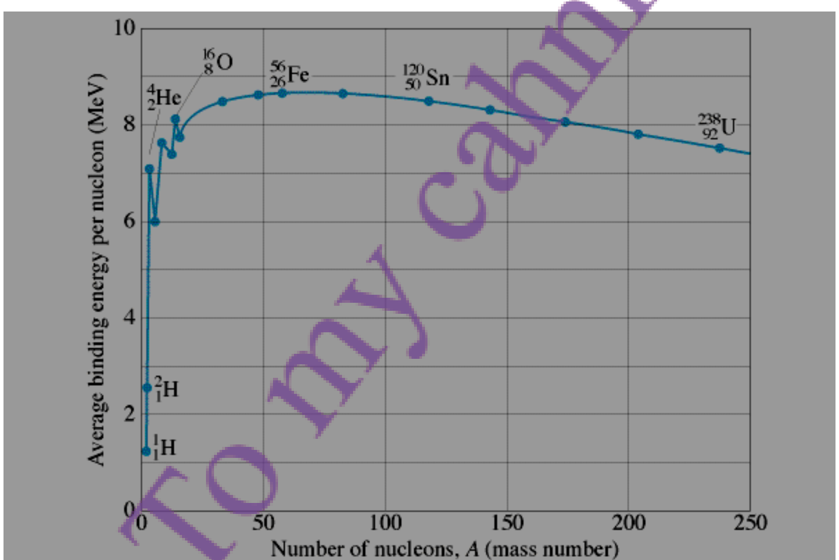
18. The most famous formula of physics, $E = mc^2$, is the basis for nuclear energy. In 1935, it was discovered by two German physicists, Otto Robert Frisch and Lise Meitner, that a heavy nucleus can fission (or break up) into two or more smaller nuclei. The total energy is, of course, conserved but the mass is not. This is completely different from the

usual situation. In the picture below you see an example of fission.



The masses of the two nuclei add up to less than the mass of the parent nucleus, and the energy released is $Q = (M_A - M_b - M_c)c^2$. This goes into kinetic energy and sends the two daughter nuclei flying apart at a large velocity. There happen to be NO completely stable nuclei above $Z = 82$, and no naturally occurring nuclei above $Z = 92$. Above these limits the nuclei decay or fall apart in some fashion to get below these limits.

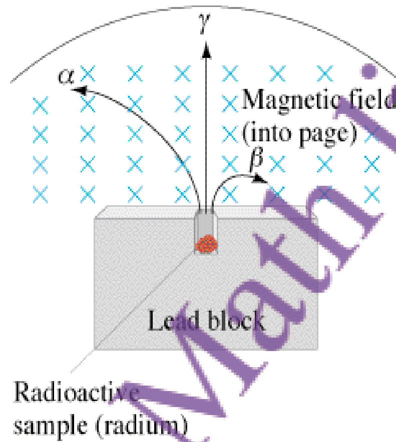
19. A very useful concept is "binding energy". Suppose you want to take a nucleon out of a nucleus. The binding energy is the amount of energy that you would have to provide to pull it on the average. Nuclei with the largest BE per nucleon are the most stable. As you



can see from the graph below, the most stable element is iron, ^{56}Fe with a BE per nucleon of about 8.6 MeV. This is why iron is the heavy element found in the largest quantity on earth and inside stars. The curve is not smmoth and you see that a ^{4}He nucleus (i.e. an α parrticle, has a relatively high binding energy and so is relatively stable. In contrast, Uranium ^{235}U or deuterium ^{2}H are much less bound and they decay.

20. Nuclei can be unstable in different ways. A nucleus can emit $\alpha, \beta,$ and γ radiations. Usually a nucleus will emit one of these three, but it is possible to emit two, or even all three of these. (In addition, as we have discussed above, a nucleus can break up into

two or more smaller fragments through the process of fission.) What is the nature of α, β, γ radiations? In the experiment shown here, you see that a piece of radium has been put in a lead block (as shielding). A magnetic field bends the charged particles emitted during the decay. We find that they are of three types:

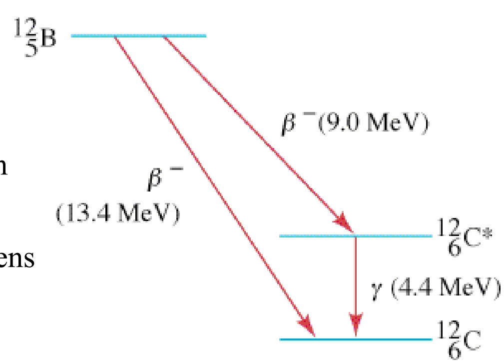


a)Heavy positively charged particles are bent to one side by the magnetic field. The amount of bending shows that they are α particles.

b)Other particles are bent much more, and in the other direction. They are electrons (β particles).

c)Some particles are not bent at all, hence must be neutral. They are γ particles (or rays, or γ photons, same thing!).

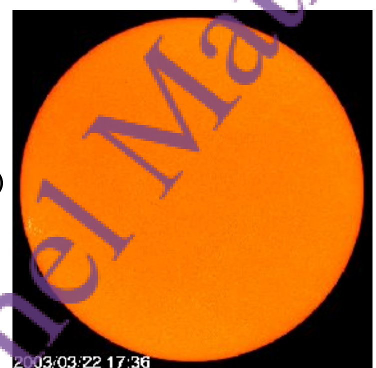
21. Let's first consider alpha decay, ${}^{A}_{Z}X \rightarrow {}^{A-4}_{Z-2}X' + {}^{4}_{2}He$. As you see A changes by 4 and Z by 2. You can think of α particles as a gang of 4 particles that always stays together in a big nucleus. Unstable nuclei simply cannot overcome the proton repulsion and an α particle ultimately succeeds in esaping the nucleus. indefinitely.
22. The simplest beta decay reaction is when a neutron decays, $n \rightarrow p + e^{-} + \bar{\nu}$. Other than a proton and electron, an anti-neutrino is also emitted. As discussed in the lecture, the (anti) neutrino is a neutral particle with a very tiny mass that interacts very weakly with matter. A nucleus can undergo beta decay with either an electron being produced, i.e ${}^{A}_{Z}X \rightarrow {}_{Z+1}{}^{A}X' + e^{-} + \bar{\nu}$ (called β^{-} decay) or with an anti-electron (positron, or positive electron), ${}^{A}_{Z}X \rightarrow {}_{Z-1}{}^{A}X' + e^{+} + \nu$, (called β^{+} decay). Beta decay involves the weak nuclear force. This is one of the four fundamental forces in the world, and its small strength means that the decay happens much more slowly than most other reactions.
23. Just as for an atom, a nucleus can only exist in certain definite energy states. When a nucleus goes from one state to the other, it can emit a photon (γ ray). Because the spacing between nuclear levels is of the order of one MeV (i.e. a million times more than in atoms), the photon is much more energetic than an optical photon. In going between levels, β emission also happens as can be seen in this diagram.



Summary of Lecture 45 – PHYSICS OF THE SUN

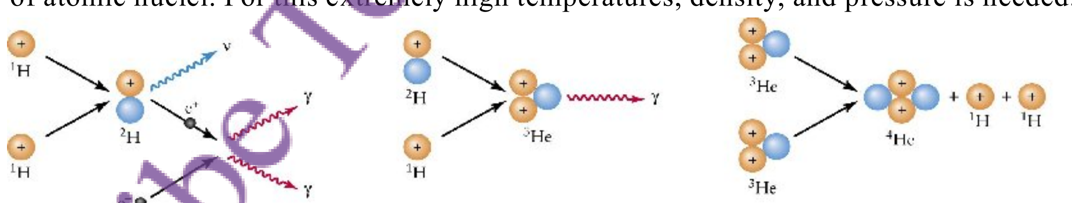
1. In this lecture I shall pull together the different parts of physics that you have learned in this course and apply them to understanding the source of all life on earth - our sun. We will learn that the sun operates according to principles that we can understand, and on the basis of this we can even predict the manner in which it will eventually die.

2. Basic solar facts:
   a) Mass of sun $= 2\times10^{30}$ kg = 333,000 Earth's
   b) Diameter of sun = 1,392,000 km = 10^{9} Earth's
   c) Age of sun = 4.6 billion years
   d) Rotation Period = 25 days at equator, 36 at poles (surface)
   e) Temperature = 15 million ^{0}K at core, 5770 ^{0}K at surface
   f) Density = $8\times$ gold at the core, average is ~ 1.5 water
   g) Composition: 72% H, 25% He, rest is metals



3. The sun puts out a huge amount of energy. In quantitative terms we measure this in by its luminosity, 3.83×10^{27} joules per second. The power output is 3.83×10^{24} kilowatts. This is equal to 8×10^{16} of the largest power plants on Earth, meaning those which produce ~5000 MW of power. Another way of expressing this: every second the sun puts out as much energy as 2.5×10^{9} (2 billion) such power plants would put out every year.

4. What powers the sun? The earth is very old (billions of years). If there was a chemical fuel (say, coal or oil) at most that would last a few million years. But the sun is many thousands of tons older than that. Only after the discovery of $E = mc^2$ did we know the real secret. The sun gets its energy from the fusion (the coming together and combining of atomic nuclei. For this extremely high temperatures, density, and pressure is needed.

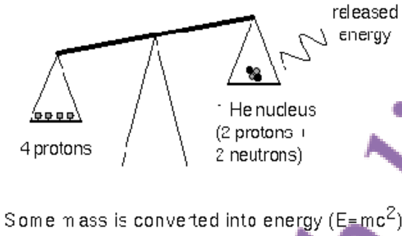


a Step 1:
- Two protons (hydrogen nuclei, ^{1}H) collide.
- One of the protons changes into a neutron (shown in blue), a neutral, nearly massless neutrino (ν), and a positron (e^{+}), a positively charged electron.
- The proton and neutron form a hydrogen isotope (^{2}H).
- The positron encounters an ordinary electron (e^{-}), annihilating both particles and converting them into gamma-ray photons (γ).

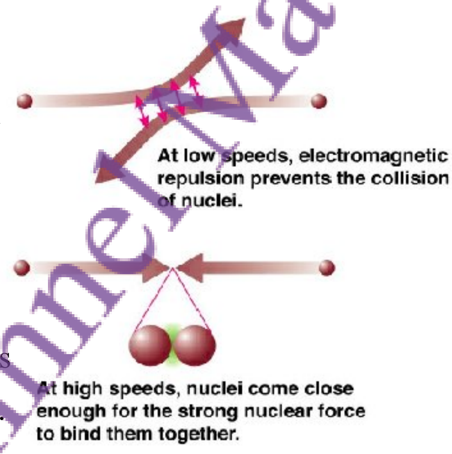
b Step 2:
- The ^{2}H nucleus from the first step collides with a third proton.
- A helium isotope (^{3}He) is formed and another gamma-ray photon is released.

c Step 3:
- Two ^{3}He nuclei collide.
- A different helium isotope with two protons and two neutrons (^{4}He) is formed and two protons are released.

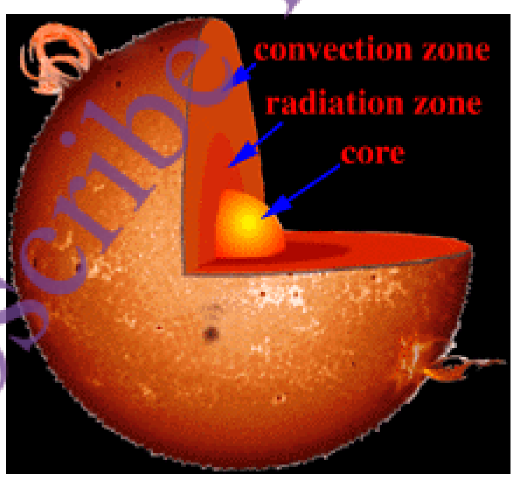
4. The basic point is that, as you can see in the picture, the combined mass of 4 protons is higher than than that of the helium nuclei into which they convert through the process shown earlier. The difference then appears in the form of kinetic energy of the released particles, which in random form is heat.

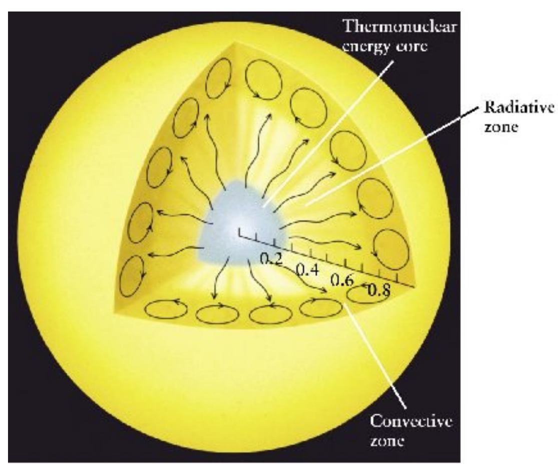


5. Protons repel protons, but the only way in which they will participate in a fusion reaction is when they can come sufficiently close. This requires that they smash into each other at sufficiently high speeds, and hence nuclear fusion in the sun requires core temperatures greater than about 8 million ^{0}K. He nuclei can also fuse with each other to release energy, but because they are heavier much higher temperatures & densities (about 100 million ^{0}K for helium fusion) are required. Thus, stars fuse hydrogen first. Each second, the sun turns 4 million tons of hydrogen into energy.



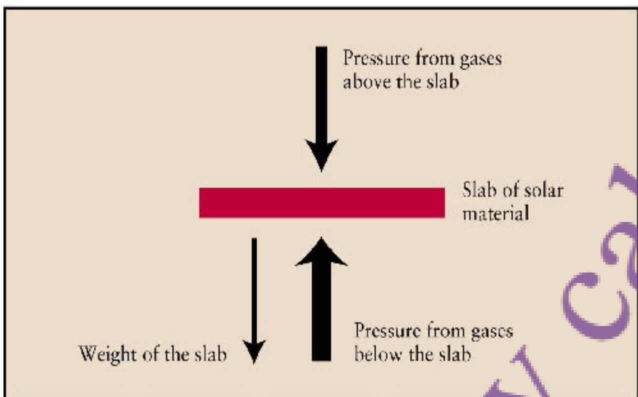
6. Fusion takes place only in the core of the sun (see diagrams below) because it is only there that the hydrogen gas is hot enough. From the core, the heat gets out by the emission of photons (radiation zone). The hot gas then exchanges heat with the sun's relatively cold exterior through convective currents. Huge columns of hot gas move from the inside towards the surface. After giving up most of the heat, the "cold" gas sinks towards the centre and the cycle goes on.

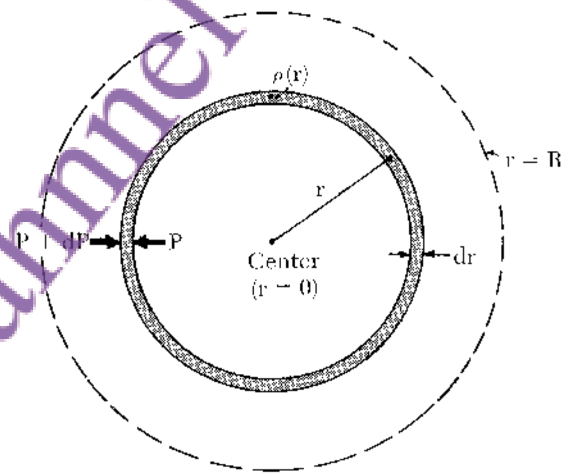




7. For over 4 billion years the sun has been nearly steady in maintaining its size. It is in a state of equilibrium between two big forces acting oppositely to each other:
   a) The hot hydrogen gas seeks to expand outwards because the thermal velocity of H atoms leads to a pressure directed outwards.
   b) Gravity tries to squeeze the star inwards because every piece of matter attracts every other piece. So the gravitational pressure is inward, increasing toward the core.

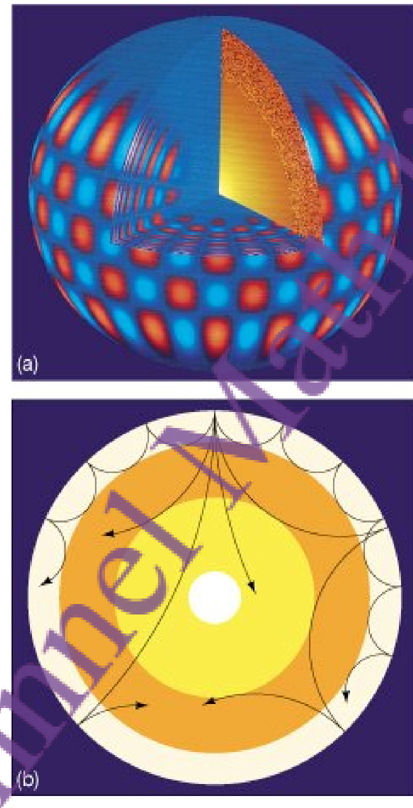
   As in the diagrams below, imagine a piece of solar material at some distance from the centre of the sun. The two forces acting upon it must exactly balance in equilibrium. If for some reason the sun cools down, the pressure of gas will decrease and the sun will contract to a new equilibrium point. Conversely, the sun will expand if the rate of fusion were to increase.



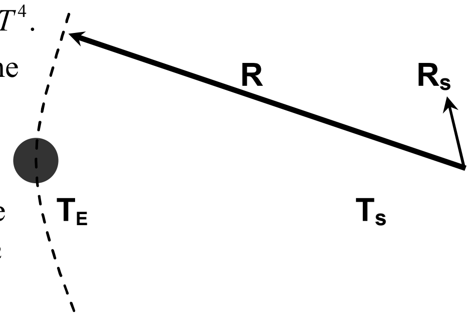


Look at the second diagram. Let $M(r)$ be the mass contained within radius r. We will not assume that the density is constant in r. First find the inward directed gravitational force on a shell of matter at radius r and thickness dr, $dF = -\dfrac{GM(r)\times\rho(r)4\pi r^2 dr}{r^2}$. There is a net pressure as shown, and we will call dP the difference of pressures. Then obviously $dF = dP\times 4\pi r^2$. Hence, $dP = -\dfrac{GM(r)\rho(r)dr}{r^2}$ or, $\dfrac{dP}{dr} = -\dfrac{GM(r)\rho(r)}{r^2}$. The total mass upto radius r is, $M(r) = \int_0^r \rho(r')4\pi r'^2 dr'$. If we knew $\rho(r)$ then $M(r)$ would also be known. By solving the differential equation, we would also then know $P(r)$. To give us a better understanding, suppose for simplicity that the sun is approximately uniform. Then the density is, $\rho \approx \dfrac{M}{(4\pi/3)R^3} \approx 1.4\text{gm cm}^{-3}$. Hence the pressure at the centre of the sun can be computed, $P_{centre} \approx \dfrac{GM\rho}{R} \approx 3\times 10^9$ atmospheres.

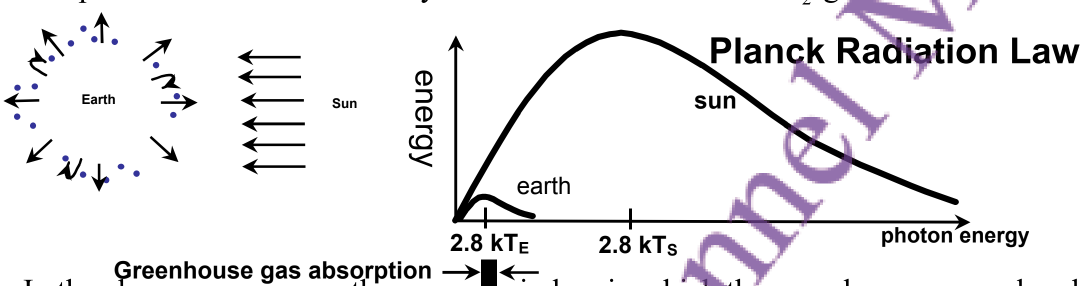
8. The equilibrium of forces is a nearly perfect one, but there are small disturbances and these cause "earthquakes" (actually sun-quakes) to occur on the surface. The study of the surface of the sun and sun-quakes is an area known as "helioseismology" (helio=sun, seismology is the study of vibrations and earthquakes). The pulsating motions of the sun can be seen by measuring the Doppler shifts of hydrogen lines across the face of the sun. Some parts are expanding towards the earth while adjacent regions contract away. This is like the modes of a ringing bell. Vibrations propagate inside the sun, and the waves travel through a hot dense gas. They experience reflection and refraction since the speed at which a wave travels changes when it goes from a region of high to low density or vice-versa. All this is being studied by astrophysicists today as they map out the sun's interior.



9. Let us calculate the surface temperature of a planet circulating the sun. We shall use the Stefan Boltzman law that you studied in the lecture of heat: the power radiated by a black body per unit surface area at temperature T is σT^4. For thermal equilibrium, we must have that all the power absorbed from the sun is re-radiated by the planet. Let P_S be the sun's flux at its surface, Then, $P_S = \sigma T_S^4$. Since radiation decreases by the distance squared, P_R = flux at earth $= \sigma T_S^4 \times \left(\frac{R_S}{R}\right)^2$ This must be multiplied by πR_E^2, which is the crossectional area of the planet. On the other hand, for emission from the planet, P_E = flux at its surface $= \sigma T_E^4$. We must multiply this by $4\pi R_E^2$, the area of planet, to get the total radiated power. Now impose the equilibrium condition: $P_R \times \pi R_E^2 = P_E \times 4\pi R_E^2$. This gives $\sigma T_S^4 \times \left(\frac{R_S}{R}\right)^2 = 4\sigma T_E^4$, and hence $T_E = \left(\frac{R_S}{2R}\right)^{1/2} T_S$. This is true for any planet, so let's see what this gives for the earth using $R_S = 7\times 10^8$ metres, $R = 1.5\times 10^{11}$ metres, and $T_S = 5800^o K$. This gives $T_E = 280K$, which is very sensible! Of course, we have assumed that the earth absorbs and emits as a black body. True?



10. Actually, the black body assumption is only approximately true. About 30% of the sun's radiation reflects off our atmosphere! In fact the average surface temperature of the earth is about 290K, or about 13 ^{0}C. The extra warming is mainly due to the "greenhouse effect". Certain gases trap the sun's radiation and it is not able to get out, thus leading to greater absorption. The main greenhouse gases are CO_2, CH_4, H_2O. (N_2, O_2 and Ar are transparent to solar and earth radiation.) CO_2 is now 360 ppm (parts per million) of the atmosphere, up from 227 ppm in 1750, before the industrial revolution. Plants turn H_2O + CO_2 into O_2 plus organics. Some estimates predict a 3 to 10 ^{0}C rise in the earth's surface temperature over the next 100 years due to the increased CO_2 greenhouse effect.

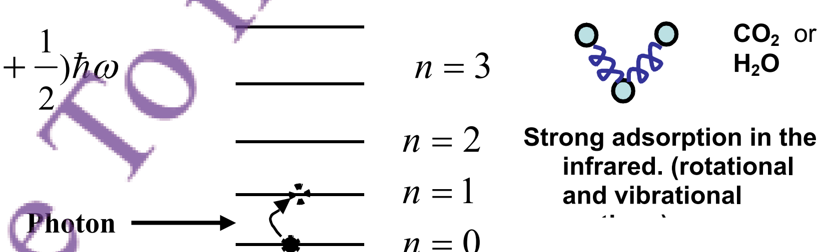


In the above, you can see the narrow window in which the greenhouse gases absorb the sun's radiation. But why are only some gases responsible, and not others?

11. The answer lies in quantum mechanics. In a previous lecture you learned that molecules can exist only in certain states that have very specific values of energy. For example, a molecule of CO_2 can oscillate and have equally spaced energy levels as shown below:

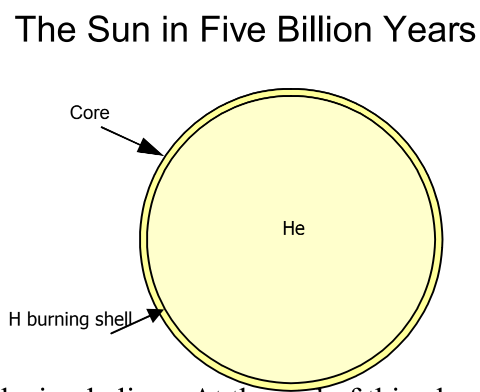
Energy levels of an oscillator:

$$\varepsilon_n = (n + \frac{1}{2})\hbar\omega$$



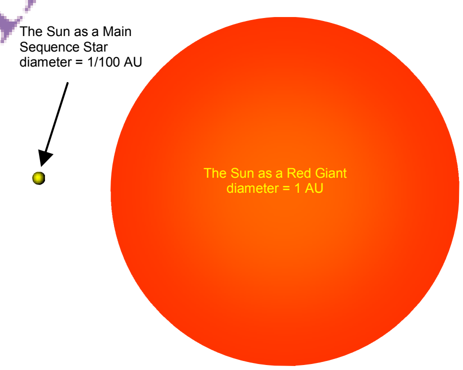
The energy at which these molecules can absorb radiation happen to lie in the spectrum of the re-radiated energy from the earth's surface. Thus, they effectively trap the out-going radiation, leading to enhanced earth temperatures. For any molecule, we can both calculate (using quantum mechanics) the frequencies at which radiation is absorbed or emitted. We can also experimentally measure the frequencies. Both of these are in very good agreement, and this is one of the reasons why we have such confidence in the correctness of quantum mechanics.

12. Finally, I have some bad news: the sun is going to die because the hydrogen supply is eventually going to run out. There is, of course, no immediate danger - the Sun can last another 5 billion years on core hydrogen fusion. During this time it will be consuming



hydrogen and producing helium. At the end of this phase, the core of the sun will have mostly helium with the little bit of hydrogen left almost entirely outside in a thin layer. The reaction rate will fall, and the core will no longer be able to balance the pull due to gravity. This will cause a shrinkage of the core. As a consequence the temperature will increase. A new phase of the sun is about to start.

13. At this point a fusion reaction in next core zone begins. This lifts the envelopes and the sun will brighten up significantly. It is now a Red Giant star, and as you can see below, it is huge! The earth will be swallowed up by the expanded sun.



14. As remarked earlier, helium can also "burn" to produce carbon, but it needs a much higher temperature. Eventually even the He will be depleted, and the reaction rate will fall. Again the small reaction rate means that the core will shrink, and heat up to the point that it crosses the carbon fusion threshold of 600 million K. Low mass stars cannot reach this temperature. Envelope expands to a supergiant, many times larger than even a Red Giant.